The local Lamperti law: point probabilities for the occupation time of on/off processes with infinite-mean periods

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Abstract

Let A_N be the number of time slots spent in the on state during the first N slots of a discrete-time on/off process whose on and off periods are independent and have regularly varying tails with indices in $(0,1)$, so that the periods have infinite mean. Equivalently, A_N counts the right steps of a lattice Lévy walk in the ballistic regime. When the two tails are comparable, the limit law of A_N/N is the generalized arcsine law of Lamperti. We prove its local form. The rescaled point probabilities $N\mathbb{P}(A_N = k)$ converge to Lamperti's density at one interior point if and only if they converge uniformly on compact sets, and this happens if and only if both period laws satisfy the strong renewal theorem. Near the ends the law is the tail of a period times a renewal mass function, which matches the bulk uniformly under Doney's local condition, and the global mode sits at bounded distance from an end for every index. We give the exact effect of an arbitrary initial period, and we prove that an initial period of length of order N turns the law into an explicit mixture. Hence the process observed from time θN on has a local aging law, with atoms of order 1 at the endpoints and end layers of the fresh shape and a smaller height. When one tail is heavier, the time on the lighter side has a local limit law at an intermediate scale, with no renewal condition when the lighter mean is finite or its index exceeds $1/2$. We classify the critical points of Lamperti's density for every index and every tail ratio, with a closed-form window for the W shape, and we count the modes of the discrete law near the ends exactly, which leaves one second-difference local limit theorem in the bulk between a lower bound and the exact count.

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1 Introduction and main results

An on/off process alternates on periods and off periods with independent lengths. Let the on periods have the law of X_+ and the off periods the law of X_-, two laws on $\{1, 2, \dots\}$ with regularly varying tails $T_\pm(n) = \mathbb{P}(X_\pm \geq n)$ of index $\alpha \in (0, 1)$, so that the periods have infinite mean, and let A_N be the number of time slots spent in the on state among the first N. Then A_N/N does not concentrate. If $T_-(n)/T_+(n) \to b \in (0, \infty)$, its limit law is the generalized arcsine law with density

$$L_{\alpha,b}(x) = \frac{\sin \pi\alpha}{\pi} \frac{b\, x^{\alpha-1}(1-x)^{\alpha-1}}{b^2 x^{2\alpha} + 2b\cos(\pi\alpha)\, x^\alpha (1-x)^\alpha + (1-x)^{2\alpha}}, \qquad 0 < x < 1, \tag{1.1}$$

which goes back to Lamperti [20] and is called the Lamperti distribution in the physics literature [21, 29]. This paper is about the local form of this law, at single points k, and about what happens at the ends of the range, under an initial period of a different law, under aging, and when one tail is heavier than the other. The same process is a lattice Lévy walk in the ballistic regime, which moves in runs of alternating direction and sits at $2A_N - N$ after N steps, so $P_N(k) = \mathbb{P}(A_N = k)$ is also the law of the walk on its natural lattice. We keep the vocabulary of the walk in the body of the paper. A period is a run, an on slot is a plus step, and the two ends of the range are the plus end $k = N$ and the minus end $k = 0$. The companion paper [23] treats the finite-mean case $1 < \alpha < 2$, where $A_N - pN$ is stable at the scale $N^{1/\alpha}$, and the two papers share the exact identities of Section 2.

1.1 Main results

The local Lamperti law. A renewal law with a tail T of index $\alpha \in (0, 1)$ satisfies the strong renewal theorem if its renewal mass function u satisfies $u(n) \sim \frac{\sin \pi\alpha}{\pi} \frac{1}{nT(n)}$. For $\alpha > 1/2$ this always holds under aperiodicity, by Garsia and Lamperti [15] and Erickson [13]. For $\alpha \leq 1/2$ it can fail, and Caravenna and Doney [8] found the exact condition. Our first theorem says that the local Lamperti law is equivalent to the strong renewal theorem for both run laws.

Theorem A (the local Lamperti law)**.** *Assume that T_+ and T_- are regularly varying with index $-\alpha$, $0 < \alpha < 1$, that $T_-(n)/T_+(n) \to b \in (0, \infty)$, and that the run laws are aperiodic. Let the walk start with a fresh run of either sign. Then the following are equivalent.*

(i) $NP_N(\lfloor x_0 N \rfloor) \to L_{\alpha,b}(x_0)$ *for one* $x_0 \in (0, 1)$.

(ii) $NP_N(k) = L_{\alpha,b}(k/N) + o(1)$ *uniformly over* k *with* k/N *in each compact subset of* $(0, 1)$.

(iii) Both run laws satisfy the strong renewal theorem.

With the theorem of Caravenna and Doney this gives a complete answer. For $\alpha > 1/2$ the local law holds under regular variation and aperiodicity alone. For $\alpha \leq 1/2$ it holds if and only if each run law satisfies their condition, which we recall in Lemma 3.1, and Doney's local condition $f_\pm(n) = O(T_\pm(n)/n)$ suffices. A single interior point already forces the strong renewal theorem for both laws. Without it the lower half of the local law still holds, $\liminf_N (NP_N(k) - L_{\alpha,b}(k/N)) \geq 0$ uniformly on compact sets, and the failure shows up as an excess at every interior point (Corollary 4.3). The proof does not need a bivariate renewal theorem. An exact identity writes $P_N(k)$ through the convolution powers of the two run laws, sorted by the number m of completed runs. For m of order $1/T_+(N)$ the local limit theorem of Gnedenko gives a Riemann sum for an explicit double integral. The key estimate, Lemma 3.5, bounds the probability that the $(m+1)$-th run

covers a level k by $CT(k)$ uniformly in m, under regular variation alone. It decouples the two signs for small m, and the contribution of small m of each sign is then exactly the quantity that Caravenna and Doney show to be negligible if and only if the strong renewal theorem holds. We identify the limit with $L_{\alpha,b}$ through its Stieltjes transform (Proposition 4.1), which gives a self-contained derivation of Lamperti's density.

The ends. At distance j from the plus end the law is the tail of a run times a renewal mass function, $P_N(N-j) = T_+(N)u_-(j) + o(T_+(N))$, under regular variation of T_+ alone. Under Doney's condition this boundary layer matches the bulk uniformly up to distance $o(N)$ from the end (Theorem B), and under the strong renewal theorem alone the upper half of the matching holds for $\alpha \neq 1/2$ (Lemma 5.1). Since $NT_+(N) \to \infty$, the end values are much larger than the bulk values, so the global mode lies at bounded distance from an end for every $\alpha \in (0,1)$ (Corollary 5.2). For $1 < \alpha < 2$ the same comparison has a golden ratio threshold [23], and here it has none.

The initial run. Wang, Schulz, Deng and Barkai [26] left "the effect of initial preparation on the rare fluctuations" for future work [26, Section VIII]. For the occupation count with $1 < \alpha < 2$ this is answered in [23], and Theorem C answers it for $0 < \alpha < 1$. Let the first run have sign σ and length d with probability $\nu_\sigma(d)$. In the finite-mean case the bulk needs no condition on the initial run on any macroscopic interior. Here it does. A first run that ends just before the k-th plus step leaves a remainder that sits in its own boundary layer, where the probabilities are of order $T(N)$ and not $1/N$. Theorem C gives the exact defect in the bulk, the necessary condition $\nu_\pm(n) = o(1/(nT_+(n)))$, the sufficient condition $n\nu_\pm(n) \to 0$ under Doney's condition, and an exact criterion. It also shows by example that an initial run which is finite almost surely can change the local law at interior points, and that the necessary condition is not sufficient. At the ends the criterion becomes a window condition.

Initial runs of length of order N, and aging. If the law of the initial run depends on N and puts mass of order 1 on lengths of order N, the local law changes to a mixture of rescaled Lamperti densities. Theorem D proves this under the strong renewal theorem for both laws. The near-diagonal terms, the initial runs that end within $o(N)$ steps of the point, are controlled by a counting bound (Lemma 7.1) and not by the end-layer theorem, which is why Doney's condition is not needed. The main application is aging. Let the process start fresh at time 0 and count the plus steps among the steps $t_a + 1, \dots, t_a + N$, with $t_a/N \to \theta$. The run in progress at time t_a is an initial run of length of order N, with the marked Dynkin–Lamperti law of Lemma 7.3, and Theorem E gives the aged law at every point. In the bulk it is the mixture L_θ of (7.4), which is the aging generalized arcsine law of Akimoto, Sera, Yamato and Yano [2], there obtained in distribution by a scaling argument. At the endpoints there are atoms of order 1, with the closed form (7.5). In the end layers the shape $u_-(j)$ of the fresh start survives and its height is multiplied by an explicit constant $c_+(\theta) < 1$, with $c_+(1) = 1/(2\sqrt{2})$ for $\alpha = 1/2$ and equal tails. The limits $\theta \to 0$ and $N \to \infty$ commute in the bulk and not at the ends.

One tail heavier. When $T_- = o(T_+)$ and $\alpha_+ < 1$, the minus side may have an infinite mean with any index $\alpha_- \in (0,1)$ or a finite mean with $\alpha_- \in (1,2)$. Then $A_N/N \to 1$, and the object is the time $N - A_N$ on the lighter side, which lives at the scale $D_N = \mu_-/T_+(N)$ when the lighter mean is finite and at the scale D_N given by $T_-(D_N) \sim T_+(N)$ when it is infinite. The limit in distribution is a Mittag-Leffler law in the first case and the law of $E^{1/\alpha_-}Z_-$ in the second, with E Mittag-Leffler and Z_- stable, and it is known in distribution through Darling and Kac [10], Cénac, Le Ny, de Loynes and Offret [9, Section 4.3] and Yamato [28]. Theorem F gives the local law. It holds under regular variation and aperiodicity alone when the lighter mean is finite or $\alpha_- > 1/2$, and for $\alpha_- \leq 1/2$ exactly when the lighter law satisfies the strong renewal theorem and the heavier law satisfies a weak condition (W_+), which holds under its own strong renewal

theorem. The strong renewal theorem of the heavier law never enters. The end layer at the heavier end is $T_+(N)u_-(j)$ and joins the bulk at the same height, and the layer at the lighter end is $T_-(N)u_+(i)$. The proof needs one new estimate, the local form of the inverse renewal count of the heavier runs (Lemma 3.8), and a uniform one-big-jump estimate for sums of a growing number of heavy-tailed summands (Lemma 3.3), which we prove by a Chernoff bound. Together with [23], this covers every pair of indices in $(0,1) \cup (1,2)$ for which T_-/T_+ has a limit.

The shape. For equal tails the density $L_{\alpha,1}$ is U-shaped for small α and W-shaped for α close to 1, with the transition at $\alpha_c = 0.5946\ldots$, the root of $\cos(\pi\alpha/2) = \alpha$. The threshold is in Baldassarri, Bouchaud, Dornic and Godrèche [3] and in Godrèche and Luck [16]. Theorem G classifies the critical points of $L_{\alpha,b}$ for every α and every b. The density has 3 critical points if and only if $|\log b| < \beta_c(\alpha)$, where β_c is given in closed form, vanishes for $\alpha \le \alpha_c$ and is infinite for $\alpha \ge \alpha_1 = 0.7365\ldots$, the root of $\sin(\pi\alpha) = \alpha$, so that beyond α_1 the W shape survives every asymmetry. The proof reduces the critical points to the level set of one function of one variable whose monotonicity we determine. At $b = 0$ the dominated regime takes over, and its ratio density gains an interior maximum exactly at α_1, so the two shape thresholds meet. The discrete law has more extrema than the density. The end profiles add local maxima of their own, and Theorem 9.3 gives the lower bound $\sigma_L + \sigma_+ + \sigma_-$ on the number of sign changes of the first differences, with $\sigma_L \in \{1,3\}$ from Theorem G and $\sigma_\pm$ from the profiles. In the layers the first differences of P_N follow those of the renewal mass function (Theorem 9.7), which rests on a difference form of Doney's local bound that is uniform in the number of summands (Lemma 9.5), and a difference form of the strong renewal theorem holds for log-convex laws (Lemma 9.1). Proposition 9.8 shows that one statement about the bulk, a second-difference local limit theorem, turns the bound into the exact count $M_+ + M_- + m_L$ of modes for all large N, with $m_L \in \{0,1\}$ the number of interior maxima of $L_{\alpha,b}$, and we leave that statement open. In every computed case the bound is attained.

1.2 Related work

Occupation times. Lamperti [20] proved an occupation time theorem for a class of processes with a regeneration state, which allows two sojourn laws with different weights, and the density $L_{\alpha,b}$ is attached to his name. Mitov and Yanev [22] study limit theorems for alternating renewal processes with infinite mean cycles, extending results of Dynkin, Lamperti and Erickson. Takács [25] and Kesten [19] proved the limit laws under finite means, and Starreveld, Bekker and Mandjes [24] give the transform of the occupation time of an alternating renewal process at an exponential time, a central limit theorem under finite second moments, and large deviations under exponential moments. Godrèche and Luck [16] give the transform of the occupation time and its limit laws for one sojourn law, including the threshold α_c for the shape. All of these are statements in distribution. We found no local limit theorem for the occupation count of an alternating renewal process with infinite means, and no treatment of the dominated regime at the local level.

Strong renewal theorems. The strong renewal theorem for $\alpha > 1/2$ is due to Garsia and Lamperti [15], Williamson [27] and Erickson [13]. Doney [12] proved it for all $\alpha \in (0,1)$ under the local condition $f(n) = O(T(n)/n)$, together with a local large deviation bound. Caravenna and Doney [8] gave the necessary and sufficient condition for $\alpha \le 1/2$, and their local large deviation bound for sums of a fixed number of variables is the main input of our key estimate. Berger [5] extends Doney's local bound, and his version is the one we quote. Berger [4] also proved multivariate strong renewal theorems. Theorem A shows that for the occupation count the bivariate problem reduces exactly to the two univariate ones.

Physics. In the physics literature the walk is the Lévy walk in the ballistic regime [29], and the occupation count is the time average of a blinking signal with power-law sojourn times [21]. Both sources use $L_{\alpha,1}$ as the limit density. Wang, Schulz, Deng and Barkai [26, Section VII.A] found an atom with a boundary layer for the occupation time when $0 < \alpha < 1$, for one sojourn law and a start of fair random sign, through a double Laplace transform, and Godrèche and Luck [16, Section 7] have the same structure. Theorem B is a lattice version of this layer at a fixed time, for two laws, with the uniform matching to the bulk. Aging of renewal processes with power-law sojourns is a standard theme there, and Akimoto, Sera, Yamato and Yano [2] proved the aged arcsine law for Brownian motion and stated its generalization to two-state renewal processes with a scaling argument. Akimoto [1] computes moments of the aged occupation time. Theorem E is the local form, with the atoms and the end layers.

The finite-mean case. For $1 < \alpha < 2$ the occupation law has a stable bulk, big-jump zones and boundary layers, and its initial-run dependence, its aging and its modes are described in [23]. We use the exact identities of that paper and recall them with proofs in Section 2.5. The two papers together describe the local law of A_N on the square $(0,1) \cup (1,2)$ of index pairs, up to the lines $\alpha_\sigma = 1$, which Section 10 leaves open.

1.3 Structure of the paper

Section 2 fixes notation and proves the exact identities. Section 3 collects the cited theorems and proves the elementary bounds, the key estimate, the one-big-jump estimate and the local inverse renewal count. Section 4 proves Theorem A, Section 5 treats the ends and the global mode, Section 6 the initial run, Section 7 the mixture law and aging, Section 8 the dominated regime, and Section 9 the shape. Section 10 lists the open problems and Section 11 describes the computations. No proof uses a computed number.

2 Setting, hypotheses and exact identities

2.1 Runs and the occupation count

The process takes N steps and each step has a sign, $+$ or $-$. The signs come in maximal runs of equal sign, and consecutive runs have opposite signs. All runs after the first are independent. A run of sign $\sigma \in \{+,-\}$ has a length distributed as X_σ, a random variable on $\{1,2,\dots\}$ with mass function $f_\sigma(n) = \mathbb{P}(X_\sigma = n)$ and tail

$$T_\sigma(x) = \mathbb{P}(X_\sigma \geq x), \qquad x \in \mathbb{R}.$$

We write $\bar\sigma$ for the sign opposite to σ. The first run is the initial run. It has a sign Σ and a length D, the pair (Σ, D) is independent of all later runs, and $D < \infty$ almost surely. We put

$$\nu_\sigma(d) = \mathbb{P}(\Sigma = \sigma,\, D = d), \qquad Q_\sigma(N) = \mathbb{P}(\Sigma = \sigma,\, D \geq N), \qquad \pi_\sigma = \mathbb{P}(\Sigma = \sigma).$$

In Sections 4 to 6 the law of the initial run does not depend on N. In Section 7 it does. A fresh start of sign σ_0 has $\nu_{\sigma_0} = f_{\sigma_0}$ and $\nu_{\bar\sigma_0} = 0$. The fair start is a fresh start of random sign, each sign with probability $\frac{1}{2}$.

Let A_N be the number of plus steps among steps $1,\dots,N$, and $P_N(k) = \mathbb{P}(A_N = k)$ for $0 \leq k \leq N$. For a fresh start of sign σ we write $F_N^\sigma(k)$ for $P_N(k)$, with $F_N^\sigma(k) = 0$ for $k \notin \{0,\dots,N\}$. We also use

$$\Pi_m^-(j) = F_m^-(m-j),$$

the probability that a fresh minus walk has exactly j minus steps among its first m steps.

2.2 Hypotheses

The hypotheses carry fixed names, and every statement says which it uses.

(RV) For each sign, T_σ is regularly varying with index $-\alpha_\sigma$, where $\alpha_\sigma \in (0,1)\cup(1,2)$.

(BAL) (RV) holds with $\alpha_+ = \alpha_- = \alpha \in (0,1)$, and $T_-(n)/T_+(n) \to b$ for some $b \in (0,\infty)$.

(DOM) (RV) holds with $\alpha_+ \in (0,1)$ and $T_-(n) = o(T_+(n))$.

(SPAN) For each sign, the support of X_σ is not contained in any arithmetic progression $c + h\mathbb{Z}$ with $h \geq 2$.

(SRT$_\sigma$) $\alpha_\sigma < 1$ and $u_\sigma(n) \sim \dfrac{\sin \pi\alpha_\sigma}{\pi}\,\dfrac{1}{nT_\sigma(n)}$ as $n \to \infty$, with u_σ the renewal mass function of Section 2.4.

(LB$_\sigma$) There is C with $f_\sigma(n) \leq C\,T_\sigma(n)/n$ for all $n \geq 1$.

(LR) For each sign, $f_\sigma(n) \sim \alpha_\sigma T_\sigma(n)/n$ as $n \to \infty$.

(LD) For each sign, $|f_\sigma(n) - f_\sigma(n+1)| = O(f_\sigma(n)/n)$.

(LD2) For each sign, $|f_\sigma(n+1) - 2f_\sigma(n) + f_\sigma(n-1)| = O(f_\sigma(n)/n^2)$.

(DSRT$_\sigma$) $\alpha_\sigma < 1$ and $u_\sigma(j) - u_\sigma(j+1) \sim (1-\alpha_\sigma)\,u_\sigma(j)/j$ as $j \to \infty$.

Under (BAL) both periods have infinite mean and both sides set the number of cycles, and this is the setting of Sections 4 to 7, where we write α for the common index. Under (DOM) the plus side has the heavier tail and an infinite mean, while the minus side may have a finite or an infinite mean. This is the setting of Section 8. Together with the finite-mean case of [23], the two regimes cover every pair of indices in $(0,1)\cup(1,2)$ for which T_-/T_+ has a limit in $[0,\infty]$, up to the exchange of the signs. If T_-/T_+ has no limit, there are only subsequential limit laws. (SPAN) is the aperiodicity needed for local limit theorems. (SRT$_\sigma$) is the strong renewal theorem for the runs of sign σ. (LB$_\sigma$) is Doney's local condition, and it implies (SRT$_\sigma$) by Lemma 3.1(iii) below. (LR) implies (LB) and (RV) with the same index. Under (LR), (LD2) implies (LD), since $f_\sigma(n+1) - f_\sigma(n) = -\sum_{i\geq n} \Delta^2 f_\sigma(i)$ with $|\Delta^2 f_\sigma(i)| \leq C f_\sigma(i)/i^2$, and $\sum_{i\geq n} f_\sigma(i)/i^2 \leq C f_\sigma(n)/n$ by Karamata's theorem. (DSRT$_\sigma$) is a difference form of the strong renewal theorem, and Lemma 9.1 shows that it holds for log-convex run laws. A hypothesis written with the subscript $\pm$ is assumed for both signs.

Constants. Unindexed C and c are positive constants that depend only on the laws of the runs and may change from one use to the next. A constant written C_δ also depends on δ.

2.3 Scales, stable densities and the Lamperti density

For a sign with $\alpha_\sigma < 1$ choose an increasing function $a_\sigma : (0,\infty) \to (0,\infty)$ with $t\,T_\sigma(a_\sigma(t)) \to 1$ as $t \to \infty$. So a_σ is regularly varying with index $1/\alpha_\sigma$. Under (BAL) we write a for a_+ and $a_m = a(m)$, and for $t > 0$ we put

$$m_N(t) = \lfloor t/T_+(N) \rfloor.$$

Then $T_+(a_{m_N(t)}) \sim T_+(N)/t$, and $a_{m_N(t)}/N \to t^{1/\alpha}$ uniformly for t in compact subsets of $(0,\infty)$ (Lemma A.1).

For $0<\alpha<1$ put $\Gamma_\alpha=\Gamma(1-\alpha)$, and let g be the density of the positive stable variable Z with $\mathbb{E}e^{-\lambda Z}=e^{-\Gamma_\alpha\lambda^\alpha}$. Its Lévy measure is $\alpha r^{-\alpha-1}dr$ on $(0,\infty)$, with tail $r^{-\alpha}$. For $t>0$ let

$$p_t(z)=t^{-1/\alpha}g\big(zt^{-1/\alpha}\big),$$

the density at time t of the stable subordinator with Laplace exponent $t\Gamma_\alpha\lambda^\alpha$. The covering densities are

$$Q_t(x)=\int_0^x p_t(z)(x-z)^{-\alpha}\,dz,\qquad Q_t^b(y)=b\int_0^y p_{bt}(z)(y-z)^{-\alpha}\,dz. \tag{2.1}$$

The density g is continuous and bounded, and $g(w)\sim\alpha w^{-1-\alpha}$ as $w\to\infty$ [14, Section XVII.6]. Hence

$$p_t(z)\le C\min\big\{t^{-1/\alpha},\,t\,z^{-1-\alpha}\big\}\qquad(t,z>0). \tag{2.2}$$

For $0<x<1$ let

$$\Lambda(x)=\int_0^\infty\big[p_{bt}(1-x)\,Q_t(x)+p_t(x)\,Q_t^b(1-x)\big]\,dt. \tag{2.3}$$

The Lamperti density $L_{\alpha,b}$ is defined in (1.1). With $t=\log((1-x)/x)$ and $\beta=\log b$ it reads

$$L_{\alpha,b}(x)=\frac{\sin\pi\alpha}{\pi}\,\frac{1+\cosh t}{\cosh(\alpha t-\beta)+\cos\pi\alpha}. \tag{2.4}$$

To check (2.4), write $w=(1-x)/x=e^t$. Then $x^{\alpha-1}(1-x)^{\alpha-1}=x^{2\alpha-2}w^{\alpha-1}$, the denominator of (1.1) is $x^{2\alpha}bw^\alpha\cdot 2[\cosh(\alpha t-\beta)+\cos\pi\alpha]$, and $x^{-2}w^{-1}=(1+w)^2/w=2+2\cosh t$. We also use, for $0\le y<1$,

$$\Phi_b(y)=\frac{b^2(1-y)^{\alpha-1}}{b^2(1-y)^{2\alpha}+2b\cos(\pi\alpha)\,(1-y)^\alpha y^\alpha+y^{2\alpha}}, \tag{2.5}$$

so $\Phi_b(0)=1$, $\Phi_b(y)=1+O(y^\alpha)$ as $y\to0$, $\Phi_b>0$, and $L_{\alpha,b}(1-y)=\frac{\sin\pi\alpha}{\pi b}\,y^{\alpha-1}\Phi_b(y)$.

2.4 Renewal kernels

For each sign let $X_1^\sigma,X_2^\sigma,\dots$ be independent copies of X_σ, $S_m^\sigma=X_1^\sigma+\cdots+X_m^\sigma$ with $S_0^\sigma=0$, and $u_m^\sigma=f_\sigma^{*m}$ the law of S_m^σ, so $u_0^\sigma=\delta_0$. The renewal mass function is

$$u_\sigma(j)=\sum_{m\ge0}u_m^\sigma(j),\qquad j\ge0,$$

so $u_\sigma(0)=1$. For $m\ge1$ and $k\ge1$ let

$$q_m^\sigma(k)=\mathbb{P}\big(S_{m-1}^\sigma<k\le S_m^\sigma\big)=\sum_{r<k}u_{m-1}^\sigma(r)\,T_\sigma(k-r), \tag{2.6}$$

the probability that the m-th run of sign σ covers level k in the time of its sign. The second form holds because X_m^σ is independent of S_{m-1}^σ. For each $k\ge1$ exactly one m has $S_{m-1}^\sigma<k\le S_m^\sigma$, so $\sum_{m\ge1}q_m^\sigma(k)=1$. We write $\omega_\sigma(i)=\min\{1,1/(iT_\sigma(i))\}$ for $i\ge1$ and $\omega_\sigma(0)=1$. Under (SRT$_\sigma$) we have $u_\sigma\le C\omega_\sigma$. For a start of sign σ_0 the boundary kernel at the plus end is the probability that a plus run starts right after the j-th minus step,

$$\kappa_+^{(+)}(j)=u_-(j)\quad(j\ge0),\qquad\kappa_+^{(-)}(j)=u_-(j)\mathbf{1}_{\{j\ge1\}}, \tag{2.7}$$

and the kernel at the minus end is defined with the signs exchanged [23, Section 2.4].

2.5 Exact identities

The following identities hold for every N and need no hypothesis. They are proved in [23, Section 2], and we include the short proofs because every section uses them. The computations of Section 11 check them against a sum over all sign strings.

Proposition 2.1 (the fresh-plus identity)**.** *For $k, j \geq 1$ with $k + j = N$,*

$$F_N^+(k) = \sum_{m\geq 1} u_m^-(j)\, q_{m+1}^+(k) + \sum_{m\geq 1} u_m^+(k)\, q_m^-(j), \tag{2.8}$$

and $F_N^+(N) = T_+(N)$, $F_N^+(0) = 0$. Exchanging the signs, for $k, j \geq 1$, $F_N^-(k) = \sum_{m\geq 1}[u_m^+(k)q_{m+1}^-(j) + u_m^-(j)q_m^+(k)]$, with $F_N^-(0) = T_-(N)$ and $F_N^-(N) = 0$.

Proof. For a fresh plus start the i-th plus run fills the steps in $(S_{i-1}^+ + S_{i-1}^-,\, S_i^+ + S_{i-1}^-]$ and the i-th minus run fills those in $(S_i^+ + S_{i-1}^-,\, S_i^+ + S_i^-]$. These intervals partition $\{1, 2, \dots\}$. If step N lies in the i-th plus run, the first N steps contain exactly S_{i-1}^- minus steps. So $A_N = k$ means $S_{i-1}^- = j$, and then step N lies in that run exactly when $S_{i-1}^+ < k \leq S_i^+$. The two events concern independent sequences, so this case has probability $u_{i-1}^-(j)q_i^+(k)$. If step N lies in the i-th minus run, then $A_N = k$ means $S_i^+ = k$, and step N lies in that run exactly when $S_{i-1}^- < j \leq S_i^-$, with probability $u_i^+(k)q_i^-(j)$. The cases are disjoint. For $j \geq 1$ the plus case with $i = 1$ vanishes, and $m = i - 1$ in the plus case and $m = i$ in the minus case give (2.8). For $k = N$ only the plus case with $i = 1$ survives, with probability $T_+(N)$, and $k = 0$ is impossible. The fresh minus start is the same with the signs exchanged. □

Proposition 2.2 (the first-run decomposition)**.** *For every initial run and every $0 \leq k \leq N$,*

$$P_N(k) = Q_+(N)\mathbf{1}_{\{k=N\}} + Q_-(N)\mathbf{1}_{\{k=0\}} + \sum_{d<N} \nu_+(d)\, F_{N-d}^-(k-d) + \sum_{d<N} \nu_-(d)\, F_{N-d}^+(k).$$

In particular $P_N(N) = Q_+(N)$ and $P_N(0) = Q_-(N)$.

Proof. We condition on the initial run. If $\Sigma = +$ and $D \geq N$, then $A_N = N$. If $\Sigma = +$ and $D = d < N$, steps $d+1, \dots, N$ are the first $N - d$ steps of a fresh minus walk independent of (Σ, D), so $A_N = d + A'$ with A' of law F_{N-d}^-. The case $\Sigma = -$ is the same. For $d < N$ Proposition 2.1 gives $F_{N-d}^-(N-d) = 0$ and $F_{N-d}^+(0) = 0$, so only the atoms survive at $k = N$ and $k = 0$. □

Proposition 2.3 (the final-run identity)**.** *For a fresh plus start, $N \geq 1$ and $0 \leq j < N$,*

$$\begin{aligned} P_N(N-j) = &\sum_{r=0}^{j} u_r^-(j)\, \mathbb{E}\big[T_+(N-j-S_r^+);\, S_r^+ \leq N-j-1\big] \\ &+ \sum_{m=1}^{j} T_-(m) \sum_{r=0}^{j-m} u_r^-(j-m)\, u_{r+1}^+(N-j). \end{aligned} \tag{2.9}$$

Proof. We split according to the run that contains step N, the final run. The $(r+1)$-th plus run starts at step $S_r^+ + S_r^- + 1$. It is the final run and $A_N = N - j$ exactly when $S_r^- = j$ and $X_{r+1}^+ \geq N - j - S_r^+ \geq 1$. This gives the term r of the first sum. The $(r+1)$-th minus run starts at step $S_{r+1}^+ + S_r^- + 1$. If it is the final run and starts at step $N - m + 1$, then $A_N = N - j$ means $S_{r+1}^+ = N - j$ and $S_r^- = j - m$, which needs $1 \leq m \leq j$, and $X_{r+1}^- \geq m$. This gives the second sum. Exactly one run is final, and $u_r^-(i) = 0$ for $r > i$. □

For the computations we use a third form of the law. For a fresh plus start let $A(k,j)$ and $B(k,j)$ be the probabilities that a plus run, respectively a minus run, starts right after exactly k plus steps and j minus steps. A plus run is preceded by m runs of each sign and a minus run by m plus runs and $m-1$ minus runs, so

$$A(k,j)=\sum_{m\ge 0} u_m^+(k)u_m^-(j), \qquad B(k,j)=\sum_{m\ge 1} u_m^+(k)u_{m-1}^-(j).$$

Splitting on the start of the final run gives, with $j=N-k$,

$$F_N^+(k)=\sum_{k'<k} A(k',j)\,T_+(k-k')+\sum_{j'<j} B(k,j')\,T_-(j-j'). \tag{2.10}$$

Every term is nonnegative, and an initial run enters by adding its weights to A and B.

We also need the Green identity of [23, Lemma 2.7]. For a fresh walk of sign σ and every $b\ge 1$,

$$\sum_{m\ge 1}\Pi_m^\sigma(b)=1+\mathbb{E}[X_{\bar\sigma}]\,u_\sigma(b), \tag{2.11}$$

where $\Pi_m^+(b)=F_m^+(b)$ and the right side is $+\infty$ when the mean of $X_{\bar\sigma}$ is infinite. The proof counts the steps m at which the walk has exactly b steps of sign σ. They form the interval between the b-th and the $(b+1)$-th such step, whose length is 1 unless the b-th step ends a run of sign σ, and then it is 1 plus the length of the following run of sign $\bar\sigma$, which is independent of the event. Section 7 uses a truncated form of this count.

3 Tools

3.1 Regular variation

We use the following facts about a function T that is regularly varying with index $-\alpha\in(-1,0)$, all from [7, Sections 1.2 to 1.5].

(R1) (Uniform convergence.) $T(\lambda x)/T(x)\to\lambda^{-\alpha}$ uniformly for λ in compact subsets of $(0,\infty)$.

(R2) (Potter.) For every $c\ge 1$ there is C_c with $T(x/c)\le C_cT(x)$ for $x\ge c$.

(R3) (Karamata.) $\sum_{n\le w}T(n)\sim wT(w)/(1-\alpha)$ and $\sum_{1\le i\le w}1/(iT(i))\sim 1/(\alpha T(w))$ as $w\to\infty$.

(R4) (Dyadic sums.) $\sum_{l\ge 1,\,2^l\le x}2^lT(2^{l-1})\le CxT(x)$ and $\sum_{l\ge 1,\,2^l>x}T(2^{l-1})\le CT(x)$ for $x\ge 1$.

(R4) follows from (R2) and Potter's theorem [7, Theorem 1.5.6], since $wT(w)$ is regularly varying with the positive index $1-\alpha$ and T with the negative index $-\alpha$, so both sums are dominated by geometric series (Lemma A.1). Under (BAL) the two tails are comparable, $cT_+\le T_-\le C\,T_+$, and $mT_+(a_m)\le C$ for all $m\ge 1$. We also use $a_{\lceil m/3\rceil}\ge c\,a_m$, which follows from (R2) for the function a.

3.2 Cited theorems

For a sign with $\alpha_\sigma<1$, under (RV) and (SPAN), $S_m^\sigma/a_\sigma(m)$ converges in law to the positive stable variable with Lévy tail $z^{-\alpha_\sigma}$, whose density we write g_σ. Under (BAL) we have $mT_-(a_mz)\to bz^{-\alpha}$, so with the common norming a_m the minus sums converge to $b^{1/\alpha}Z$, whose density is $g_b(z)=b^{-1/\alpha}g(zb^{-1/\alpha})$. Note that $t^{-1/\alpha}g_b(yt^{-1/\alpha})=p_{bt}(y)$. The next lemma is stated for one sign and applies to each sign whose index is in $(0,1)$.

Lemma 3.1 (classical inputs)**.** *Let $T = T_\sigma$ be regularly varying with index $-\alpha \in (-1, 0)$ and let $f = f_\sigma$ satisfy (SPAN). Write u_m, u and a_m for the objects of this sign.*

(i) (Gnedenko's local limit theorem [17, Theorem 4.2.1].) $\sup_x |a_m u_m(x) - g_\sigma(x/a_m)| \to 0$ *as $m \to \infty$. Hence $\sup_x u_m(x) \le C/a_m$ for all $m \ge 1$. Under (BAL) the same holds for the minus sums with the norming a_m and the density g_b.*

(ii) (Local large deviations [8, Theorem 1.1, Eq. (1.6)].) There is C with $u_m(x) \le C\,\dfrac{m}{a_m}\,T(x)$ for all $m \ge 1$ and $x \ge 1$.

(iii) (The strong renewal theorem.) If $\alpha > 1/2$, (SRT$_\sigma$) holds [15, 13], see [8, Theorem 1.4]. If $\alpha \le 1/2$, (SRT$_\sigma$) holds if and only if

$$\lim_{\delta\to 0} \limsup_{x\to\infty}\; xT(x) \sum_{1\le z\le \delta x} \frac{f(x-z)}{zT(z)^2} = 0$$

[8, Theorem 1.4]. In every case (LB$_\sigma$) implies (SRT$_\sigma$) [12], see [8, Eq. (1.15)].

(iv) (The strong renewal theorem as a small-m condition [8, Section 4.1, Eq. (4.1)].) (SRT$_\sigma$) holds if and only if

$$\lim_{\delta\to 0} \limsup_{x\to\infty} I_\sigma(\delta, x) = 0, \qquad I_\sigma(\delta, x) = xT(x) \sum_{1\le m\le \delta/T(x)} u_m(x). \tag{3.1}$$

(v) (Doney's local bound [5, Theorem 2.4], extending [12, Theorem A].) Under (LB$_\sigma$) there is C with $u_m(x) \le C\, mT(x)/x$ for all $m \ge 1$ and $x \ge a_m$.

We comment on the form of these statements. In (ii) and (iv) Caravenna and Doney use the tail $\mathbb{P}(X > x)$ and a function A with $A(x) \sim 1/\mathbb{P}(X > x)$ for renewal laws. The difference between $\mathbb{P}(X > x)$ and $T(x) = \mathbb{P}(X \ge x)$ is $f(x) = o(T(x))$. In (iv) their condition reads $\lim_{\delta\to 0} \limsup_x (x/A(x)) \sum_{1\le m\le A(\delta x)} u_m(x) = 0$. Since $A(\delta x) \sim \delta^\alpha / T(x)$ and the sum increases with its upper limit, this is equivalent to (3.1). (SPAN) makes the span 1, so their $\mathbb{P}(S_m \in x + (-1, 0])$ is $u_m(x)$. In (v) Berger uses a scale a'_m that may differ from ours by a constant factor. For $c\,a_m \le x \le a_m$ the bound follows from (ii), since $m/a_m \le m/x$ there, so (v) holds for $x \ge a_m$ with our scale. For a sign with index in $(1, 2)$ we use the local limit theorem and the local large deviation bound in the form of [23, Lemma A.1]: with $\mu = \mathbb{E}X$ and the scale a of that sign, $\sup_x u_m(x) \le C/a(m)$, and $u_m(x) \le C(m/a(m))T(x - m\mu)$ for $x - m\mu \ge 1$.

3.3 Elementary bounds

The following bounds use only (RV) with index in $(0, 1)$ and (SPAN) for one sign, and we drop the sign from the notation. They hold for each sign with $\alpha_\sigma < 1$.

Lemma 3.2 (sums of runs)**.** *(i) $\mathbb{P}(S_m \ge x) \le C\, mT(x)$ for all $m, x \ge 1$.*

(ii) $\mathbb{P}(S_m < x) \le e^{-mT(x)}$ for all $m \ge 0$ and $x \ge 1$.

(iii) $u_m(x) \le C\, a_m^{-1} e^{-mT(x+1)/3}$ for all $m \ge 2$ and $x \ge 0$.

Proof. (i) If $S_m \ge x$, then some $X_i \ge x$, or $\sum_i X_i \mathbf{1}_{\{X_i < x\}} \ge x$. By the union bound and Markov's inequality, $\mathbb{P}(S_m \ge x) \le mT(x) + m\,\mathbb{E}[X; X < x]/x$. Now $\mathbb{E}[X; X < x] = \sum_{1\le n<x} \mathbb{P}(n \le X < x) \le \sum_{n<x} T(n) \le CxT(x)$ by (R3).

(ii) $S_m < x$ forces $X_i < x$ for every $i \le m$, which has probability $(1 - T(x))^m \le e^{-mT(x)}$.

(iii) Let $h = \lfloor m/2 \rfloor \ge m/3$. Conditioning on S_h, $u_m(x) = \sum_{y\le x} u_h(y) u_{m-h}(x - y) \le \mathbb{P}(S_h \le x) \sup_z u_{m-h}(z)$. By (ii) $\mathbb{P}(S_h \le x) = \mathbb{P}(S_h < x + 1) \le e^{-mT(x+1)/3}$, and by Lemma 3.1(i) $\sup_z u_{m-h}(z) \le C/a_{m-h} \le C/a_m$, since $m - h \ge m/2$. □

The crude bound (i) has a sharp form on the range where one big summand carries the sum. We need it uniformly in the number of summands.

Lemma 3.3 (one big jump)**.** *For $\varepsilon > 0$ and $x \geq 1$ put $\varrho(\varepsilon, x) = \sup\{|\mathbb{P}(S_m \geq y)/(mT(y)) - 1| : \ y \geq x/2, \ m \geq 1, \ mT(y) \leq \varepsilon\}$. Then $\lim_{\varepsilon \to 0} \limsup_{x \to \infty} \varrho(\varepsilon, x) = 0$.*

Proof. Let $y \geq x/2$ and $mT(y) \leq \varepsilon$. For the lower bound, $\mathbb{P}(S_m \geq y) \geq \mathbb{P}(\max_{i \leq m} X_i \geq y) = 1 - (1 - T(y))^m \geq mT(y) - \frac{1}{2}(mT(y))^2 \geq mT(y)(1 - \varepsilon/2)$. For the upper bound fix $0 < \delta < \eta < \frac{1}{2}$ with $\eta/\delta \geq 2$. On $\{S_m \geq y\}$ one of the following holds: some $X_i \geq (1-\eta)y$; at least two summands are at least δy; exactly one summand is at least δy, it is below $(1-\eta)y$, and the sum W of the summands below δy is at least ηy; no summand is at least δy and $W \geq y$. The first event has probability at most $mT((1-\eta)y) \leq mT(y)((1-\eta)^{-\alpha} + o(1))$ by (R1), the $o(1)$ being uniform in m and $y \geq x/2$ as $x \to \infty$. The second has probability at most $m^2 T(\delta y)^2 \leq C_\delta \varepsilon \, mT(y)$ by (R2). For the last two we bound W by a Chernoff estimate. Each summand of W is at most δy, so $e^{\lambda X} \leq 1 + (e^{\lambda \delta y} - 1)X/(\delta y)$ on $\{X < \delta y\}$, and with $\mathbb{E}[X; X < \delta y] \leq C\delta y T(\delta y) \leq C_\delta \, \delta \, yT(y)$ by (R3) and (R2),

$$\mathbb{P}(W \geq z) \leq \exp\Big(-\lambda z + m \, \frac{e^{\lambda \delta y} - 1}{\delta y} \, C_\delta \delta y T(y)\Big) \leq \Big(\frac{eC_\delta \delta \, mT(y) \, y}{z}\Big)^{z/(\delta y)}$$

for $z \geq eC_\delta \delta \, mT(y) y$, by the choice $e^{\lambda \delta y} = z/(C_\delta \delta m T(y) y)$. At $z = \eta y$ the condition holds for small ε, the base is $C_{\delta,\eta} mT(y) \leq 1$ and the exponent is $\eta/\delta \geq 2$, so $\mathbb{P}(W \geq \eta y) \leq (C_{\delta,\eta} mT(y))^2 \leq C_{\delta,\eta} \varepsilon \, mT(y)$. The third event has probability at most $mT(\delta y)\mathbb{P}(W' \geq \eta y)$ with W' a sum of $m-1$ truncated summands, which is at most $C_\delta mT(y) \cdot C_{\delta,\eta}\varepsilon$. The fourth has probability at most $\mathbb{P}(W \geq y) \leq \mathbb{P}(W \geq \eta y)$. Adding up, $\mathbb{P}(S_m \geq y) \leq mT(y)[(1-\eta)^{-\alpha} + o(1) + C_{\delta,\eta}\varepsilon]$. Let $x \to \infty$, then $\varepsilon \to 0$, then $\eta \to 0$. □

Lemma 3.4 (the covering kernel at scale N)**.** *Assume (BAL) and (SPAN), and fix $\delta \in (0,1)$. Uniformly for $t \in [\delta, 1/\delta]$ and $x \in [\delta, 1]$,*

$$q^+_{m_N(t)+1}(\lfloor xN \rfloor) = T_+(N)\big[Q_t(x) + o(1)\big], \qquad q^-_{m_N(t)}(\lfloor xN \rfloor) = T_+(N)\big[Q^b_t(x) + o(1)\big].$$

Proof. Write $m = m_N(t)$ and $k = \lfloor xN \rfloor$, and fix $\theta \in (0, \delta/2)$. Split $q^+_{m+1}(k) = \sum_{r<k} u^+_m(r) T_+(k-r)$ at $r = k - \theta N$. On $r \leq k - \theta N$ we have $(k-r)/N \in [\theta, 1]$, so by (R1) $T_+(k-r)/T_+(N) = ((k-r)/N)^{-\alpha} + o(1)$ uniformly. By Lemma 3.1(i), $u^+_m(r) = a_m^{-1}[g(r/a_m) + o(1)]$ uniformly in r. The error terms contribute at most $o(1/a_m) \cdot k \cdot T_+(\theta N) = o(T_+(N))$, since N/a_m is bounded. The main term is $T_+(N) \sum_{r \leq k - \theta N} a_m^{-1} g(r/a_m)((k-r)/N)^{-\alpha}$, a Riemann sum with mesh $1/N$ in $z = r/N$. Since $a_m/N \to t^{1/\alpha}$ uniformly and g is bounded and uniformly continuous, it converges to $\int_0^{x-\theta} p_t(z)(x-z)^{-\alpha} dz$, uniformly in t and x. On $r > k - \theta N$, Lemma 3.1(i) and (R3) give at most $(C/a_m) \sum_{i < \theta N + 1} T_+(i) \leq C_\delta \, \theta^{(1-\alpha)/2} T_+(N)$ for large N, using (R1) and (R2) to compare $\theta N T_+(\theta N)$ with $N T_+(N)$. By the first bound in (2.2) the integral over $[x-\theta, x]$ is at most $C_\delta \theta^{1-\alpha}$. Let $N \to \infty$ and then $\theta \to 0$. The minus kernel is the same with u^-_{m-1} in place of u^+_m, the density g_b in place of g, and $T_-(N) \sim bT_+(N)$. The shift from m to $m-1$ does not change the limit. □

The next bound is the key estimate of the paper. It bounds the covering kernel by the tail at the level, uniformly in the number of runs, under regular variation alone.

Lemma 3.5 (the key bound)**.** *For a sign with $\alpha_\sigma < 1$ there is C with $q^\sigma_{m+1}(k) \leq C \, T_\sigma(k)$ for all $m \geq 0$ and $k \geq 1$.*

Proof. We drop the sign. For $m = 0$, $q_1 = T$. Let $m \geq 1$. By (2.6), $q_{m+1}(k) = \sum_{r<k} u_m(r) T(k-r)$. The part with $r < k/2$ is at most $T(k/2) \leq CT(k)$ by (R2). For

$r \ge k/2$ put $i = k - r \in [1, k/2]$, and split the range of i into dyadic blocks $[w/2, w)$ with $w = 2^l$, $l \ge 1$, $w \le k$. On a block,

$$\sum_{w/2 \le i < w} u_m(k-i)T(i) \le T(w/2)\,\mathbb{P}\big(S_m \in (k-w, k-w/2]\big).$$

The window has at most w lattice points, all at least $k/2$. By Lemma 3.1(ii) and (R2) each point has probability at most $C(m/a_m)T(k)$. By Lemma 3.2(i) and (R2) the whole window has probability at most $C\,mT(k)$. So the block is at most $C\,mT(k)\,T(w/2)\min\{1, w/a_m\}$. Summing with (R4), the blocks with $w \le a_m$ give at most $C\frac{mT(k)}{a_m}\sum_{w \le a_m} wT(w/2) \le C\,mT(k)T(a_m)$, and the blocks with $w > a_m$ give at most $C\,mT(k)\sum_{w > a_m} T(w/2) \le C\,mT(k)T(a_m)$. Since $mT(a_m) \le C$, the total is at most $CT(k)$. □

Note that Lemma 3.1(ii) is used here at points $x \ge k/2$ that may lie far below a_m as well as above it, and it holds for all x.

Lemma 3.6 (decay in m)**.** *For a sign with $\alpha_\sigma < 1$ there are C, c with $q_m^\sigma(j) \le C\,T_\sigma(j)\,e^{-c\,mT_\sigma(j)}$ for all $m, j \ge 1$.*

Proof. We drop the sign. If $mT(j) \le 9$, this is Lemma 3.5 with $c = 1/9$. Let $mT(j) > 9$, so $m \ge 10$, and put $h = \lfloor (m-1)/2 \rfloor \ge m/3$. Splitting $S_{m-1} = S_h + (S_{m-1} - S_h)$ gives $q_m(j) = \sum_{s<j} u_h(s)\,q_{m-h}(j-s)$. For $s < j/2$ we use $q_{m-h}(j-s) \le CT(j-s) \le CT(j)$ (Lemma 3.5 and (R2)) and $\mathbb{P}(S_h < j/2) \le e^{-hT(j)} \le e^{-mT(j)/3}$ (Lemma 3.2(ii)). For $j/2 \le s < j$ we use

$$\sum_{j/2 \le s < j} u_h(s)\,q_{m-h}(j-s) \le \sup_{s<j} u_h(s) \sum_{i \le j/2} CT(i) \le C\,a_h^{-1} e^{-hT(j)/3}\,jT(j),$$

by Lemma 3.2(iii) and (R3). Since $hT(j) \ge mT(j)/3 > 3$, we have $T(j) \ge 3/h \ge T(a_h)$ for large h, so $j \le Ca_h$ and $jT(j)/a_h \le CT(j)$. Both parts are at most $CT(j)e^{-mT(j)/9}$. □

The next lemma collects the Fourier bounds we need for the differences of the convolution powers, in space and in the index.

Lemma 3.7 (Fourier bounds)**.** *Let $\alpha_\sigma < 1$, assume (RV) and (SPAN) for the sign σ, and drop the sign. Let $\varphi(t) = \mathbb{E}e^{itX}$, and for $y \ge 1$ let $V_m^{(y)}(x) = \mathbb{P}(S_m = x,\ \max_{i \le m} X_i < y)$.*

(i) There is $c > 0$ with $|\varphi(t)| \le e^{-cT(1/|t|)}$ for $0 < |t| \le \pi$.

(ii) For all $m \ge 1$, $y \ge 1$ and x, $|V_m^{(y)}(x-1) - V_m^{(y)}(x)| \le Ce^{CmT(y)}a_m^{-2}$. In particular $\sup_x |u_m(x-1) - u_m(x)| \le Ca_m^{-2}$, and $|u_m(x-1) - u_m(x)| \le Ca_m^{-2}e^{-cmT(x)}$ for all $m \ge 1$ and $x \ge 1$.

(iii) $\sup_x |u_{m+1}(x) - u_m(x)| \le C/(ma_m)$ for all $m \ge 1$.

Proof. (i) Let X' be an independent copy of X. Then $|\varphi(t)|^2 = \mathbb{E}\cos(t(X - X'))$, and for $|t| \le 1$, using $1 - \cos\vartheta \ge \vartheta^2/4$ on $|\vartheta| \le 1$, a point n_0 of the support of f, and Karamata's theorem,

$$1 - |\varphi(t)|^2 \ge \frac{t^2}{4}\mathbb{E}\big[(X-X')^2;\ |X-X'| \le 1/|t|\big] \ge \frac{t^2}{4}f(n_0) \sum_{2n_0 \le n \le 1/(2|t|)} \frac{n^2}{4}f(n) \ge c\,T(1/|t|)$$

for $|t| \le t_0$. Hence $|\varphi(t)| \le (1 - cT(1/|t|))^{1/2} \le e^{-cT(1/|t|)/2}$ there. On $t_0 \le |t| \le \pi$, (SPAN) gives $|\varphi(t)| < 1$, so by continuity $|\varphi(t)| \le 1 - c_0$, which is at most $e^{-cT(1/|t|)}$ for c small since $T \le 1$.

(ii) The generating function of $V_m^{(y)}$ is φ_y^m with $\varphi_y(t) = \sum_{n<y} f(n)e^{itn}$, and $|\varphi_y| \le |\varphi| + T(y) \le e^{-cT(1/|t|)}(1 + e^c T(y))$, so $|\varphi_y(t)|^m \le e^{CmT(y)}e^{-cmT(1/|t|)}$. Fourier inversion gives

$$|V_m^{(y)}(x-1) - V_m^{(y)}(x)| = \Big|\frac{1}{2\pi}\int_{-\pi}^{\pi} \varphi_y(t)^m (e^{it}-1)e^{-itx}\,dt\Big| \le \frac{e^{CmT(y)}}{2\pi}\int_{-\pi}^{\pi} |t|\, e^{-cmT(1/|t|)}\,dt.$$

Substitute $t = s/a_m$. For $s \le 1$, $mT(a_m/s) \ge mT(a_m) \ge c$. For $1 \le s \le a_m/x_0$, Potter's bounds give $mT(a_m/s) \ge cs^{\alpha-\delta}$. For $|t| \ge 1/x_0$ the integrand is at most $e^{-cmT(x_0)} \le e^{-c'm}$. So the integral is at most $a_m^{-2}\big[\int_0^1 s\,ds + \int_1^\infty se^{-cs^{\alpha-\delta}}ds\big] + Ce^{-c'm} \le Ca_m^{-2}$, since $a_m \le Cm^{1/\alpha+\delta}$. The case $y = \infty$ is the bound for u_m. For the last claim let $m \ge 2$, $k = \lceil m/2\rceil$ and write $u_m = u_{m-k} * u_k$. Then $|u_m(x-1) - u_m(x)| \le \sup_z |u_k(z-1) - u_k(z)|\,\mathbb{P}(S_{m-k} \le x) \le Ca_k^{-2}e^{-(m-k)T(x+1)}$ by Lemma 3.2(ii), and $a_k \ge ca_m$, $(m-k)T(x+1) \ge cmT(x)$.

(iii) We have $u_{m+1}(x) - u_m(x) = (2\pi)^{-1}\int_{-\pi}^{\pi}\varphi(t)^m(\varphi(t)-1)e^{-itx}dt$ and $|\varphi(t)-1| \le \mathbb{E}\min(2, |t|X) \le 2T(1/|t|) + |t|\,\mathbb{E}[X; X < 1/|t|] \le CT(1/|t|)$ by Karamata's theorem. So the left side is at most $C\int_0^\pi T(1/t)e^{-cmT(1/t)}dt$. Substitute $t = s/a_m$ again. For $s \le 1$, $T(a_m/s) \le T(a_m) \le C/m$. For $1 \le s \le a_m/x_0$, $T(a_m/s) \le Cs^{\alpha+\delta}/m$ and the exponential is at most $e^{-cs^{\alpha-\delta}}$. For $s \ge a_m/x_0$ the integrand is at most $e^{-c'm}$ on an interval of length πa_m. Hence the integral is at most $a_m^{-1}[C/m + C/m + Ca_m e^{-c'm}] \le C/(ma_m)$. □

The last lemma of this section is the local form of the inverse renewal count. For a sign with $\alpha_\sigma < 1$ let $E_\sigma = Z_\sigma^{-\alpha_\sigma}$, where Z_σ is the positive stable variable with Lévy tail $z^{-\alpha_\sigma}$, and let ϕ_σ be its density. Since $\mathbb{P}(T_\sigma(n)\#\{m : S_m^\sigma < n\} \ge t) = \mathbb{P}(S^\sigma_{\lfloor t/T_\sigma(n)\rfloor} < n) \to \mathbb{P}(Z_\sigma < t^{-1/\alpha_\sigma})$, the variable E_σ is the limit of $T_\sigma(n)$ times the number of runs of sign σ completed within n steps of that sign. It is a rescaled Mittag-Leffler variable, with $\mathbb{E}E_\sigma^n = n!/(\Gamma(1+n\alpha_\sigma)\Gamma(1-\alpha_\sigma)^n)$ and $\phi_\sigma(0) = 1$, because $\mathbb{P}(E_\sigma \le \varepsilon) = \mathbb{P}(Z_\sigma \ge \varepsilon^{-1/\alpha_\sigma}) \sim \varepsilon$. Its density is decreasing from 0 when $\alpha_\sigma < 1/2$ and has an interior maximum when $\alpha_\sigma > 1/2$, since $\phi'_\sigma(0) = -\Gamma(1-\alpha_\sigma)^2/\Gamma(1-2\alpha_\sigma)$, and at $\alpha_\sigma = 1/2$ it is $\phi_\sigma(x) = e^{-\pi x^2/4}$.

Lemma 3.8 (the local inverse renewal count)**.** *Let $\alpha_\sigma < 1$, assume (RV) and (SPAN) for the sign σ, and drop the sign.*

(i) $q_{m+1}(k) \ge (1 - CmT(k))\,T(k)$ *for all* $m \ge 0$ *and* $k \ge 1$.

(ii) For every $K > 0$, $\sup_{m\ge 0:\ mT(k)\le K}\big|q_{m+1}(k)/T(k) - \phi(mT(k))\big| \to 0$ *as* $k \to \infty$.

Proof. (i) Since $T(k-r) \ge T(k)$ for $r \ge 0$, (2.6) gives $q_{m+1}(k) \ge \mathbb{P}(S_m < k)T(k) \ge (1 - CmT(k))T(k)$ by Lemma 3.2(i).

(ii) First let $t = mT(k) \in [\varepsilon, K]$. Then $m \to \infty$ with k, and $a_m/k \to t^{1/\alpha}$ uniformly, as in Lemma A.1(iii). Fix $\eta \in (0, 1/2)$ and split $q_{m+1}(k) = \sum_{r<k} u_m(r)T(k-r)$ at $r = (1-\eta)k$. As in the proof of Lemma 3.4, the part $r < (1-\eta)k$ is a Riemann sum that converges, uniformly in $t \in [\varepsilon, K]$, to $T(k)\int_0^{(1-\eta)t^{-1/\alpha}} g(w)(1 - t^{1/\alpha}w)^{-\alpha}dw$, and the part $r \ge (1-\eta)k$ is at most $(C/a_m)\sum_{i\le\eta k} T(i) \le C_\varepsilon \eta^{1-\alpha}T(k)$. Letting $\eta \to 0$ gives the limit $\int_0^1 p_t(x)(1-x)^{-\alpha}dx$, and this is $\phi(t)$. Indeed, let $(\sigma_s)_{s\ge0}$ be the subordinator with Lévy measure $\alpha r^{-\alpha-1}dr$ and no drift, so that σ_t has the density p_t, and let $\tau = \inf\{s : \sigma_s > 1\}$. The compensation formula [6, Proposition III.2] gives τ the density $t \mapsto \int_0^1 p_t(x)\,(1-x)^{-\alpha}\,dx$, since $(1-x)^{-\alpha}$ is the Lévy tail at $1-x$. Moreover $\{\tau > t\} = \{\sigma_t \le 1\} = \{t^{1/\alpha}Z \le 1\} = \{E \ge t\}$, so τ has the law of E.

Now let $t = mT(k) \le \varepsilon$. The lower bound (i) gives $q_{m+1}(k)/T(k) \ge 1 - C\varepsilon$, and $\phi(t) \ge 1 - C\varepsilon$ on $[0, \varepsilon]$. For the upper bound fix $0 < \theta, \eta < \frac12$ and write $\delta_k = \sup_{1/2\le\lambda\le1}|T(\lambda k)/T(k) - \lambda^{-\alpha}|$, which tends to 0 by (R1). Split $q_{m+1}(k)$ into three ranges. The range $r < \eta k$ contributes at most $T((1-\eta)k) \le ((1-\eta)^{-\alpha} + \delta_k)T(k)$. The range $\eta k \le r \le (1-\eta)k$ contributes at most $T(\eta k)\mathbb{P}(S_m \ge \eta k) \le C_\eta \varepsilon\, T(k)$ by

Lemma 3.2(i) and (R2). For the range $r > (1-\eta)k$ put $i = k - r < \eta k$ and use dyadic blocks $[w/2, w)$. A block contributes at most $T(w/2)\mathbb{P}(S_m \in [k-w, k))$, and the window probability is at most the smaller of $Cw(m/a_m)T(k)$, by Lemma 3.1(ii), and

$$\mathbb{P}(S_m \ge k-w) - \mathbb{P}(S_m \ge k) \le mT(k)\Big[\frac{T(k-w)}{T(k)} - 1 + 2\varrho(\varepsilon,k)\frac{T(k-w)}{T(k)}\Big] \\ \le mT(k)\big[Cw/k + \delta_k + 3\varrho(\varepsilon,k)\big],$$

by Lemma 3.3, where we used $w \le \eta k$ and $T(k-w)/T(k) \le 2$ for large k. The blocks with $w \le \theta a_m$ contribute at most $C(mT(k)/a_m)\sum_{w\le\theta a_m} wT(w/2) \le C\theta^{1-\alpha'}mT(a_m)T(k) \le C\theta^{1-\alpha'}T(k)$ for any $\alpha' \in (\alpha, 1)$, by (R4) and Potter's bounds. The blocks with $\theta a_m < w \le \eta k$ contribute at most $mT(k)\sum_w T(w/2)[Cw/k + \delta_k + 3\varrho] \le CmT(k)[\eta^{1-\alpha}kT(\eta k)/k + (\delta_k + 3\varrho)T(\theta a_m)]$, and $mT(\theta a_m) \le C\theta^{-\alpha'}$, so this is at most $C[\eta^{1-\alpha}\varepsilon + (\delta_k + 3\varrho(\varepsilon,k))\theta^{-\alpha'}]T(k)$. Hence

$$\sup_{mT(k)\le\varepsilon} \frac{q_{m+1}(k)}{T(k)} \le (1-\eta)^{-\alpha} + \delta_k + C\theta^{1-\alpha'} + C_\eta\varepsilon \\ + C\theta^{-\alpha'}\big(\delta_k + 3\varrho(\varepsilon,k)\big).$$

Let $k \to \infty$, then $\varepsilon \to 0$, then $\theta \to 0$, then $\eta \to 0$. Together with the lower bound and $\phi(t) = 1 + O(\varepsilon)$ on $[0,\varepsilon]$ this proves (ii) on $[0,\varepsilon]$, and with the first part on $[\varepsilon, K]$. □

4 The local Lamperti law

4.1 The limit density

Proposition 4.1 (identification of the limit)**.** *The integral Λ of (2.3) is finite and continuous on $(0,1)$, and $\Lambda = L_{\alpha,b}$. Moreover, for every $\theta > -1$,*

$$\int_0^1 \frac{L_{\alpha,b}(v)\,dv}{1+\theta v} = \frac{(1+\theta)^{\alpha-1} + b}{(1+\theta)^\alpha + b}, \tag{4.1}$$

so $L_{\alpha,b}$ is a probability density with mean $1/(1+b)$.

Proof. Finiteness and continuity. Let x range over a compact subset K of $(0,1]$. Split $Q_t(x)$ at $z = x/2$. The part $z < x/2$ is at most $(x/2)^{-\alpha}$, and by (2.2) the part $z \ge x/2$ is at most $C\min\{t^{-1/\alpha}, t\}\int_0^{x/2} w^{-\alpha}dw$. So $Q_t(x) \le C_K(1 + \min\{t^{-1/\alpha}, t\})$, and the same holds for Q_t^b. By (2.2), $p_t(x) \le C_K \min\{t^{-1/\alpha}, t\}$. The integrand of (2.3) is therefore at most $C_K \min\{t^{-1/\alpha}, t\}$ for $x \in K$, which is integrable on $(0,\infty)$ because $1/\alpha > 1$. The integrand is continuous in x, so Λ is finite and continuous by dominated convergence.

Homogeneity. For $x, y > 0$ put $\lambda(x,y) = \int_0^\infty [p_{bt}(y)Q_t(x) + p_t(x)Q_t^b(y)]\,dt$, so $\Lambda(x) = \lambda(x, 1-x)$. For $c > 0$ the definitions give $p_t(cz) = c^{-1}p_{tc^{-\alpha}}(z)$, and the substitution $z = cw$ in (2.1) gives $Q_t(cx) = c^{-\alpha}Q_{tc^{-\alpha}}(x)$ and $Q_t^b(cy) = c^{-\alpha}Q^b_{tc^{-\alpha}}(y)$. Substituting $s = tc^{-\alpha}$ in the integral over t then gives $\lambda(cx, cy) = c^{-1}\lambda(x,y)$. Hence

$$\lambda(x,y) = \frac{1}{x+y}\,\Lambda\Big(\frac{x}{x+y}\Big).$$

The double Laplace transform. For $\lambda, \mu > 0$, $\int_0^\infty e^{-\mu y}p_{bt}(y)\,dy = e^{-bt\Gamma_\alpha\mu^\alpha}$, and since Q_t is the convolution of p_t with $w^{-\alpha}$ on $(0,\infty)$,

$$\int_0^\infty e^{-\lambda x}Q_t(x)\,dx = e^{-t\Gamma_\alpha\lambda^\alpha}\,\Gamma_\alpha\lambda^{\alpha-1}, \qquad \int_0^\infty e^{-\mu y}Q_t^b(y)\,dy = b\,e^{-bt\Gamma_\alpha\mu^\alpha}\,\Gamma_\alpha\mu^{\alpha-1}.$$

All functions are nonnegative, so by Tonelli's theorem

$$\int_0^\infty\int_0^\infty e^{-\lambda x-\mu y}\lambda(x,y)\,dx\,dy=\int_0^\infty \Gamma_\alpha\big(\lambda^{\alpha-1}+b\mu^{\alpha-1}\big)e^{-t\Gamma_\alpha(\lambda^\alpha+b\mu^\alpha)}\,dt=\frac{\lambda^{\alpha-1}+b\mu^{\alpha-1}}{\lambda^\alpha+b\mu^\alpha}.$$

Now change variables to $x=sv$, $y=s(1-v)$ with $s>0$ and $v\in(0,1)$, so $dx\,dy=s\,ds\,dv$. By homogeneity the left side is

$$\int_0^1\Lambda(v)\int_0^\infty e^{-s(\lambda v+\mu(1-v))}\,ds\,dv=\int_0^1\frac{\Lambda(v)\,dv}{\lambda v+\mu(1-v)}.$$

With $\mu=1$ and $\lambda=1+\theta$ this is (4.1) with Λ in place of $L_{\alpha,b}$. At $\theta=0$ it gives $\int_0^1\Lambda=1$.

Inversion. For $z\in\mathbb{C}\setminus[0,1]$ let $G(z)=\int_0^1\Lambda(v)\,dv/(z-v)$, and let $H(w)=(w^{\alpha-1}+b)/(w^\alpha+b)$ with principal powers on $\mathbb{C}\setminus(-\infty,0]$. The denominator does not vanish there, since $|\arg w^\alpha|<\pi\alpha<\pi$. For real $z>1$ put $\theta=-1/z\in(-1,0)$. Then $1/(z-v)=z^{-1}/(1+\theta v)$, so the transform gives $G(z)=z^{-1}H(1-1/z)$. Since $1-1/z\in(-\infty,0]$ exactly when $z\in(0,1]$, the right side is analytic on $\mathbb{C}\setminus[0,1]$, as is G. The two agree on $(1,\infty)$, hence on all of $\mathbb{C}\setminus[0,1]$. Since Λ is continuous and integrable, $-\frac{1}{\pi}\operatorname{Im}G(x+i\varepsilon)$ tends to $\Lambda(x)$ as $\varepsilon\downarrow0$ for every $x\in(0,1)$ (Lemma A.2).

We compute the boundary value. Let $x\in(0,1)$ and $w=(1-x)/x$. As $\varepsilon\downarrow0$, $1-1/(x+i\varepsilon)$ tends to $-w$ from the upper half plane, so it tends to $we^{i\pi}$ on the principal branch. With $E=e^{i\pi\alpha}$ this gives $(we^{i\pi})^\alpha=w^\alpha E$ and $(we^{i\pi})^{\alpha-1}=-w^{\alpha-1}E$, so

$$G(x+i0)=\frac{1}{x}\,\frac{b-w^{\alpha-1}E}{b+w^\alpha E}=\frac{1}{x}\,\frac{(b-w^{\alpha-1}E)(b+w^\alpha\bar E)}{b^2+2bw^\alpha\cos\pi\alpha+w^{2\alpha}}.$$

The imaginary part of the numerator is $-b\sin(\pi\alpha)\,w^{\alpha-1}(1+w)$. Hence

$$-\frac{1}{\pi}\operatorname{Im}G(x+i0)=\frac{\sin\pi\alpha}{\pi}\,\frac{b\,w^{\alpha-1}(1+w)}{x\,(b^2+2bw^\alpha\cos\pi\alpha+w^{2\alpha})}.$$

Since $1+w=1/x$, we have $w^{\alpha-1}(1+w)/x=x^{-1-\alpha}(1-x)^{\alpha-1}$, and the denominator is $x^{-2\alpha}[b^2x^{2\alpha}+2b\cos(\pi\alpha)x^\alpha(1-x)^\alpha+(1-x)^{2\alpha}]$. So the right side is $L_{\alpha,b}(x)$, and $\Lambda=L_{\alpha,b}$. Finally, differentiating (4.1) at $\theta=0$ gives $-\int_0^1 vL_{\alpha,b}(v)\,dv=-1/(1+b)$. □

The mean $1/(1+b)$ has the expected direction. If $b<1$, the plus runs are heavier and the plus steps occupy more than half of the time on average.

4.2 Proof of Theorem A

We first describe the structure of the proof. By the fresh-plus identity, $P_N(k)$ is a sum over the number m of completed minus runs (first sum) or plus runs (second sum) of a product of a point probability and a covering probability. Typical configurations have m of order $1/T_+(N)$, since m runs of a law with tail T have total length of order a_m, and $a_m\asymp N$ exactly when $mT_+(N)\asymp1$. On this range Gnedenko's theorem and Lemma 3.4 turn the sum into a Riemann sum for Λ. Large m is negligible because N steps rarely contain many more runs than $1/T(N)$, which is Lemma 3.2(iii). Small m is the delicate range. There a few runs have total length of order N, so one run is of order N, and the probability that a sum of few runs lands exactly on a given point is the quantity controlled by the strong renewal theorem. The key bound of Lemma 3.5 removes the covering probability from this range at the cost of a constant, and Lemma 3.1(iv) then says that what is left is negligible if and only if the strong renewal theorem holds. Since every term is nonnegative, the same comparison runs in the other direction and gives the necessity.

We prove the theorem for a fresh plus start. Fix $0 < x_1 < x_2 < 1$ and let k range over $[x_1 N, x_2 N]$. Put $j = N - k$, $x = k/N$ and $y = 1 - x$. By Proposition 2.1,

$$P_N(k) = \underbrace{\sum_{m \ge 1} u_m^-(j)\, q_{m+1}^+(k)}_{\mathrm{I}} + \underbrace{\sum_{m \ge 1} u_m^+(k)\, q_m^-(j)}_{\mathrm{II}}.$$

Every term is nonnegative. Fix $\delta \in (0,1)$ and split the sums over m into small $m < \delta/T_+(N)$, medium $\delta/T_+(N) \le m \le 1/(\delta T_+(N))$ and large $m > 1/(\delta T_+(N))$. We write $\mathrm{I_s}$, $\mathrm{I_m}$, $\mathrm{I_l}$ and so on. Let

$$\Lambda_\delta(x) = \int_\delta^{1/\delta} \big[p_{bt}(1-x)Q_t(x) + p_t(x)Q_t^b(1-x)\big]\, dt.$$

By Proposition 4.1, Λ_δ increases to $L_{\alpha,b}$ as $\delta \to 0$, and by Dini's theorem the convergence is uniform on $[x_1, x_2]$.

Medium m. Every medium m is $m_N(t)$ for $t = mT_+(N) \in [\delta, 1/\delta]$, and $a_m = Nt^{1/\alpha}(1 + o(1))$ uniformly. By Lemma 3.1(i), $Nu_m^-(j) = p_{bt}(y) + o(1)$ and $Nu_m^+(k) = p_t(x) + o(1)$ uniformly. By Lemma 3.4, $q_{m+1}^+(k) = T_+(N)[Q_t(x) + o(1)]$ and $q_m^-(j) = T_+(N)[Q_t^b(y) + o(1)]$ uniformly. Put $h(t,x) = p_{bt}(1-x)Q_t(x) + p_t(x)Q_t^b(1-x)$. Since p_t, Q_t and Q_t^b are bounded on the relevant compact sets, the products of the error terms are $o(1)$ uniformly, so

$$N(\mathrm{I_m} + \mathrm{II_m}) = \sum_{\delta/T_+(N) \le m \le 1/(\delta T_+(N))} T_+(N)\big[h(mT_+(N), x) + o(1)\big].$$

The number of terms is at most $1/(\delta T_+(N))$, so the error terms add up to $o(1)$. The function h is continuous on $[\delta, 1/\delta] \times [x_1, x_2]$. Indeed g is continuous and bounded, so $(t,z) \mapsto p_t(z)$ is continuous on $(0,\infty)^2$, and $(t,x) \mapsto Q_t(x)$ is continuous by dominated convergence, after the substitution $z = xw$ in (2.1), which gives $Q_t(x) = x^{1-\alpha}\int_0^1 p_t(xw)(1-w)^{-\alpha} dw$. So h is uniformly continuous there, and the sum is a Riemann sum with mesh $T_+(N)$ that converges to $\int_\delta^{1/\delta} h(t,x)\, dt$ uniformly in x. Hence

$$\sup_{x_1 N \le k \le x_2 N} \big|N(\mathrm{I_m} + \mathrm{II_m}) - \Lambda_\delta(k/N)\big| \to 0. \tag{4.2}$$

Large m. By Lemma 3.5 and (R2), $q_{m+1}^+(k) \le CT_+(k) \le C_{x_1} T_+(N)$. By Lemma 3.2(iii) for the minus sign, $u_m^-(j) \le C a_m^{-1} e^{-mT_-(j+1)/3}$, and $T_-(j+1) \ge cT_+(N)$. For $m \ge 1/T_+(N)$ we have $a_m \ge cN$. Hence

$$N\,\mathrm{I_l} \le C\,T_+(N) \sum_{m > 1/(\delta T_+(N))} e^{-cmT_+(N)} \le C\, e^{-c/\delta}.$$

For $\mathrm{II_l}$, Lemma 3.6 gives $q_m^-(j) \le CT_+(N)e^{-cmT_+(N)}$, and $u_m^+(k) \le C/a_m \le C/N$, so $N\,\mathrm{II_l} \le Ce^{-c/\delta}$ as well.

Small m, sufficiency. By Lemma 3.5, $q_{m+1}^+(k) \le C_{x_1} T_+(N)$ and $q_m^-(j) \le CT_-(j) \le C_{x_2} T_+(N)$. So

$$N\,\mathrm{I_s} \le C\,NT_+(N) \sum_{m < \delta/T_+(N)} u_m^-(j), \qquad N\,\mathrm{II_s} \le C\,NT_+(N) \sum_{m < \delta/T_+(N)} u_m^+(k).$$

Since $j \ge (1 - x_2)N$, (R2) gives $NT_+(N) \le CjT_-(j)$ and $\delta/T_+(N) \le C\delta/T_-(j)$, so $N\,\mathrm{I_s} \le C\,I_-(C\delta, j)$ in the notation of (3.1). In the same way $N\,\mathrm{II_s} \le C\,I_+(C\delta, k)$. Assume ($\mathrm{SRT}_+$) and ($\mathrm{SRT}_-$). Since j and k are at least $\min(x_1, 1 - x_2)N$, Lemma 3.1(iv) gives

$$\lim_{\delta \to 0} \limsup_{N \to \infty} \sup_{x_1 N \le k \le x_2 N} N(\mathrm{I_s} + \mathrm{II_s}) = 0.$$

Together with (4.2), the bound on large m and the uniform convergence of Λ_δ, letting $N \to \infty$ and then $\delta \to 0$ gives $\sup_{x_1 N \le k \le x_2 N} |NP_N(k) - L_{\alpha,b}(k/N)| \to 0$. This proves (iii)⇒(ii), and (ii)⇒(i) is trivial.

Small m, necessity. Assume (i), let $k_N = \lfloor x_0 N \rfloor$ and $j_N = N - k_N$, and take $[x_1, x_2] \ni x_0$. For $m < \delta/T_+(N)$ the part $r < k/2$ of (2.6) gives $q^+_{m+1}(k) \ge \mathbb{P}(S^+_m < k/2)\, T_+(k)$, and by Lemma 3.2(i) $\mathbb{P}(S^+_m \ge k/2) \le CmT_+(k/2) \le C_{x_1}\delta$. So $q^+_{m+1}(k) \ge \frac{1}{2}T_+(N)$ for $\delta \le \delta_0$. Since every term is nonnegative,

$$\tfrac{1}{2}NT_+(N) \sum_{m<\delta/T_+(N)} u^-_m(j_N) \le N\,\mathrm{I_s} \le NP_N(k_N) - N(\mathrm{I_m} + \mathrm{II_m}).$$

By (i) and (4.2) the right side tends to $\varepsilon(\delta) = L_{\alpha,b}(x_0) - \Lambda_\delta(x_0)$, and $\varepsilon(\delta) \to 0$ as $\delta \to 0$. The map $N \mapsto j_N$ is nondecreasing with increments 0 or 1, so it takes every large integer value j. For $j = j_N$, (R2) gives $jT_-(j) \le CNT_+(N)$ and $\delta/(CT_-(j)) \le \delta/T_+(N)$. Hence $\limsup_{j\to\infty} I_-(\delta/C, j) \le C\varepsilon(\delta)$, and (SRT$_-$) follows from Lemma 3.1(iv). The same argument with $\mathrm{II_s}$, the bound $q^-_m(j) \ge \mathbb{P}(S^-_{m-1} < j/2)T_-(j) \ge \frac{1}{2}T_-(j)$ for small m, and the sequence k_N, which also takes every large value, gives (SRT$_+$). This proves (i)⇒(iii).

For a fresh minus start the identity of Proposition 2.1 has q^-_{m+1} and q^+_m in place of q^-_m and q^+_{m+1}. This index shift changes none of the limits and none of the bounds, and the proof is the same. □

Corollary 4.2. *Assume (BAL) and (SPAN) and a fresh start of either sign. If $\alpha > 1/2$, the local Lamperti law (ii) of Theorem A holds. If $\alpha \le 1/2$, it holds if and only if the condition of Lemma 3.1(iii) holds for both run laws, and (LB$_\pm$) suffices.*

Proof. Combine Theorem A with Lemma 3.1(iii). □

Without the strong renewal theorem, half of the local law survives. The failure can only show up as an excess.

Corollary 4.3 (the lower half of the local law)**.** *Assume (BAL) and (SPAN), and let the start be fresh of either sign. Then for all $0 < x_1 < x_2 < 1$,*

$$\liminf_{N\to\infty} \inf_{x_1 N \le k \le x_2 N} \big(NP_N(k) - L_{\alpha,b}(k/N)\big) \ge 0.$$

If (SRT$_+$) or (SRT$_-$) fails, then $\limsup_N NP_N(\lfloor xN \rfloor) > L_{\alpha,b}(x)$ for every $x \in (0,1)$.

Proof. The medium and large ranges of the proof of Theorem A use neither (SRT$_+$) nor (SRT$_-$). Every term of the fresh-plus identity is nonnegative, so $NP_N(k) \ge N(\mathrm{I_m} + \mathrm{II_m}) = \Lambda_\delta(k/N) + o(1)$ uniformly by (4.2), and $\Lambda_\delta \to L_{\alpha,b}$ uniformly on $[x_1, x_2]$ as $\delta \to 0$. This gives the first claim. If $\limsup_N NP_N(\lfloor xN \rfloor) \le L_{\alpha,b}(x)$ for some x, the first claim gives convergence at x, and Theorem A gives (SRT$_\pm$). □

Theorem A transfers at once to mixed fresh starts, such as the fair start, since P_N is then a convex combination of F^+_N and F^-_N. It also gives the local law of the walk itself. The position after N steps is $W_N = 2A_N - N$. So for a fresh or mixed fresh start under (BAL), (SPAN) and (SRT$_\pm$),

$$N\,\mathbb{P}(W_N = v) = L_{\alpha,b}\Big(\frac{1 + v/N}{2}\Big) + o(1)$$

uniformly over $v \equiv N \pmod 2$ with v/N in each compact subset of $(-1, 1)$. The limit density of W_N/N on $(-1,1)$ is $\frac{1}{2}L_{\alpha,b}((1+\xi)/2)$, and the factor $\frac{1}{2}$ is compensated by the spacing 2 of the lattice of W_N. For $b = 1$ the density $\frac{1}{2}L_{\alpha,1}((1+\xi)/2)$ is the scaling function of the ballistic Lévy walk in [29, Eq. (61)], as one checks by putting $|\xi + 1| = 2x$ and $|\xi - 1| = 2(1 - x)$ there.

Remark 4.4 (the global law)**.** Under (SRT$_\pm$), Theorem A gives $\mathbb{P}(sN \le A_N \le s'N) \to \int_s^{s'} L_{\alpha,b}$ for $0 < s < s' < 1$ by Riemann sums, and since $L_{\alpha,b}$ has total mass 1 this gives the convergence of A_N/N in law. Proposition 4.1 identifies the limit without any input from [20]. This is a derivation of the global law only under (SRT$_\pm$). The convergence in law is the classical statement [20], and we do not use it. The same argument shows that $\mathbb{P}(A_N \le \varepsilon_N N) + \mathbb{P}(A_N \ge N - \varepsilon_N N) \to 0$ for every $\varepsilon_N \to 0$ under (SRT$_\pm$), since the mass of P_N on $[\delta N, (1-\delta)N]$ tends to $\int_\delta^{1-\delta} L_{\alpha,b}$. So the end layers of Section 5 dominate pointwise but carry no mass in the limit.

Remark 4.5 (where the bivariate structure went)**.** The count A_N is a functional of the bivariate renewal process (S_m^+, S_m^-), and one might expect the local law to need a bivariate strong renewal theorem. The proof shows that it does not. Lemma 3.5 bounds the covering kernel by $CT(k)$ uniformly in m, which decouples the two signs for small m. The small-m part of each sign is then the univariate quantity of Lemma 3.1(iv), and a single interior point controls both.

5 The end layers and the global mode

Near the plus end, $A_N = N - j$ with j small means that the run in progress at time N is a plus run of length of order N. Its probability is of order $T_+(N)$, and the only question is where it started. It starts right after the j-th minus step exactly when a minus run ends there, which gives the renewal mass function $u_-(j)$. The next theorem makes this precise at fixed j under regular variation alone, and uniformly up to distance $(1-\delta)N$ from the end under Doney's condition.

Theorem B (the end layers)**.** *Assume (BAL) and (SPAN).*

(a) For each fixed $j \ge 0$, a fresh plus start has $P_N(N-j) = T_+(N)u_-(j) + o(T_+(N))$, and a fresh minus start has $P_N(N-j) = T_+(N)u_-(j)\mathbf{1}_{\{j\ge 1\}} + o(T_+(N))$.

(b) Assume moreover (LB$_+$) and (SRT$_-$). Then for a fresh plus start and every $\delta \in (0,1)$, uniformly in $0 \le j \le (1-\delta)N$,

$$P_N(N-j) = T_+(N)\big[u_-(j)\,\Phi_b(j/N) + o(\omega_-(j))\big],$$

with Φ_b as in (2.5)*. For a fresh minus start the same holds with $u_-(j)\mathbf{1}_{\{j\ge1\}}$ in place of $u_-(j)$.*

(c) Assume (LB$_\pm$) and $T_\pm(n) \sim c_\pm n^{-\alpha}$ with $c_\pm > 0$. Then for a fresh start of either sign and every sequence $J_N \to \infty$ with $J_N \le N/2$,

$$P_N(k) = N^{-1}L_{\alpha,b}(k/N)\,(1+o(1)) \qquad \text{uniformly over } J_N \le k \le N - J_N.$$

The same statements hold at the minus end with the signs exchanged.

Part (a) has a simple reading. The event $A_N = N - j$ with j fixed requires a plus run of length of order N. On this event the first N steps contain at most $2j+2$ runs, since the j minus steps lie in at most $j+1$ minus runs, and the proof shows that the long run is, with probability close to 1, the run in progress at time N. The tail of that run gives the factor $T_+(N)$. Before it, the minus steps that have been taken are exactly j, and they are the union of complete minus runs. So j must be a renewal epoch of the minus runs, which has probability $u_-(j)$. For a minus start the epoch $j = 0$ is excluded, since the walk begins with a minus run. In the continuous-time model with one law and a fair start, Wang, Schulz, Deng and Barkai [26, Section VII.A] found an atom with a boundary layer for the occupation time, and (a) is a lattice version of this layer for two laws at a fixed time.

In (b) the factor Φ_b is what makes the layer and the bulk agree. Under (SRT$_-$), uniformly for $\varepsilon N \le j \le (1-\delta)N$,

$$T_+(N)\, u_-(j)\, \Phi_b(j/N) = N^{-1} L_{\alpha,b}(1-j/N)\,(1+o(1)). \tag{5.1}$$

Indeed, with $y = j/N$, (SRT$_-$) and (R1) give $T_+(N)u_-(j) = \frac{\sin \pi\alpha}{\pi b}\, y^{\alpha-1}N^{-1}(1+o(1))$, and the identity after (2.5) finishes the check. For j bounded, $\Phi_b(j/N) \to 1$ and (b) reduces to (a). Without pure power tails the ratio $T_-(j)/T_-(N)$ keeps a slowly varying factor in the range $1 \ll j \ll N$, and (b) is the correct form.

5.1 Proof of Theorem B(a)

Let the start be fresh plus and fix j. We use Proposition 2.3. Fix $r \le j$. We show that $\mathbb{E}[T_+(N-j-S_r^+); S_r^+ \le N-j-1] = T_+(N)(1+o(1))$. On $\{S_r^+ \le N/2\}$ the ratio $T_+(N-j-S_r^+)/T_+(N)$ is at most $T_+(N/2-j)/T_+(N) \le C$ and tends to 1 almost surely, so this part is $T_+(N)(1+o(1))$ by dominated convergence. On $\{S_r^+ > N/2\}$ put $i = N-j-S_r^+$ and fix $I \ge 1$. For fixed r and $n \ge 1$, $u_r^+(n) \le r \sup_{s \ge n/r} f_+(s)$, because one of the r summands is at least n/r. Since $f_+(s) = T_+(s) - T_+(s+1) = o(T_+(s))$ by (R1), the part $i \le I$ is at most $I \max_{n \ge N/2} u_r^+(n) = o(T_+(N))$. The part $i > I$ is at most $T_+(I)\mathbb{P}(S_r^+ > N/2) \le C_r T_+(I) T_+(N)$ by Lemma 3.2(i) and (R2). Let $N \to \infty$ and then $I \to \infty$. The second sum in (2.9) has finitely many terms, each at most $C_j \max_{r \le j} u_{r+1}^+(N-j) = o(T_+(N))$ by the same bound on u_{r+1}^+. Hence $P_N(N-j) = T_+(N)\sum_{r \le j} u_r^-(j) + o(T_+(N)) = T_+(N)u_-(j) + o(T_+(N))$.

For a fresh minus start we use Proposition 2.2 with $\nu_- = f_-$. Since $F_{N-d}^+(N-j) = 0$ for $d > j$, it gives

$$P_N(N-j) = \sum_{1 \le d \le j} f_-(d)\, F_{N-d}^+(N-j).$$

By the fresh plus case, $F_{N-d}^+(N-j) = T_+(N)u_-(j-d) + o(T_+(N))$ for each $d \le j$, and $\sum_{1 \le d \le j} f_-(d)u_-(j-d) = u_-(j)$ for $j \ge 1$ by the renewal equation. For $j = 0$ the sum is empty. □

Only (RV) of T_+ is used in (a). The minus law enters only through u_-.

5.2 Proof of Theorem B(b)

Let the start be fresh plus. By (iii) of Lemma 3.1, (LB$_+$) implies (SRT$_+$), so Theorem A applies. For $\varepsilon N \le j \le (1-\delta)N$ the claim follows from Theorem A and (5.1), since $u_-(j) \asymp \omega_-(j)$ there. For $j \le j_0$ with j_0 fixed it is (a), since $\omega_-(j) \ge c_{j_0}$ and $|\Phi_b(j/N) - 1| \to 0$. Since $\Phi_b(y) = 1 + O(y^\alpha)$ and $u_-(j) \ge c\,\omega_-(j)$ for $j \ge j_0$ under (SRT$_-$), it remains to show the following. There is a function ζ with $\zeta(\eta) \to 0$ as $\eta \to 0$ such that for every small $\eta > 0$ there are $\varepsilon > 0$ and j_0 with

$$\limsup_{N \to \infty} \sup_{j_0 \le j \le \varepsilon N} \Big| \frac{P_N(N-j)}{T_+(N)u_-(j)} - 1 \Big| \le \zeta(\eta), \tag{5.2}$$

Let $k = N - j$ and write $P_N(k) = \mathrm{I} + \mathrm{II}$ as in Section 4.2. Put $M = \eta/T_+(N)$ and $\theta = \eta^{1/(2+2\alpha)}$. For small η we have $\eta^{1/\alpha} \le \theta/4$, since $1/\alpha > 1 > 1/(2+2\alpha)$, so $a_M \le \theta N$ for large N.

Term I, $m \le M$. We show that $q_{m+1}^+(k) = T_+(N)(1 + O(\eta^{1/2} + \theta + \varepsilon))$ uniformly in $m \le M$ and $j \le \varepsilon N$. For the lower bound, the part $r < \theta N$ of (2.6) gives $q_{m+1}^+(k) \ge \mathbb{P}(S_m^+ < \theta N)T_+(k)$, and $\mathbb{P}(S_m^+ \ge \theta N) \le CmT_+(\theta N) \le C\eta\theta^{-\alpha} \le C\eta^{1/2}$ by Lemma 3.2(i) and (R1). For the upper bound, the part $r < \theta N$ is at most $T_+(k-\theta N) \le (1+o(1))(1-\varepsilon-\theta)^{-\alpha}T_+(N)$. For $r \ge \theta N$ we have $r \ge a_M \ge a_m$, so Lemma 3.1(v) gives $u_m^+(r) \le CmT_+(r)/r \le C\eta\theta^{-1-\alpha}/N$, and with (R3) this part is at most $C\eta\theta^{-1-\alpha}N^{-1}\sum_{i \le N} T_+(i) \le C\eta\theta^{-1-\alpha}T_+(N) = C\eta^{1/2}T_+(N)$.

Next, the tail of the renewal sum. For $m \ge 1/T_-(j)$ we have $a_m \ge cj$, so by Lemma 3.2(iii) for the minus sign and $T_-(j+1) \ge cT_-(j)$,

$$\sum_{m>M} u_m^-(j) \le \frac{C}{j} \sum_{m>M} e^{-cmT_-(j)} \le \frac{C}{jT_-(j)}\, e^{-cMT_-(j)}$$

whenever $MT_-(j) \ge 1$. Now $MT_-(j) \ge \eta T_-(\varepsilon N)/T_+(N) \to \eta b \varepsilon^{-\alpha}$. We choose $\varepsilon = \varepsilon(\eta) \le \eta$ so small that $e^{-c\eta b\varepsilon^{-\alpha}/2} \le \eta$. Since $u_-(j) \ge c/(jT_-(j))$ for $j \ge j_0$, this gives $\sum_{m>M} u_m^-(j) \le C\eta\, u_-(j)$. Hence $\sum_{1\le m\le M} u_m^-(j) q_{m+1}^+(k) = T_+(N)u_-(j)(1+O(\eta^{1/2}))$.

Term I, $m > M$. By Lemma 3.5, $q_{m+1}^+(k) \le CT_+(k) \le CT_+(N)$, so this part is at most $CT_+(N)\sum_{m>M} u_m^-(j) \le C\eta\, T_+(N)u_-(j)$.

Term II. By Lemma 3.6, $q_m^-(j) \le CT_-(j)e^{-cmT_-(j)}$. Choose $c' > 0$ with $a_{c'/T_+(N)} \le N/2$ for large N. For $m \le c'/T_+(N)$ we have $k \ge N/2 \ge a_m$, and Lemma 3.1(v) gives $u_m^+(k) \le CmT_+(N)/N$. This part is at most

$$C\,T_-(j)\frac{T_+(N)}{N}\sum_{m\ge 1} me^{-cmT_-(j)} \le C\frac{T_+(N)}{NT_-(j)} \le C\,\frac{j}{N}\,T_+(N)u_-(j) \le C\varepsilon\,T_+(N)u_-(j).$$

For $m > c'/T_+(N)$ we have $u_m^+(k) \le C/a_m \le C/N$, and this part is at most $CN^{-1}e^{-c''s}$ with $s = T_-(j)/T_+(N)$, because $T_-(j)\sum_{m>c'/T_+(N)} e^{-cmT_-(j)} \le Ce^{-cc's}$. Relative to $T_+(N)u_-(j) \ge c/(Nsj/N)$ this is at most $C(j/N)\,s\,e^{-c''s} \le C\varepsilon$.

Collecting the terms gives (5.2) with $\zeta(\eta) = C(\eta^{1/2} + \theta + \varepsilon + \eta) \le C\eta^{1/(2+2\alpha)}$. For a fresh minus start, Proposition 2.1 gives $P_N(k) = \sum_{m\ge1}[u_m^+(k)q_{m+1}^-(j) + u_m^-(j)q_m^+(k)]$. The second sum is treated as I and gives $T_+(N)\sum_{m\ge1} u_m^-(j) = T_+(N)u_-(j)$ for $j \ge 1$, and the first is treated as II. □

5.3 Proof of Theorem B(c)

By (iii) of Lemma 3.1, (LB$_\pm$) implies (SRT$_\pm$). Apply (b) at the plus end with $\delta = 1/2$. For $J_N \le j \le N/2$, (SRT$_-$) gives $u_-(j) \asymp \omega_-(j)$, so the error $o(\omega_-(j))$ is $o(u_-(j))$. With pure power tails, $T_+(N)u_-(j) = \frac{\sin\pi\alpha}{\pi}\frac{c_+N^{-\alpha}}{c_-j^{1-\alpha}}(1+o(1)) = \frac{\sin\pi\alpha}{\pi b}\,N^{-1}(j/N)^{\alpha-1}(1+o(1))$ uniformly in $j \ge J_N$, with $b = c_-/c_+$. Multiplying by $\Phi_b(j/N)$ gives $N^{-1}L_{\alpha,b}(1-j/N)(1+o(1))$. The minus end is the same with the signs exchanged. □

5.4 An upper bound under the strong renewal theorem

Part (b) of Theorem B uses (LB$_+$). Its upper half, the bound $P_N(N-j) \le CT_+(N)\omega_-(j)$ on $j \le N/2$, holds under the strong renewal theorem alone, except on the line $\alpha = 1/2$, where we need the slowly varying factor to be bounded. Section 7 uses this bound for the remainder walk after a long initial run.

Lemma 5.1 (the end layer from above)**.** *Assume (BAL) and (SPAN), and one of the following: $\alpha > 1/2$; $\alpha < 1/2$ and (SRT$_\pm$); $\alpha = 1/2$, (SRT$_\pm$), and $nT_+(n)^2$ bounded above and away from zero. Then there is C such that for all $n \ge 1$ and $0 \le i \le n/2$,*

$$F_n^-(i) \le C\,T_+(n)\,\omega_+(i), \qquad F_n^+(n-i) \le C\,T_+(n)\,\omega_-(i).$$

Proof. We prove the first bound, and the second is its mirror. At $i = 0$, $F_n^-(0) = T_-(n) \le CT_+(n)$. Let $i \ge 1$ and $j = n - i \ge n/2$. By Proposition 2.1 for a fresh minus start, $F_n^-(i) = \sum_m u_m^+(i)q_{m+1}^-(j) + \sum_m u_m^-(j)q_m^+(i)$. In the first sum Lemma 3.5 gives $q_{m+1}^-(j) \le CT_-(j) \le CT_+(n)$, and the sum of $u_m^+(i)$ over m is $u_+(i) \le C\omega_+(i)$ by (SRT$_+$), which is automatic for $\alpha > 1/2$. In the second sum Lemma 3.6 gives $q_m^+(i) \le CT_+(i)e^{-cmT_+(i)}$. If $\alpha > 1/2$, Lemma 3.1(ii) gives $u_m^-(j) \le CmT_-(j)/a_m \le CT_+(n)m/a_m$, and since m/a_m is regularly varying with index $1 - 1/\alpha \in (-1, 0)$, $\sum_m (m/a_m)e^{-cmT_+(i)} \le CM^2/a_M$ with

$M = 1/T_+(i)$, and $a_M \asymp i$, so the second sum is at most $CT_+(n)/(iT_+(i))$. If $\alpha < 1/2$, we bound $e^{-cmT_+(i)}$ by 1, so the second sum is at most $CT_+(i)u_-(j) \le CT_+(i)/(nT_+(n))$ by (SRT$_-$), and $T_+(i)/(nT_+(n)) \le CT_+(n)/(iT_+(i))$ because $iT_+(i)^2 \le CnT_+(n)^2$ for $i \le n$, as $xT_+(x)^2$ is regularly varying with the positive index $1 - 2\alpha$ [7, Theorem 1.5.3]. If $\alpha = 1/2$, the same route works when $xT_+(x)^2$ is bounded above and away from zero. □

5.5 The global mode

Corollary 5.2 (the global mode)**.** *Assume (BAL), (SPAN) and (LB$_\pm$), and let the start be fresh of either sign. Let $\kappa_+(j) = \lim_N P_N(N-j)/T_+(N)$ and $\kappa_-(j) = \lim_N P_N(j)/T_-(N)$, given by Theorem B(a). There is J such that for all large N every global mode of P_N lies in $\{0,\dots,J\} \cup \{N-J,\dots,N\}$. If moreover $\max_j \kappa_+(j) > b \max_j \kappa_-(j)$ and the maximum of κ_+ is attained at a single j_+, then the global mode is $N - j_+$ for all large N.*

Proof. Both profiles tend to 0 and are not identically 0, so their maxima are positive and attained. By Theorem A, $P_N \le C/N$ on $[\varepsilon N, (1-\varepsilon)N]$. By Theorem B(b), $P_N(N-j) \le CT_+(N)\omega_-(j) \le CT_+(N)\omega_-(J)$ for $J \le j \le \varepsilon N$, and the same at the minus end. Since $NT_+(N) \to \infty$ and $\omega_\pm(J) \to 0$ as $J \to \infty$, for J large and then N large every value of P_N on $[J, N-J]$ is less than half of $\min\{T_+(N)\max\kappa_+,\, T_-(N)\max\kappa_-\}$. The values at distance less than J from the ends converge after division by $T_+(N)$ or $T_-(N)$, and $T_-(N) \sim bT_+(N)$. This gives both claims. □

For a fresh plus start, $\kappa_+ = u_-$ has $\kappa_+(0) = 1$, and the mode is at an end for every $\alpha \in (0,1)$. For $1 < \alpha < 2$ the comparison between the atoms and the bulk has a golden ratio threshold [23]. Here the end always wins, because the bulk values are of order $1/N$ and the end values of order $T(N) \gg 1/N$.

Remark 5.3 (what (b) needs)**.** We use (LB$_+$) only through Lemma 3.1(v), in the range $j \le \varepsilon N$. Under (SRT$_\pm$) alone Lemma 5.1 gives the upper half of (b) when $\alpha \ne 1/2$, and we do not know whether the lower half holds. It would follow from the bound

$$\sum_{i \le \theta N} T_+(i) \sum_{m \le \eta/T_+(N)} u_m^-(j)\, u_m^+(N-j-i) \le \zeta(\theta,\eta)\, T_+(N) u_-(j)$$

for $j \le \varepsilon N$, with $\zeta(\theta,\eta) \to 0$ as $\theta, \eta \to 0$. The bound of Lemma 3.1(ii) loses a factor of order N/j here.

6 The initial run

In this section the initial run has an arbitrary law (ν_+, ν_-) that does not depend on N, with $D < \infty$ almost surely. For $0 \le k \le N$ let

$$\mathcal{D}_N^+(k) = N \sum_{k/2 \le d \le \min(k, N-1)} \nu_+(d)\, F_{N-d}^-(k-d),$$

$$\mathcal{D}_N^-(k) = N \sum_{(N-k)/2 \le d \le \min(N-k, N-1)} \nu_-(d)\, F_{N-d}^+(k).$$

The term $\mathcal{D}_N^+(k)$ is the event that the initial run is a plus run that ends after at least half of the k plus steps. The rest of the walk is then a fresh minus walk that has few plus steps, so it sits in its own end layer, where the probabilities are of order $T(N)$ and not $1/N$.

Theorem C (the initial run)**.** *Assume (BAL), (SPAN) and (SRT$_\pm$).*

(a) (The defect.) Uniformly over k with k/N in each compact subset of $(0,1)$,

$$NP_N(k) = L_{\alpha,b}(k/N) + \mathcal{D}_N^+(k) + \mathcal{D}_N^-(k) + o(1).$$

(b) (A necessary condition.) If $NP_N(\lfloor x_0 N\rfloor) \to L_{\alpha,b}(x_0)$ *for one* $x_0 \in (0,1)$*, then* $nT_+(n)\,\nu_\pm(n) \to 0$ *as* $n \to \infty$*.*

(c) (A sufficient condition.) Assume moreover (LB$_\pm$*). If* $n\nu_\pm(n) \to 0$*, then* $NP_N(k) = L_{\alpha,b}(k/N) + o(1)$ *uniformly over* k *with* k/N *in each compact subset of* $(0,1)$*. This holds in particular when* ν_+ *and* ν_- *are eventually nonincreasing.*

(d) (An exact criterion.) Assume moreover (LB$_\pm$*) and* $u_\pm(i) > 0$ *for all* $i \ge 0$*. Then the conclusion of (c) holds if and only if, for all* $0 < x_1 < x_2 < 1$*,*

$$\sup_{x_1 N \le k \le x_2 N} NT_+(N) \sum_{0 \le i \le k/2} \nu_+(k-i)\,\omega_+(i) \to 0 \qquad \text{and}$$

$$\sup_{x_1 N \le k \le x_2 N} NT_+(N) \sum_{0 \le i \le (N-k)/2} \nu_-(N-k-i)\,\omega_-(i) \to 0.$$

(e) (A finite initial run that changes the bulk.) Let $c > 0$*,* $n_i = 2^{2^i}$*, and let* $\nu_+(n_i) = c/(n_i T_+(n_i))$ *for* $i \ge i_0$*, with the rest of the mass of the initial run placed anywhere. Then for every* $x_0 \in (0,1)$*, along the* N *with* $\lfloor x_0 N\rfloor \in \{n_i\}$*,*

$$\liminf NP_N(\lfloor x_0 N\rfloor) \ge L_{\alpha,b}(x_0) + c\,b\,x_0^{\alpha-1}(1-x_0)^{-\alpha}.$$

Under the hypotheses of (d) the condition of (b) is not sufficient. An initial run with a cluster of $\lfloor\sqrt{n_i}\rfloor$ *equal masses* $T_+(\sqrt{n_i})/(n_i T_+(n_i))$ *just below each* n_i*, for* $i \ge i_0$*, and the rest of its mass at* $d = 1$*, satisfies the condition of (b), while the local law fails along the* N *with* $\lfloor x_0 N\rfloor \in \{n_i\}$*.*

(f) (The end layer.) Assume only (BAL) and (SPAN). For each fixed $j \ge 0$*,*

$$P_N(N-j) = Q_+(N)\mathbf{1}_{\{j=0\}} + T_+(N)\kappa_+(j) + \sum_{1 \le m < N/2} \nu_+(N-m)\,\Pi^-_m(j) + o(T_+(N)),$$

where

$$\kappa_+(j) = \pi_+ u_-(j)\mathbf{1}_{\{j\ge 1\}} + \sum_{d=1}^{j} \nu_-(d)\,u_-(j-d). \tag{6.1}$$

Hence $P_N(N-j)/T_+(N) \to \kappa_+(j)$ *for every* $j \ge 1$ *if and only if*

$$W_N^+ := \sum_{1 \le m < N/2} \nu_+(N-m)\,T_+(m) = o(T_+(N)).$$

This holds if $n\nu_+(n) \to 0$*, and it holds for every fresh start.*

The same statements hold with the signs exchanged.

Part (b) is the first place where the infinite mean shows. For $1 < \alpha < 2$ the bulk needs no condition on the initial run on any macroscopic interior [23]. Here a first run of length close to k leaves a fresh walk with very few plus steps, whose probability is of order $T(N) \gg 1/N$. In (f) the window condition replaces the condition $\nu_+ = o(T_+)$ of the finite-mean case, because now $\sum_{m<N/2} T_+(m)$ grows like $NT_+(N)$. At $j = 0$ the end value is $P_N(N) = Q_+(N)$ exactly. So in (f) the profile value at $j = 0$ is $\lim_N Q_+(N)/T_+(N)$ when this limit exists, and not $\kappa_+(0) = 0$.

We compare the conditions. The necessary condition (b) asks that $\nu_\pm(n)$ be $o(1/(nT_+(n)))$, and the sufficient condition (c) asks that $\nu_\pm(n)$ be $o(1/n)$. The gap between them is the factor $T_+(n)$, which tends to 0. The exact criterion (d) says that what matters is a weighted window sum of $\nu_\pm$ just below the point, with the weight

$\omega_\pm(i) \asymp 1/(iT(i))$, whose partial sums up to w grow like $1/T(w)$. A single spike of size $c/(nT(n))$ at n fails (b). A cluster of w equal masses $T(w)/(nT(n))$ just below n satisfies (b) and still fails (d), because the weights $\omega(i)$ over the cluster add up to about $1/(\alpha T(w))$. Under (LB$_\pm$) every initial run with a regularly varying tail $\mathbb{P}(D \geq n)$ of negative index and an eventually monotone mass function satisfies (c), whatever its index, so it changes neither the bulk nor, by (f), the end layers. Note that the initial run may be much heavier than the other runs. For instance $\nu_\pm(n) \propto n^{-1-\gamma}$ with $0 < \gamma < \alpha$ gives an initial run with a heavier tail than every later run, and under (LB$_\pm$) the local Lamperti law still holds. The effect of such a run is only felt at speed $\mathbb{P}(D \geq \varepsilon N)$, as the computations of Section 11 show for $\gamma = 0.2$ and $\alpha = 1/2$.

6.1 Proofs

Proof of (a). Fix $0 < x_1 < x_2 < 1$ and let $k \in [x_1 N, x_2 N]$. In Proposition 2.2 the atoms vanish. Consider the plus part $\sum_{d<N} \nu_+(d) F^-_{N-d}(k-d)$, where only $d \leq k$ contributes. Fix a small $\varepsilon > 0$. For $d < \varepsilon N$ the horizon is $n = N - d \geq (1-\varepsilon)N$ and $(k-d)/n \in [x_1/2, x_2]$. Theorem A for the fresh minus walk holds uniformly in the horizon $n \to \infty$ and in the ratio in a compact set. Since $|(k-d)/n - k/N| \leq 2\varepsilon$ and $L_{\alpha,b}$ is uniformly continuous on $[x_1/2, x_2]$, this gives $NF^-_{N-d}(k-d) = L_{\alpha,b}(k/N) + O(\varepsilon) + o(1)$. So the terms with $d < \varepsilon N$ contribute $\mathbb{P}(\Sigma = +, D < \varepsilon N) L_{\alpha,b}(k/N) + O(\varepsilon) + o(1) = \pi_+ L_{\alpha,b}(k/N) + O(\varepsilon) + o(1)$. For $\varepsilon N \leq d < k/2$ the ratio $(k-d)/(N-d)$ lies in $[x_1/2, x_2]$ and $N - d \geq N/2$, so $NF^-_{N-d}(k-d) \leq C$ by Theorem A, and these terms contribute at most $C\mathbb{P}(D \geq \varepsilon N) \to 0$. The terms with $k/2 \leq d \leq k$ form $\mathcal{D}^+_N(k)$. The minus part $\sum_{d<N} \nu_-(d) F^+_{N-d}(k)$ is the same, with $d \leq N - k$ and the ratio $k/(N-d) \leq 2x_2/(1+x_2) < 1$ for $d < (N-k)/2$. Since $\pi_+ + \pi_- = 1$, letting $N \to \infty$ and then $\varepsilon \to 0$ proves (a). □

Proof of (b). All terms of $\mathcal{D}^\pm_N$ are nonnegative. Since $F^-_{N-k}(0) = T_-(N-k)$, the term $d = k$ of $\mathcal{D}^+_N(k)$ is $N\nu_+(k)T_-(N-k)$. In the same way the term $d = N - k$ of $\mathcal{D}^-_N(k)$ is $N\nu_-(N-k)T_+(k)$. By (a) and the hypothesis both tend to 0 along $k = k_N = \lfloor x_0 N \rfloor$. Both k_N and $N - k_N$ take every large integer value, and $NT_-(N-k_N)$ and $NT_+(k_N)$ are comparable with $k_N T_+(k_N)$ and $(N-k_N)T_+(N-k_N)$ by (R2). Hence $nT_+(n)\nu_\pm(n) \to 0$. □

Proof of (c) and (d). Write $i = k - d$, so $\mathcal{D}^+_N(k) = N \sum_{0 \leq i \leq k/2} \nu_+(k-i) F^-_{N-k+i}(i)$. The horizon $n = N - k + i$ lies between $(1-x_2)N$ and N, and $i/n \leq (k/2)/(N - k/2) \leq x_2/(2-x_2) < 1$. Theorem B(b) at the minus end, with the signs exchanged, applies to the fresh minus walk under (LB$_-$) and (SRT$_+$). It gives, uniformly in these i,

$$F^-_n(i) = T_-(n)\big[u_+(i)\Phi_{1/b}(i/n) + o(\omega_+(i))\big].$$

Since $u_+ \leq C\omega_+$, $\Phi_{1/b}$ is bounded on $[0, x_2/(2-x_2)]$ and $T_-(n) \leq CT_+(N)$, we get $\mathcal{D}^+_N(k) \leq C\, NT_+(N) \sum_{0 \leq i \leq k/2} \nu_+(k-i)\omega_+(i)$. If moreover $u_+(i) > 0$ for all i, then $u_+ \geq c\,\omega_+$ by (SRT$_+$), and $\Phi_{1/b}$ is bounded below on that interval, so the reverse inequality holds with a smaller constant for large N. The same holds for $\mathcal{D}^-_N$. Since $NP_N(k) - L_{\alpha,b}(k/N) = \mathcal{D}^+_N(k) + \mathcal{D}^-_N(k) + o(1)$ uniformly with $\mathcal{D}^\pm_N \geq 0$, this proves (d).

For (c), let $\nu_+(d) \leq \epsilon/d$ for $d \geq x_1 N/2$. Then by (R3)

$$NT_+(N) \sum_{0 \leq i \leq k/2} \nu_+(k-i)\omega_+(i) \leq \frac{2\epsilon N T_+(N)}{k}\Big(1 + \sum_{1 \leq i \leq k} \frac{1}{iT_+(i)}\Big) \leq C\epsilon\, \frac{NT_+(N)}{kT_+(k)} \leq C_{x_1}\epsilon.$$

The same bound holds for the minus sum, so $\mathcal{D}^\pm_N \to 0$ uniformly and (a) gives the local law. If ν_+ is eventually nonincreasing, then $\frac{n}{2}\nu_+(n) \leq \sum_{n/2 < d \leq n} \nu_+(d) \to 0$. □

Proof of (e). The masses $c/(n_i T_+(n_i))$ are summable, since $1/(nT_+(n))$ is regularly varying with the negative index $\alpha - 1$ and n_i grows doubly exponentially. So the initial run is well defined for i_0 large. Let $k = \lfloor x_0 N \rfloor = n_i$. By (a) and the nonnegativity of the other terms, $NP_N(k) \geq L_{\alpha,b}(k/N) + N\nu_+(k)T_-(N-k) + o(1)$. Now

$$N\nu_+(k)T_-(N-k) = c\,\frac{N}{k}\,\frac{T_-(N-k)}{T_+(k)} \to c\,x_0^{-1}\,b\Big(\frac{1-x_0}{x_0}\Big)^{-\alpha} = c\,b\,x_0^{\alpha-1}(1-x_0)^{-\alpha},$$

by (R1). For the second example put $n = n_i$ and $w = \lfloor \sqrt{n} \rfloor$, and let $\nu_+(d) = T_+(w)/(nT_+(n))$ for $n - w < d \leq n$. The mass of the cluster is $wT_+(w)/(nT_+(n))$, which is regularly varying in n with index $-(1-\alpha)/2 < 0$, so the clusters have summable mass. For d in the cluster, $dT_+(d)\nu_+(d) \leq CT_+(w) \to 0$, so the condition of (b) holds. At $k = n$ the first sum in (d) is at least

$$NT_+(N)\,\frac{T_+(w)}{nT_+(n)} \sum_{0 \leq i < w} \omega_+(i) \geq c\,\frac{NT_+(N)}{nT_+(n)} \cdot T_+(w) \cdot \frac{1}{\alpha T_+(w)}(1 + o(1)) \geq c' > 0$$

by (R3), so the criterion of (d) fails and the local law fails along these N. □

Proof of (f). By Proposition 2.2, with $F^-_{N-d}(N-j-d) = \Pi^-_{N-d}(j)$,

$$P_N(N-j) = Q_+(N)\mathbf{1}_{\{j=0\}} + \sum_{d<N} \nu_+(d)\,\Pi^-_{N-d}(j) + \sum_{1 \leq d \leq j} \nu_-(d)\,F^+_{N-d}(N-j),$$

since $F^+_{N-d}(N-j) = 0$ for $d > j$. By Theorem B(a) for a fresh plus walk, the last sum is $T_+(N)\sum_{d \leq j} \nu_-(d)u_-(j-d) + o(T_+(N))$. In the middle sum, split at $d = N/2$. For $d \leq N/2$, Theorem B(a) for a fresh minus walk gives $\Pi^-_n(j) = T_+(n)[u_-(j)\mathbf{1}_{\{j \geq 1\}} + o(1)]$ uniformly in $n \geq N/2$. Since $1 \leq T_+(N-d)/T_+(N) \leq C$ for $d \leq N/2$ and the ratio tends to 1 for each fixed d, dominated convergence gives $\sum_{d \leq N/2} \nu_+(d)T_+(N-d) = \pi_+ T_+(N)(1 + o(1))$. So the terms with $d \leq N/2$ give $T_+(N)\pi_+ u_-(j)\mathbf{1}_{\{j \geq 1\}} + o(T_+(N))$. The terms with $d > N/2$ are exactly $\sum_{1 \leq m < N/2} \nu_+(N-m)\Pi^-_m(j)$ with $m = N - d$. This proves the formula.

For the criterion, let $j \geq 1$. By Theorem B(a) for a fresh minus walk, $\Pi^-_m(j) \leq C_j T_+(m)$ for all m, so $W^+_N = o(T_+(N))$ makes the sum $o(T_+(N))$. Conversely, suppose that $P_N(N-j)/T_+(N) \to \kappa_+(j)$ for every $j \geq 1$. Then the sum, which is nonnegative, is $o(T_+(N))$ for every $j \geq 1$. Pick $j^* \geq 1$ with $u_-(j^*) > 0$. Then $\Pi^-_m(j^*) \geq cT_+(m)$ for $m \geq m^*$, so $\sum_{m^* \leq m < N/2} \nu_+(N-m)T_+(m) = o(T_+(N))$. For each $m < m^*$ take $j = m$. Then $\Pi^-_m(m) = T_-(m) > 0$, so $\nu_+(N-m) = o(T_+(N))$. Together these give $W^+_N = o(T_+(N))$.

If $n\nu_+(n) \to 0$, then $\nu_+(N-m) = o(1/N)$ uniformly in $m < N/2$, and $W^+_N \leq o(1/N)\sum_{m < N/2} T_+(m) = o(T_+(N))$ by (R3). For a fresh minus start $\nu_+ = 0$. For a fresh plus start, fix M. The terms $m < M$ give at most $M \max_{n \geq N-M} f_+(n) = o(T_+(N))$, and the terms $M \leq m < N/2$ give at most $T_+(M)\mathbb{P}(N/2 < X_+ \leq N-M) \leq T_+(M)T_+(N/2) \leq CT_+(M)T_+(N)$. Let $N \to \infty$ and then $M \to \infty$. □

If the law of the initial run depends on N and puts mass of order 1 on lengths of order N, the local law itself changes. A first run of length zN is followed by a fresh walk of length $(1-z)N$, and the local law becomes a mixture of rescaled Lamperti densities. Section 7 proves this, under (SRT$_\pm$) and with no condition of the type (LB), and applies it to the process observed from a late time on.

7 Initial runs of length of order N, and aging

Theorem C fixes the law of the initial run. If that law depends on N and puts mass of order 1 on lengths of order N, the local law changes. A first run of length zN is followed

by a fresh walk of length $(1-z)N$, and the law becomes a mixture of rescaled Lamperti densities. The first theorem of this section proves the mixture under (SRT$_\pm$). The second applies it to the process observed from step t_a+1 on, with $t_a/N \to \theta$, which is the aged occupation time of the physics literature, and gives its atoms and its end layers.

7.1 The mixture law

Let the law $\nu^{(N)}$ of the initial run depend on N. We write $D^{(N)}$ for its length and keep the notation $\nu_\pm$, $Q_\pm$, $\pi_\pm$, with the dependence on N understood. We consider three conditions.

(H0) The limit $\pi^0 = \lim_{\varepsilon\to 0}\lim_{N\to\infty}\mathbb{P}(D^{(N)} < \varepsilon N)$ exists.

(H1) For each sign, the measures $\mathbb{P}(\Sigma=\sigma,\, D^{(N)}/N \in \cdot\,)$ restricted to $(0,1)$ converge vaguely to $\rho_\sigma(z)\,dz$, with ρ_σ continuous on $(0,1)$.

(H2) For each $\varepsilon \in (0,\frac12)$, $\sup_N \sup_{\varepsilon N \le d \le (1-\varepsilon)N} N\nu_\pm(d) < \infty$.

For $0<x<1$ put

$$\mathcal{L}(x) = \pi^0 L_{\alpha,b}(x) + \int_0^x \rho_+(z)\, L_{\alpha,b}\Big(\frac{x-z}{1-z}\Big)\frac{dz}{1-z} + \int_0^{1-x} \rho_-(z)\, L_{\alpha,b}\Big(\frac{x}{1-z}\Big)\frac{dz}{1-z}. \quad (7.1)$$

The first integral is a plus initial run of length zN followed by a fresh minus walk of length $(1-z)N$ that takes $(x-z)N$ plus steps, and the second is the mirror.

Theorem D (initial runs of length of order N)**.** *Assume (BAL), (SPAN) and (SRT$_\pm$), and let the law of the initial run satisfy (H0), (H1) and (H2). Then $NP_N(\lfloor xN\rfloor) \to \mathcal{L}(x)$ uniformly for x in compact subsets of $(0,1)$.*

The proof needs a bound on the near-diagonal terms of the first-run decomposition, the initial runs that end within δN steps of the point k. The remainder is then a fresh walk with few steps of its own sign, which sits in its own end layer, where the probabilities are of order $T(N)$ and not $1/N$. Under (LB$_\pm$), Theorem B(b) bounds each such term. Under (SRT$_\pm$) we bound their sum instead, by a truncated form of the count behind the Green identity (2.11).

Lemma 7.1 (the counting bound)**.** *For a fresh minus walk and integers $n_0 \ge 1$, $I \ge 0$ and $m_0 \ge 1$,*

$$\sum_{i=0}^{I} F^-_{n_0+i}(i) \le C\, m_0 T_-(n_0) + e^{-m_0 T_+(I+1)} + u_-(n_0)\sum_{n=1}^{I} T_+(n),$$

and the mirror bound holds for $\sum_{i\le I} F^+_{n_0+i}(n_0)$ with the signs exchanged. The constant C is that of Lemma 3.2(i), and the bound needs only (RV).

Proof. Let C_m be the number of minus steps among the first m steps of the fresh minus walk, and t_b the step of its b-th minus step. Then $F^-_{n_0+i}(i) = \mathbb{P}(C_{n_0+i} = n_0)$, so by Tonelli's theorem the left side is $\mathbb{E}\#\{m \in [n_0, n_0+I] : C_m = n_0\}$. The set $\{m : C_m = n_0\}$ is the interval $[t_{n_0}, t_{n_0+1})$. Its length is 1 unless step t_{n_0} ends the r-th minus run, that is $S^-_r = n_0$, and then it is $1 + X^+_r$, where X^+_r is independent of the event $\{S^-_r = n_0\}$, which is a function of the minus runs alone. Since the window starts at $n_0 \le t_{n_0}$, the count is at most $\mathbf{1}_{\{t_{n_0} \le n_0+I\}} + \sum_r \mathbf{1}_{\{S^-_r = n_0\}} \min(X^+_r, I)$, whose expectation is at most $\mathbb{P}(t_{n_0} - n_0 \le I) + u_-(n_0)\,\mathbb{E}\min(X_+, I)$, and $\mathbb{E}\min(X_+,I) = \sum_{n=1}^{I} T_+(n)$. Now $t_{n_0} - n_0 = S^+_{J-1}$, where $J-1$ is the number of complete minus runs before the n_0-th minus step, a function of the minus runs. The event $\{J-1 < m_0\} = \{S^-_{m_0} \ge n_0\}$ has probability at most $Cm_0T_-(n_0)$ by Lemma 3.2(i). On $\{J-1 \ge m_0\}$ we have $S^+_{J-1} \ge S^+_{m_0}$, and by independence $\mathbb{P}(S^+_{m_0} \le I) \le (1-T_+(I+1))^{m_0} \le e^{-m_0T_+(I+1)}$. □

Corollary 7.2. *Assume (BAL) and the upper half of (SRT$_-$), $u_- \le C\omega_-$. Fix $c \in (0,1)$ and $\delta \in (0,1)$. Then for $n_0 \ge cN$ and all large N,*

$$\sum_{i \le \delta N} F^-_{n_0+i}(i) \le \zeta(\delta) := C_c\big(\delta^{\alpha/4} + e^{-\delta^{-\alpha/4}} + \delta^{(1-\alpha)/2}\big),$$

and the mirror bound holds under the upper half of (SRT$_+$).

Proof. Apply Lemma 7.1 with $I = \lfloor \delta N \rfloor$ and $m_0 = \lceil \eta/T_+(N) \rceil$, $\eta = \delta^{\alpha/4}$. By (R2), $m_0 T_-(n_0) \le C_c(\eta + T_+(N))$. By (R1), $m_0 T_+(\delta N + 1) \ge \eta\, \delta^{-\alpha/2} = \delta^{-\alpha/4}$ for large N. Finally $u_-(n_0) \le C_c/(NT_+(N))$ and, by (R3) and (R2), $\sum_{n \le \delta N} T_+(n) \le C\delta N T_+(\delta N) \le C\delta^{(1-\alpha)/2} N T_+(N)$. □

Proof of Theorem D. Fix $0 < x_1 < x_2 < 1$, let $k = k_N$ with $x = k/N \in [x_1, x_2]$, put $j = N - k$, and fix $0 < \delta < \varepsilon < \min(x_1, 1 - x_2)/4$. In Proposition 2.2 the atoms vanish. We split the plus part $\sum_{d \le k} \nu_+(d) F^-_{N-d}(k-d)$ into three ranges.

Range 1, $d < \varepsilon N$. The horizon is $n = N - d \ge (1-\varepsilon)N$ and $(k-d)/n$ lies in a compact subset of $(0,1)$, at distance at most 2ε from x. Theorem A(ii) for the fresh minus walk is uniform in the ratio as $n \to \infty$, and $L_{\alpha,b}$ is uniformly continuous on compacts, so $N F^-_{N-d}(k-d) = L_{\alpha,b}(x) + O(\varepsilon) + o(1)$. This range contributes $\mathbb{P}(\Sigma = +, D < \varepsilon N)\,[L_{\alpha,b}(x) + O(\varepsilon) + o(1)]/N$.

Range 2, $\varepsilon N \le d \le k - \delta N$. Here $(k-d)/(N-d) \in [\delta, (x_2 - \varepsilon)/(1-\varepsilon)]$ and the horizon is at least $(1-x_2)N$, so by Theorem A(ii) $N F^-_{N-d}(k-d) = g_x(d/N) + o(1)$ uniformly, with $g_x(z) = L_{\alpha,b}((x-z)/(1-z))/(1-z)$, and $N F^-_{N-d}(k-d) \le C$. The function g_x is continuous on $[\varepsilon, x - \delta]$, and the family (g_x) is equicontinuous in $x \in [x_1, x_2]$ there. By (H1), with a limit measure that has a density, the weighted sum converges: $N \sum_{\varepsilon N \le d \le k - \delta N} \nu_+(d) F^-_{N-d}(k-d) = \int_\varepsilon^{x-\delta} \rho_+(z) g_x(z)\, dz + o(1)$, uniformly in x.

Range 3, $k - \delta N < d \le k$. Put $i = k - d \le \delta N$. The terms are $\nu_+(k-i) F^-_{j+i}(i)$ with $j = N - k \ge (1-x_2)N$, and $d \in [\varepsilon N, (1-\varepsilon)N]$ by the choice of ε, so (H2) gives $N\nu_+(k-i) \le C_\varepsilon$. By Corollary 7.2 with $n_0 = j$ and $c = 1 - x_2$, this range contributes at most $C_\varepsilon \zeta(\delta)/N$.

The minus part $\sum_{d \le j} \nu_-(d) F^+_{N-d}(k)$ is the same, with $F^+_{N-d}(k) = F^+_{k+i'}(k)$, $i' = j - d$, and the mirror of Corollary 7.2 with $n_0 = k \ge x_1 N$. Adding the ranges, letting $N \to \infty$, then $\delta \to 0$, then $\varepsilon \to 0$, and using (H0) for the mass of range 1 and the integrability of $\rho_\pm$ near 0 and of g_x near $z = x$, where $g_x(z) \le C(x-z)^{\alpha-1}$, gives the claim. □

Theorem D needs no condition of the type (LB). Theorem B(b) would bound each near-diagonal term by $CT(N)\omega(k-d)$, and the sum of these bounds over $i \le \delta N$ is of order $T(N)/T(\delta N)$, which is not small. The counting bound replaces the pointwise estimate by the expected length of the level set $\{m : C_m = n_0\}$, which costs only the upper half of the strong renewal theorem at n_0.

7.2 The aged process

Let the walk start fresh with either sign at step 1, and observe it from step $t_a + 1$ on, for an aging time $t_a = t_a(N)$ with $t_a/N \to \theta \in (0, \infty)$. Let A_N now be the number of plus steps among the steps $t_a + 1, \dots, t_a + N$, and $P_N(k) = \mathbb{P}(A_N = k)$. The run in progress at step $t_a + 1$ is the initial run of the observed walk. Let Σ be its sign and D the number of its steps from step $t_a + 1$ on. Given (Σ, D), the later runs are fresh and alternate, so P_N is the law of Proposition 2.2 with an initial run that depends on N. Let u_c be the renewal mass function of the cycle law $f_c = f_+ * f_-$ and $v_c = f_+ * u_c$, so that for a fresh plus start $u_c(s)$ and $v_c(s)$ are the probabilities that a plus run, respectively a minus run,

starts at step $s+1$. A run that starts at step $s+1 \le t_a+1$ with length l straddles step t_a+1 exactly when $l \ge t_a - s + 1$, and then $D = s + l - t_a$. Hence

$$\nu_+(d) = \sum_{s=0}^{t_a} u_c(s)\, f_+(t_a - s + d), \qquad \nu_-(d) = \sum_{s=0}^{t_a} v_c(s)\, f_-(t_a - s + d), \tag{7.2}$$

and $Q_\sigma(N)$ is the same sum with $T_\sigma(t_a - s + N)$ in place of $f_\sigma(t_a - s + d)$. For a fresh minus start the roles of u_c and v_c are exchanged. Put $p_+ = 1/(1+b)$ and $p_- = b/(1+b)$, the limit shares of the two signs, and for $z > 0$

$$\rho_\sigma(z) = p_\sigma \, \frac{\sin \pi\alpha}{\pi} \, \frac{\theta^\alpha z^{-\alpha}}{\theta + z}. \tag{7.3}$$

Lemma 7.3 (the marked Dynkin–Lamperti law)**.** *Assume (BAL). For every $r > 0$, $\mathbb{P}(\Sigma = \sigma,\, D \ge rN) \to \int_r^\infty \rho_\sigma(z)\, dz$, $\mathbb{P}(\Sigma = \sigma) \to p_\sigma$, and $\lim_{\varepsilon \to 0} \limsup_N \mathbb{P}(D < \varepsilon N) = 0$. In particular the aged initial run satisfies (H0) with $\pi^0 = 0$ and (H1) with the densities* (7.3)*.*

Proof. The cycle tail is $T_c(n) = \mathbb{P}(X_+ + X_- \ge n) \sim (1+b) T_+(n)$, since the two summands are independent with regularly varying tails of the same index. The renewal measure $U_c(n) = \sum_{s \le n} u_c(s)$ satisfies $U_c(n) T_c(n) \to 1/(\Gamma(1+\alpha)\Gamma(1-\alpha)) = \sin(\pi\alpha)/(\pi\alpha)$ by Karamata's Tauberian theorem [14, Section XIV.3]. So $U_c(N\,dy)/U_c(N)$ converges vaguely to $d(y^\alpha)$ on $[0,\infty)$, and by (R1), $T_+(t_a - y + rN)/T_+(N) \to (\theta - y/N + r)^{-\alpha}$ uniformly in $y \in [0, t_a]$ after the substitution $y = Ny'$. Hence

$$\mathbb{P}(\Sigma = +,\, D \ge rN) = \sum_{s \le t_a} u_c(s) T_+(t_a - s + rN) \to \frac{\sin \pi\alpha}{\pi\alpha}\, p_+ \int_0^\theta (\theta + r - y)^{-\alpha}\, d(y^\alpha).$$

Differentiating in r and using the identity $\int_0^\theta y^{\alpha-1}(\theta + r - y)^{-\alpha-1} dy = \theta^\alpha/(\alpha r^\alpha(\theta + r))$, which the substitution $v = y/(\theta + r - y)$ turns into $(\theta+r)^{-1} \int_0^{\theta/r} v^{\alpha-1} dv$, gives the density $\rho_+(r)$. The minus sign is the same with v_c, whose partial sums are asymptotic to U_c. Since $\int_0^\infty \theta^\alpha z^{-\alpha}(\theta+z)^{-1} dz = \pi/\sin\pi\alpha$, the densities ρ_σ have total mass p_σ, and $p_+ + p_- = 1$, so no mass escapes to 0 and $\mathbb{P}(\Sigma = \sigma) \to p_\sigma$. □

In the scaling limit the run boundaries form the range of a stable subordinator whose jumps carry the sign $+$ with probability p_+, independently of their lengths. So the sign of the straddling run is asymptotically independent of its length. This is the statement after Eq. (18) of Akimoto, Sera, Yamato and Yano [2], there without proof. For $0 < x < 1$ let

$$L_\theta(x) = \int_0^x \rho_+(z)\, L_{\alpha,b}\Big(\frac{x-z}{1-z}\Big) \frac{dz}{1-z} + \int_0^{1-x} \rho_-(z)\, L_{\alpha,b}\Big(\frac{x}{1-z}\Big) \frac{dz}{1-z}, \tag{7.4}$$

the mixture (7.1) with $\pi^0 = 0$, and put

$$Q_\sigma^\infty(\theta) = \int_1^\infty \rho_\sigma(z)\, dz = p_\sigma \, \frac{\sin\pi\alpha}{\pi}\, B\Big(\frac{\theta}{1+\theta}; \alpha, 1-\alpha\Big), \qquad c_\sigma(\theta) = \int_0^1 \rho_\sigma(z)(1-z)^{-\alpha}\, dz, \tag{7.5}$$

with $B(x; a, b) = \int_0^x u^{a-1}(1-u)^{b-1} du$ the incomplete beta function. The substitutions $z = \theta w$ and $u = 1/(1+w)$ give the closed form of Q_σ^∞, and Euler's integral gives $c_\sigma(\theta) = p_\sigma \frac{\sin\pi\alpha}{\pi}\, \theta^{\alpha-1} B(1-\alpha, 1-\alpha)\, {}_2F_1(1, 1-\alpha; 2-2\alpha; -1/\theta)$. At $\alpha = 1/2$, $\theta = 1$ and $b = 1$ this is $c_+(1) = 1/(2\sqrt{2})$. We also need a condition on the cycle renewal sequences,

$$u_c(n) \le C\, \omega_+(n) \quad \text{and} \quad v_c(n) \le C\, \omega_+(n) \qquad (n \ge 1), \tag{U}$$

with the roles of u_c and v_c exchanged for a fresh minus start at step 1.

Lemma 7.4 (when (U) holds)**.** *Assume (BAL) and (SPAN). Then (U) holds under (LB$_\pm$), for every α. It holds under no further condition when $\alpha > 1/2$. For $\alpha \le 1/2$ it holds when f_c and f_+ satisfy the strong renewal theorem.*

Proof. Under (SPAN) for the plus sign the support of $X_+ + X_-$ lies in no progression $c + h\mathbb{Z}$ with $h \ge 2$, so f_c is aperiodic. Under (LB$_\pm$), splitting the convolution at $n/2$ gives $f_c(n) \le CT_c(n)/n$, so Doney's theorem (Lemma 3.1(iii)) gives the strong renewal theorem for f_c, that is $u_c \le C\omega_+$ since $T_c \asymp T_+$. For $\alpha > 1/2$ the same follows from Lemma 3.1(iii) without (LB), and for $\alpha \le 1/2$ it is the assumption on f_c. For $v_c = f_+ * u_c$ split at $n/2$. The part $\sum_{i \le n/2} f_+(i)u_c(n-i)$ is at most $C\omega_+(n/2) \le C\omega_+(n)$. The part $\sum_{i>n/2} f_+(i)u_c(n-i) = f_+(n) + \sum_{1\le z<n/2} f_+(n-z)u_c(z)$ is at most $f_+(n) + C\sum_{1\le z<n/2} f_+(n-z)/(zT_+(z)^2)$, since $\omega_+(z) \le 1/(zT_+(z)^2)$. Under (LB$_+$) this is at most $CT_+(n)/n + C(T_+(n)/n)U_c(n/2) \le C/n \le C\omega_+(n)$. Otherwise the sum is the quantity of Lemma 3.1(iii) for the plus law, which is $O(1/(nT_+(n)))$ when f_+ satisfies the strong renewal theorem, and $f_+(n) \le T_+(n) \le \omega_+(n)$ for large n. □

We split the aged initial run by the start of the straddling run. Write $\nu_\sigma = \nu_\sigma^{\mathrm{y}} + \nu_\sigma^{\mathrm{o}}$, where ν_σ^{y} collects the terms of (7.2) with $s > t_a/2$, the young runs, and ν_σ^{o} those with $s \le t_a/2$, the old runs.

Lemma 7.5 (pointwise bounds)**.** *Assume (BAL) and (SPAN), and fix $\varepsilon \in (0,1)$.*

(a) Under (U), $N\nu_\pm^{\mathrm{y}}(d) \le C_{\varepsilon,\theta}$ for all $d \ge \varepsilon N$ and large N.

(b) Under (LB$_\pm$), $N\nu_\pm(d) \le C_{\varepsilon,\theta}$ for all $d \ge \varepsilon N$ and large N.

Proof. (a) For $s > t_a/2$, (U) gives $u_c(s) \le C\omega_+(t_a/2) \le C_\theta/(NT_+(N))$, and $\sum_s f_+(t_a - s + d) \le T_+(d) \le C_\varepsilon T_+(N)$ by (R2). The minus sign is the same with v_c. (b) For $s \le t_a/2$ the argument of f_+ in (7.2) is at least $t_a/2 + d \ge cN$, so (LB$_+$) gives $f_+ \le CT_+(N)/N$, and $\sum_{s\le t_a/2} u_c(s) = U_c(t_a/2) \le C/T_+(N)$ by the proof of Lemma 7.3. So the old part is at most C/N, and the young part is covered by (a), since (LB$_\pm$) implies (U) by Lemma 7.4. □

Theorem E (the aged law)**.** *Let the walk start fresh with either sign at step 1, and let $t_a/N \to \theta \in (0,\infty)$. Assume (BAL) and (SPAN).*

(a) (The bulk.) Assume one of the following: (LB$_\pm$); $\alpha > 1/2$; $\alpha < 1/2$, (SRT$_\pm$) and (U); $\alpha = 1/2$, (SRT$_\pm$), (U), and $nT_+(n)^2$ bounded above and away from zero. Then $NP_N(\lfloor xN \rfloor) \to L_\theta(x)$ uniformly for x in compact subsets of $(0,1)$.

(b) (The atoms.) Under (BAL) alone, $P_N(N) = Q_+(N) \to Q_+^\infty(\theta)$ and $P_N(0) \to Q_-^\infty(\theta)$.

(c) (The end layers.) Under (U), for every fixed $j \ge 1$,

$$P_N(N-j) = T_+(N)\big[c_+(\theta)\,u_-(j) + o(1)\big], \qquad P_N(j) = T_-(N)\big[c_-(\theta)\,u_+(j) + o(1)\big].$$

The atoms are of order 1, against $T_+(N)$ and 0 for a fresh start, and $\int_0^1 L_\theta + Q_+^\infty(\theta) + Q_-^\infty(\theta) = 1$, since the integral of $\rho_+ + \rho_-$ over $(0,\infty)$ is 1. The density L_θ is the aging generalized arcsine law of Akimoto, Sera, Yamato and Yano [2, Theorem 2], who obtain the convergence in distribution by a scaling argument. Part (a) is its local form, and the mixture of Theorem D with Lemma 7.3 gives the convergence in distribution under (BAL) alone, by continuous mapping from Lamperti's theorem for the remainder walk. The end layers keep the fresh shape $u_-(j)$ and lose the factor $c_+(\theta) < p_+ < 1$. As $\theta \to 0$ the densities $\rho_\sigma\,dz$ converge to $p_\sigma\delta_0$, so $L_\theta \to L_{\alpha,b}$ uniformly on compacts and $Q_\sigma^\infty(\theta) \to 0$, while $c_+(\theta) \to p_+$, since $\int_0^1 z^{-\alpha}(1-z)^{-\alpha}(\theta+z)^{-1}dz \sim \theta^{-\alpha}\pi/\sin\pi\alpha$. The fresh layer is

$u_-(j)$ for either starting sign, so the limits $\theta \to 0$ and $N \to \infty$ do not commute at the ends. For small θ the straddling run still has length of order $\theta N \gg j$, and only a plus straddler can feed the plus layer. For Brownian motion this is the remark after Eq. (15) of [2], with $c(r) \to \frac{1}{2}$, which is the case $b = 1$. As $\theta \to \infty$, $Q_\sigma^\infty(\theta) \to p_\sigma$ and $L_\theta = O(\theta^{\alpha-1})$.

Proof of Theorem E(b). This is Lemma 7.3 at $r = 1$, since $P_N(N) = Q_+(N) = \mathbb{P}(\Sigma = +, D \geq N)$ by Proposition 2.2. □

Proof of Theorem E(a). Under ($\mathrm{LB}_\pm$), Lemma 7.3 gives (H0) with $\pi^0 = 0$ and (H1), Lemma 7.5(b) gives (H2), and ($\mathrm{LB}_\pm$) gives ($\mathrm{SRT}_\pm$). So Theorem D applies and gives L_θ. In the other three cases ($\mathrm{SRT}_\pm$) holds, by Lemma 3.1(iii) when $\alpha > 1/2$ and by assumption otherwise, and (U) holds by Lemma 7.4 or by assumption. Ranges 1 and 2 of the proof of Theorem D used only (H0) and (H1). In range 3 we split $\nu_+ = \nu_+^{\mathrm{y}} + \nu_+^{\mathrm{o}}$. The young part satisfies (H2) by Lemma 7.5(a), and Corollary 7.2 bounds its contribution by $C_\varepsilon \zeta(\delta)/N$ as before. It remains to bound the old part

$$E_{\mathrm{o}} = \sum_{i \leq \delta N} \nu_+^{\mathrm{o}}(k-i)\, F_{j+i}^-(i),$$

with $k/N \in [x_1, x_2]$, $j = N - k$ and $\delta \leq 1 - x_2$, and its mirror. Lemma 5.1 applies to $F_{j+i}^-(i)$, because $j + i \geq (1 - x_2)N$ and $i \leq \delta N \leq (j + i)/2$, and gives $F_{j+i}^-(i) \leq C T_+(N) \omega_+(i)$. Write $T = T_+$ and $\omega = \omega_+$. With $M = t_a + k$ and $y = i + s$, (7.2) gives

$$E_{\mathrm{o}} \leq C T(N) \sum_{y \geq 0} f_+(M - y)\, W(y), \qquad W(y) = \sum_{\substack{i+s=y \\ i \leq \delta N,\ s \leq t_a/2}} \omega(i)\, u_c(s),$$

and on the range of the sum $M - y \geq t_a/2 + k - \delta N \geq cN$. By (U), $u_c \leq C\omega$ with $\omega(0) = 1$, so $W(y) \leq C(\omega * \omega)(y) \leq C/(yT(y)^2)$ for $y \geq 1$, by splitting the convolution at $y/2$ and using $\sum_{i \leq Y} \omega(i) \leq C/T(Y)$ from (R3). For $y > 2\delta N$ every term has $s \geq y/2$, so $W(y) \leq C\omega(y) \sum_{i \leq \delta N} \omega(i) \leq C\omega(y)/T(\delta N)$.

The range $y \leq 2\delta N$. If $\alpha > 1/2$, the function $1/(yT(y)^2)$ is regularly varying with the positive index $2\alpha - 1$, so $\sup_{y \leq 2\delta N} W(y) \leq C/(\delta N T(\delta N)^2)$, and since $\sum_y f_+(M - y) \leq 1$ this range contributes at most $CT(N)^2/(\delta N T(\delta N)^2) \leq C\delta^{2\alpha-1-\epsilon'}/N$ by Potter's bounds, for any $\epsilon' > 0$. If $\alpha \leq 1/2$, this range contributes at most $CT(N)[f_+(M) + \sum_{1 \leq y \leq 2\delta N} f_+(M - y)/(yT(y)^2)]$. By Lemma 3.1(iii), ($\mathrm{SRT}_+$) says that $MT(M) \sum_{z \leq \delta' M} f_+(M - z)/(zT(z)^2) \leq \zeta_{\mathrm{CD}}(\delta') + o(1)$ with $\zeta_{\mathrm{CD}}(\delta') \to 0$ as $\delta' \to 0$, uniformly over M in a range of order N, and its term $z = 1$ gives $f_+(M) = o(1/(MT(M)))$. Since $2\delta N \leq C\delta M$ and $MT(M) \asymp NT(N)$, this range contributes at most $C(\zeta_{\mathrm{CD}}(C\delta) + o(1))/N$.

The range $2\delta N < y \leq \eta N$, with $\eta = \delta^{1-\alpha}$. Using $W(y) \leq C\omega(\delta N)/T(\delta N)$ there, and $\sum_{2\delta N < y \leq \eta N} f_+(M - y) = T(M - \eta N) - T(M - 2\delta N) \leq T(N)(C\eta + o(1))$ by (R1) at the scale N, this range contributes at most $C(\eta + o(1))T(N)^2/(\delta N T(\delta N)^2) \leq C(\delta^{\alpha-\epsilon'} + o(1))/N$.

The range $y > \eta N$. On a dyadic block $(Y, 2Y]$ with $Y \geq \eta N/2$ we have $W(y) \leq C\omega(Y)/T(\delta N)$, and $\sum_{Y < y \leq 2Y} f_+(M - y) = T(M - 2Y) - T(M - Y) \leq T(N)(CY/N + o(1))$ by (R1). Summing over the blocks, with $\sum_Y Y\omega(Y) = \sum_Y 1/T(Y) \leq C/T(N)$ and $\sum_{Y \geq \eta N/2} \omega(Y) \leq C\omega(\eta N) \log(1/\eta)$, this range contributes at most $C\frac{T(N)^2}{T(\delta N)}\Big[\frac{1}{NT(N)} + o(1)\,\omega(\eta N) \log\frac{1}{\eta}\Big] \leq C\frac{\delta^{\alpha-\epsilon'}}{N} + o\Big(\frac{1}{N}\Big)$.

Hence $NE_{\mathrm{o}} \leq C\delta^{\min(\alpha, 2\alpha-1)-\epsilon'} + o(1)$ for $\alpha > 1/2$ and $NE_{\mathrm{o}} \leq C\zeta_{\mathrm{CD}}(C\delta) + C\delta^{\alpha-\epsilon'} + o(1)$ for $\alpha \leq 1/2$, which tends to 0 as $N \to \infty$ and then $\delta \to 0$. The minus side is the same with v_c, f_-, the condition of Lemma 3.1(iii) for f_-, and the mirror of Lemma 5.1. For $\alpha > 1/2$ every input is automatic, for $\alpha < 1/2$ the inputs are ($\mathrm{SRT}_\pm$) and (U), and at $\alpha = 1/2$ Lemma 5.1 adds the condition on $nT_+(n)^2$. □

Proof of Theorem E(c). Fix $j \geq 1$. Proposition 2.2 at $k = N - j$ gives

$$P_N(N-j) = \sum_{m=1}^{N-1} \nu_+(N-m)\,\Pi_m^-(j) + \sum_{d=1}^{j} \nu_-(d)\,F_{N-d}^+(N-j).$$

In the second sum $F_{N-d}^+(N-j) \leq C_j T_+(N)$ by Theorem B(a), and $\nu_-(d) \leq \sup_{s>t_a/2} v_c(s) + T_-(t_a/2) \to 0$, since $v_c(s) \to 0$ by the renewal theorem for the aperiodic cycle law with infinite mean. So it is $o(T_+(N))$. Fix $\eta \in (0,1)$. For $m > \eta N$, Theorem B(a) for a fresh minus walk gives $\Pi_m^-(j) = T_+(m)[u_-(j) + o(1)]$ uniformly, and Lemma 7.3 with the bounded continuous integrand $(1-z)^{-\alpha}$ on $[0, 1-\eta]$ gives $\sum_{d<(1-\eta)N} \nu_+(d) T_+(N-d)/T_+(N) \to \int_0^{1-\eta} \rho_+(z)(1-z)^{-\alpha} dz$. So the terms with $m > \eta N$ give $T_+(N) u_-(j) \int_0^{1-\eta} \rho_+(z)(1-z)^{-\alpha} dz + o(T_+(N))$. For $m \leq \eta N$ we use $\Pi_m^-(j) \leq C_j T_+(m)$. The young part contributes at most $(C/N) \sum_{m \leq \eta N} T_+(m) \leq C\eta^{1-\alpha-\epsilon'} T_+(N)$ by Lemma 7.5(a) and (R3). The old part is at most $C_j \sum_y f_+(M'-y) W'(y)$ with $M' = t_a + N$ and $W'(y) = \sum_{m+s=y,\ m \leq \eta N,\ s \leq t_a/2} T_+(m) u_c(s)$. By (U), splitting at $y/2$, $W' \leq C$ everywhere, and $W'(y) \leq C\omega_+(y)\,\eta N T_+(\eta N)$ for $y > 2\eta N$. The range $y \leq 2\eta N$ costs at most $C T_+(N)(\eta + o(1))$, a drop of T_+ over a window of length $2\eta N$ at the scale N, and the dyadic blocks above $2\eta N$ cost at most $C\eta N T_+(\eta N) T_+(N)[1/(N T_+(N)) + o(1)\omega_+(\eta N)] \leq C\eta^{1-\alpha-\epsilon'} T_+(N) + o(T_+(N))$, as in the proof of (a). Let $N \to \infty$ and then $\eta \to 0$. The minus end is the mirror. □

A uniform matching of the aged layer to the bulk, the analogue of Theorem B(b), is not proved. As $x \to 1$ the plus integral of L_θ behaves like $\frac{\sin \pi\alpha}{\pi b} c_+(\theta)(1-x)^{\alpha-1}$, which under (SRT$_-$) and pure power tails is what $T_+(N) c_+(\theta) u_-(j)$ gives for $1 \ll j \ll N$, so the two pieces are consistent. Section 11 reports the checks of all three parts against the exact aged law.

8 One tail heavier: the dominated regime

Throughout this section (DOM) holds. The plus runs have the heavier tail, $T_- = o(T_+)$, with $\alpha_+ \in (0,1)$, and the minus runs have an index $\alpha_- \in (0,1) \cup (1,2)$, so their mean $\mu_- = \mathbb{E}X_-$ is finite exactly when $\alpha_- > 1$. We assume (RV) and (SPAN) for both signs. Then $A_N/N \to 1$ in probability, and the object is the time $N - A_N$ spent on the lighter side. The plus side alone sets the number of cycles. By time N about $E_+/T_+(N)$ plus runs are complete, with E_+ the Mittag-Leffler variable of Section 3.3, and the minus time is the sum of that many minus runs. So the scale and the limit are

$$\begin{aligned} \mu_- < \infty: &\qquad D_N = \frac{\mu_-}{T_+(N)}, \qquad \rho = \phi_+, \qquad \omega(j) = 1, \\ \alpha_- < 1: &\qquad D_N = a_-\big(1/T_+(N)\big), \qquad \rho = \text{the density of } E_+^{1/\alpha_-} Z_-, \qquad \omega = \omega_-, \end{aligned}$$

where Z_- is the positive stable variable with Lévy tail $z^{-\alpha_-}$, independent of E_+, so that $T_-(D_N) \sim T_+(N)$ in the second case. In both cases D_N is nondecreasing, regularly varying in N with a positive index below 1, and $D_N = o(N)$, so $T_+(N-j) \sim T_+(N)$ uniformly in $j \leq K D_N$. We write $\phi = \phi_+$, and we put

$$\Psi(y) = \rho(y) \quad (\mu_- < \infty), \qquad \Psi(y) = \rho(y) \Big/ \Big(\frac{\sin \pi\alpha_-}{\pi}\, y^{\alpha_- - 1} \Big) \quad (\alpha_- < 1),$$

so that $\Psi(0+) = 1$ in both cases. For $\mu_- < \infty$ this is $\phi(0) = 1$. For $\alpha_- < 1$, the substitution $x = (y/v)^{\alpha_-}$ in $\rho(y) = \int_0^\infty \phi(x) x^{-1/\alpha_-} g_-(y x^{-1/\alpha_-})\, dx$ gives $\rho(y) = \alpha_- y^{\alpha_- - 1} \int_0^\infty \phi((y/v)^{\alpha_-}) g_-(v) v^{-\alpha_-} dv$, and as $y \to 0$ dominated convergence gives $\rho(y) \sim \alpha_- y^{\alpha_- - 1} \mathbb{E} Z_-^{-\alpha_-} = \frac{\sin \pi\alpha_-}{\pi} y^{\alpha_- - 1}$. When $\alpha_+ = \alpha_- = \alpha$ and $b = 0$ the limit is

the ratio Z_-/Z_+ of two independent copies of the same stable law, with the explicit density

$$\rho(y) = \frac{\sin \pi\alpha}{\pi} \, \frac{y^{\alpha-1}}{1 + 2y^{\alpha} \cos \pi\alpha + y^{2\alpha}}, \tag{8.1}$$

which is also the limit of $b^{-1/\alpha} L_{\alpha,b}(1 - b^{1/\alpha} y)$ as $b \to 0$. So the balanced and the dominated regime join continuously at $b = 0$. We also use the condition

$$D_n T_+(n) \sum_{m \le \delta / T_+(n)} u_m^+(n) \to 0 \qquad (n \to \infty) \quad \text{for some } \delta > 0. \tag{W_+}$$

For $\delta < \delta'$ the extra terms number at most $\delta'/T_+(n)$ and each is at most $C/a_+(\delta/T_+(n)) \le C_\delta/n$ by Lemma 3.1(i), so the difference is $O(D_n/n) \to 0$, and (W_+) for one δ is (W_+) for every δ.

Theorem F (the dominated regime)**.** *Assume (DOM) and (SPAN), and let the start be fresh with either sign.*

(a) (The global law.) $(N - A_N)/D_N$ *converges in law to the law with density* ρ*.*

(b) (The lower half.) For $0 < y_1 < y_2 < \infty$, $\liminf_N \inf_{y_1 D_N \le j \le y_2 D_N} \big(D_N P_N(N - j) - \rho(j/D_N) \big) \ge 0$.

(c) (The local law.) If $\mu_- < \infty$ *or* $\alpha_- > 1/2$*, then*

$$\sup_{y_1 D_N \le j \le y_2 D_N} \big| D_N P_N(N - j) - \rho(j/D_N) \big| \to 0.$$

If $\alpha_- \le 1/2$*, this holds if and only if (SRT$_-$) and (W$_+$) hold. (W$_+$) holds under (SRT$_+$), and whenever* $\alpha_+ \ge 1/2$ *or* $\alpha_- > 1/2$*.*

(d) (The end layers.) For each fixed $j \ge 0$, $P_N(N - j)/T_+(N) \to \kappa_+^{(\sigma_0)}(j)$ *of* (2.7)*, under (RV) for the plus sign alone, and for each fixed* $i \ge 0$, $P_N(i)/T_-(N) \to \kappa_-^{(\sigma_0)}(i)$ *under (RV) for the minus sign alone. Under the condition of (c), and if moreover* $\alpha_- \ne 1/2$*, uniformly over* $0 \le j \le y_2 D_N$*,*

$$P_N(N - j) = T_+(N) \big[\kappa_+^{(\sigma_0)}(j) \, \Psi(j/D_N) + o(\omega(j)) \big].$$

At $\alpha_- = 1/2$ *the same holds under (SRT$_+$) with* $\alpha_+ < 1/2$*.*

Part (a) is known in distribution. For $\mu_- < \infty$ it is a Darling–Kac law [10] for the time spent in a part of a null recurrent system that is left at a positive rate, and for the lattice walk it is stated without proof by Cénac, Le Ny, de Loynes and Offret [9, Section 4.3]. Yamato [28] proves limits of this type for processes time-changed by inverse subordinators. Lamperti's theorem [20] covers unequal tails but gives only $A_N/N \to 1$ there. We include (a) because it follows from (b) in two lines. Parts (b) to (d) are local, and we did not find them in the literature. The local law needs two inputs only: the local limit theorem for the lighter sums and the local form of the inverse renewal count of the heavier runs, which is Lemma 3.8. The strong renewal theorem enters through the lighter law alone, and only when its mean is infinite with index at most $1/2$. In that case the heavier law must also satisfy (W_+). We do not know whether (W_+) can fail when $\alpha_+ < 1/2$ and (SRT_+) fails.

Throughout the proofs we write $n = N - j$ and, by Proposition 2.1 for a fresh plus start,

$$P_N(N-j) = \mathrm{I}_N(j) + \mathrm{R}_N(j), \qquad \mathrm{I}_N(j) = \sum_{m \ge 1} u_m^-(j) \, q_{m+1}^+(n), \qquad \mathrm{R}_N(j) = \sum_{m \ge 1} u_m^+(n) \, q_m^-(j).$$

For a fresh minus start the kernels are q_m^+ and q_{m+1}^-, which changes no bound and no limit below. We first record where the mass of $u_-(j)$ sits in m.

Lemma 8.1 (the index of the minus runs)**.** *Assume (RV) and (SPAN) for the minus sign.*

(i) If $\mu_- < \infty$, then $\lim_{A\to\infty} \limsup_{j\to\infty} \sum_{m:\ |m-j/\mu_-|>Aa_-(j)} u_m^-(j) = 0$, where a_- is the scale of the minus sums.

(ii) If $\alpha_- < 1$, then $\sum_{m>K/T_-(j)} u_m^-(j) \le CK^{1-1/\alpha_-}/(jT_-(j))$ for all $K \ge 1$ and $j \ge 1$.

Proof. (i) By the local limit theorem for the minus sums [23, Lemma A.1], $u_m^-(j) = a_-(m)^{-1}[g_-((j-m\mu_-)/a_-(m))+o(1)]$ uniformly in j, so the sum over $|m-j/\mu_-| \le Aa_-(j)$ is a Riemann sum that converges to $\mu_-^{-1}\int_{|z|\le A/\mu_-} g_-(z)\,dz$, while $u_-(j) \to 1/\mu_-$ by the renewal theorem. (ii) By Lemma 3.1(i), $u_m^-(j) \le C/a_-(m)$, and since $1/a_-(m)$ is regularly varying with index $-1/\alpha_- < -1$, $\sum_{m>M} 1/a_-(m) \le CM/a_-(M)$. With $M = K/T_-(j)$ we have $a_-(M) \ge cK^{1/\alpha_-}j$ by (R2) for a_-, which gives the bound. □

8.1 The bulk: proof of parts (a), (b) and (c)

Let $j = yD_N$ with $y \in [y_1, y_2]$, so that $T_+(n) \sim T_+(N)$ uniformly.

The main term, $\mu_- < \infty$. On the window $|m - j/\mu_-| \le Aa_-(j)$ we have $mT_+(n) = y + O(Aa_-(j)/D_N) + o(1) = y + o(1)$, since $a_-(j)/j \to 0$ when $\alpha_- > 1$. So Lemma 3.8(ii) and the uniform continuity of ϕ give $q_{m+1}^+(n) = T_+(N)[\phi(y) + o(1)]$ on the window, and by Lemma 8.1(i) the weights $u_m^-(j)$ on the window add up to $1/\mu_- + o_A(1)$. Off the window Lemma 3.5 gives $q_{m+1}^+(n) \le CT_+(N)$, and the weights add up to $o_A(1)$. Hence $\mathrm{I}_N(j) = (T_+(N)/\mu_-)(\phi(y)+o(1)) = (\phi(y)+o(1))/D_N$, with no condition beyond (RV) and (SPAN).

The main term, $\alpha_- < 1$. Now $T_-(j) \sim y^{-\alpha_-}T_+(N)$. Split the sum over m at $\varepsilon/T_+(N)$ and $K/T_+(N)$. On the middle range, Lemma 3.1(i) gives $u_m^-(j) = a_-(m)^{-1}[g_-(j/a_-(m)) + o(1)]$ and Lemma 3.8(ii) gives $q_{m+1}^+(n) = T_+(N)[\phi(mT_+(N)) + o(1)]$, both uniformly. Since $a_-(x/T_+(N)) \sim x^{1/\alpha_-}D_N$ uniformly for $x \in [\varepsilon, K]$, the sum is a Riemann sum with mesh $T_+(N)$ in $x = mT_+(N)$, and

$$T_+(N)\sum_m \frac{1}{a_-(m)}\, g_-\Big(\frac{j}{a_-(m)}\Big)\phi\big(mT_+(N)\big) \to \frac{1}{D_N}\int_\varepsilon^K \phi(x)\, x^{-1/\alpha_-} g_-\big(yx^{-1/\alpha_-}\big)\,dx,$$

uniformly in y, and the integral tends to $\rho(y)$ as $\varepsilon \to 0$ and $K \to \infty$. On the large range Lemma 3.5 and Lemma 8.1(ii) give at most $CT_+(N)K^{1-1/\alpha_-}/(jT_-(j)) \le CK^{1-1/\alpha_-}/D_N$, since $jT_-(j) \sim y^{1-\alpha_-}D_N T_+(N)$. On the small range Lemma 3.5 gives at most

$$CT_+(N) \sum_{m<\varepsilon'/T_-(j)} u_m^-(j) = \frac{CT_+(N)}{jT_-(j)}\, I_-(\varepsilon', j), \tag{8.2}$$

with I_- as in (3.1) and $\varepsilon' = C\varepsilon$. The prefactor is of order $1/D_N$, and by Lemma 3.1(iv) the factor $I_-(\varepsilon', j)$ tends to 0 as $j \to \infty$ and then $\varepsilon' \to 0$ if and only if (SRT$_-$) holds.

The second term, $\alpha_- < 1$. By Lemma 3.5 for the minus sign, $q_m^-(j) \le CT_-(j) \le CT_+(N)$. For $m > \delta/T_+(n)$, Lemma 3.1(i) gives $u_m^+(n) \le C/a_+(\delta/T_+(n)) \le C_\delta/N$, and $\sum_m q_m^-(j) = 1$, so this part is $O(1/N) = o(1/D_N)$. The part $m \le \delta/T_+(n)$ is at most $CT_+(N)\sum_{m\le\delta/T_+(n)} u_m^+(n)$, which is $o(1/D_N)$ exactly under (W$_+$), since $D_n \sim D_N$. Under (SRT$_+$), $\sum_m u_m^+(n) = u_+(n) \le C/(nT_+(n))$, so the left side of (W$_+$) is at most $CD_n/n \to 0$. In general Lemma 3.1(ii) gives $\sum_{m\le M} u_m^+(n) \le CT_+(n)\sum_{m\le M} m/a_+(m)$ with $M = \delta/T_+(n)$. The sequence $m/a_+(m)$ is regularly varying with index $1 - 1/\alpha_+$. If $\alpha_+ < 1/2$ the series converges and the sum is at most $CT_+(n)$, so the left side of (W$_+$) is at most $CD_nT_+(n)^2 \sim CD_nT_-(D_n)^2$, which tends to 0 when $\alpha_- > 1/2$, because $xT_-(x)^2$ is regularly varying with index $1 - 2\alpha_- < 0$. If $\alpha_+ \ge 1/2$ the partial sums are at most $CM^2\lambda(M)/a_+(M)$ with λ slowly varying ($\lambda = 1$ for $\alpha_+ > 1/2$, by Karamata's

theorem [7, Proposition 1.5.8 and 1.5.9a]), and $a_+(\delta/T_+(n)) \geq c_\delta n$, so the sum is at most $C_\delta \lambda(1/T_+(n))/(nT_+(n))$ and the left side of (W_+) is at most $C_\delta \lambda(1/T_+(n))D_n/n \to 0$, since D_n/n is regularly varying with a negative index. This proves the last sentence of (c).

The second term, $\mu_- < \infty$. Here $q^-_m(j)$ is not $O(T_-(j))$ uniformly in m, and we count directly. Split at $m = \delta j$ with $\delta\mu_- \leq 1/4$. For $m > \delta j$, Lemma 3.1(i) gives $u^+_m(n) \leq C/a_+(\delta j)$, and $a_+(\delta j) \geq cN$ because j is of order $1/T_+(N)$, so with $\sum_m q^-_m(j) = 1$ this part is $O(1/N) = o(T_+(N))$. For $m \leq \delta j$ write $q^-_m(j) = \sum_{r<j} u^-_{m-1}(r)T_-(j-r)$. The part $r < j/2$ is at most $T_-(j/2) \leq CT_-(j)$. For $r \geq j/2$ we have $r - (m-1)\mu_- \geq j/4$, so the local large deviation bound for the minus sums [23, Lemma A.1(iii)] gives $u^-_{m-1}(r) \leq C(m/a_-(m))T_-(j)$, and $\sum_r T_-(j-r) \leq \mu_-$. Hence $q^-_m(j) \leq CT_-(j)(1 + m/a_-(m))$ for $m \leq \delta j$. With $u^+_m(n) \leq C(m/a_+(m))T_+(n)$ from Lemma 3.1(ii), the part $m \leq \delta j$ of $\mathrm{R}_N(j)$ is at most

$$CT_+(N)\,T_-(j) \sum_{m\leq \delta j} \frac{m}{a_+(m)}\Big(1 + \frac{m}{a_-(m)}\Big) \\ \leq CT_+(N)\,T_-(j)\,\lambda(j)\Big[\max\Big(1, \frac{j^2}{a_+(j)}\Big) + \max\Big(1, \frac{j^3}{a_+(j)a_-(j)}\Big)\Big]$$

with λ slowly varying. The two products with $T_-(j)$ are regularly varying in j with indices $-\alpha_- + \max(0, 2 - 1/\alpha_+)$ and $-\alpha_- + \max(0, 3 - 1/\alpha_+ - 1/\alpha_-)$, and both are negative, since $2 - 1/\alpha_+ < 1 < \alpha_-$ and $3 - 1/\alpha_+ - 1/\alpha_- - \alpha_- < 2 - \alpha_- - 1/\alpha_- \leq 0$. So $\mathrm{R}_N(j) = o(T_+(N)) = o(1/D_N)$ as $j \to \infty$, with no condition on j/D_N. In the mixed region the bulk sits at the height $T_+(N)$ of the end layer, and a minus run in progress at time N costs a polynomial factor against it.

Part (b). Every term is nonnegative. Dropping the small range of I_N and all of R_N leaves the middle and large ranges, which give $D_N P_N(N-j) \geq \rho(y) - \eta(\varepsilon, K) + o(1)$ uniformly, with $\eta(\varepsilon, K) \to 0$ as $\varepsilon \to 0$ and $K \to \infty$.

Part (c), sufficiency. If $\mu_- < \infty$ the two terms give the claim with no condition. If $\alpha_- < 1$, under (SRT$_-$) and (W_+) the small range of (8.2) and the second term are $o(1/D_N)$, and the rest converges. If $\alpha_- > 1/2$ both conditions are automatic, by Lemma 3.1(iii) and the last paragraph on (W_+).

Part (c), necessity for $\alpha_- \leq 1/2$. Suppose (SRT$_-$) fails. Since $I_-(\varepsilon', j)$ is nondecreasing in ε', Lemma 3.1(iv) gives $c > 0$ such that for every $\varepsilon' > 0$, $I_-(\varepsilon', j) \geq c$ along a sequence of $j \to \infty$ that may depend on ε'. Fix $y \in (y_1, y_2)$ and a small $\eta > 0$. For each such j choose N with $j/D_N \in [y - \eta, y + \eta]$, which is possible because D_N is nondecreasing and $D_{N+1}/D_N \to 1$. By Lemma 3.8(i), $q^+_{m+1}(n) \geq \frac{1}{2}T_+(N)$ for $mT_+(n) \leq \delta_0$, which covers $m < \varepsilon'/T_-(j)$ once ε' is small. So the small range of $\mathrm{I}_N(j)$ is at least $\frac{1}{2}T_+(N)\sum_{m<\varepsilon'/T_-(j)} u^-_m(j) \geq c'/D_N$, with $c' > 0$ independent of ε', while the middle range is at least $(\rho(y) - \eta(\varepsilon', K) - o(1))/D_N$. Choosing ε' and K so that $\eta(\varepsilon', K) < c'/4$, and η so that $|\rho(j/D_N) - \rho(y)| < c'/4$, gives $D_N P_N(N-j) - \rho(j/D_N) \geq c'/4$ along the sequence, so the local law fails. Suppose instead that (W_+) fails, so that $D_n T_+(n) \sum_{m\leq \delta/T_+(n)} u^+_m(n) \geq c$ along a sequence of n, for a δ so small that $mT_-(j) \leq \delta_0$ on that range for $j \leq y_2 D_N$. Lemma 3.8(i) for the minus sign gives $q^-_m(j) \geq \frac{1}{2}T_-(j)$ there, so $\mathrm{R}_N(j) \geq \frac{1}{2}T_-(j)\sum_{m\leq\delta/T_+(n)} u^+_m(n) \geq c''/D_N$ along the sequence, for j with j/D_{n+j} near y and $N = n + j$, where $D_{n+j} \sim D_n$ because $j = o(n)$. Together with (b) this gives $D_N P_N(N-j) \geq \rho(y) + c'' - o(1)$, and the local law fails again.

Part (a). By (b) and a Riemann sum, $\liminf_N \mathbb{P}((N - A_N)/D_N \in [y_1, y_2]) \geq \int_{y_1}^{y_2} \rho$ for all $0 < y_1 < y_2$. Since ρ is a probability density on $(0, \infty)$, the mass of $(N - A_N)/D_N$ outside $[y_1, y_2]$ is eventually at most $1 - \int_{y_1}^{y_2} \rho + o(1)$, which is small for y_1 small and y_2 large. Lower bounds on every compact interval together with tightness give convergence in law. □

8.2 The ends: proof of part (d)

Fixed j. In $\mathrm{I}_N(j)$ only $m \le j$ occur, and for fixed m, $q^+_{m+1}(n) = \mathbb{P}(S^+_{m+1} \ge n) - \mathbb{P}(S^+_m \ge n) \sim T_+(n)$, by Lemma 3.3 or by the subexponentiality of a regularly varying tail. So $\mathrm{I}_N(j) \sim T_+(N) \sum_{m \le j} u^-_m(j) = T_+(N) u_-(j)$. In $\mathrm{R}_N(j)$ only $m \le j+1$ occur, and for fixed m, $u^+_m(n) = o(T_+(n))$, because $f_+(x) = T_+(x) - T_+(x+1) = o(T_+(x))$ under (RV), and for $m \ge 2$ one of the summands is at least n/m. This uses (RV) for the plus sign only. The minus end is the mirror, with $q^-_m(N-i) \sim T_-(N)$ and $u^-_m(N-i) = o(T_-(N))$ for fixed m, and the start corrections are those of (2.7).

The range $1 \ll j \ll D_N$. We show that $P_N(N-j) = T_+(N) u_-(j)(1+o(1))$ there, under the condition of (c), $\alpha_- \ne 1/2$, and $u_- \le C\omega$, which holds by (SRT$_-$) when $\alpha_- < 1$ and trivially when $\mu_- < \infty$. For the main term, every m that carries the mass of $u_-(j)$ has $mT_+(N) \to 0$: if $\mu_- < \infty$ the mass sits at $m \approx j/\mu_- \ll 1/T_+(N)$ by Lemma 8.1(i), and if $\alpha_- < 1$ the mass beyond $m = K/T_-(j)$ is at most $CK^{1-1/\alpha_-} u_-(j)$ by Lemma 8.1(ii) and the lower half of (SRT$_-$), while $m \le K/T_-(j)$ gives $mT_+(N) \le KT_+(N)/T_-(j) \to 0$ because $T_-(j) \gg T_-(D_N) \sim T_+(N)$. Lemma 3.8(ii) at $t \to 0$ gives $q^+_{m+1}(n) = T_+(N)(1+o(1))$ on that range, and Lemma 3.5 bounds the rest, so $\mathrm{I}_N(j) = T_+(N) u_-(j)(1 + O(K^{1-1/\alpha_-}) + o(1))$, and K is arbitrary.

For the second term with $\alpha_- < 1$, Lemma 3.6 gives $q^-_m(j) \le CT_-(j) e^{-cmT_-(j)}$. For $m > \delta/T_+(n)$ we have $u^+_m(n) \le C_\delta/N$, so this part is at most C_δ/N, and relative to $T_+(N)\omega(j) = T_+(N)/(jT_-(j))$ it is at most $CjT_-(j)/(NT_+(N)) \le CD_N/N \to 0$, since $xT_-(x)$ has a positive index. For $m \le \delta/T_+(n)$ use $u^+_m(n) \le C(m/a_+(m))T_+(N)$. If $\alpha_+ \ge 1/2$, then $\sum_m (m/a_+(m)) e^{-cmT_-(j)} \le C\lambda(M)M^2/a_+(M)$ with $M = 1/T_-(j)$ and λ slowly varying, so the ratio to $T_+(N)\omega(j)$ is at most $C\lambda(1/T_-(j))\, j/a_+(1/T_-(j))$, which tends to 0 because $a_+(1/T_-(j))/j$ is regularly varying in j with the positive index $\alpha_-/\alpha_+ - 1$ when $\alpha_+ < \alpha_-$, and $\alpha_+ \ge 1/2$ with $\alpha_- \ne 1/2$ forces $\alpha_+ < \alpha_-$. If $\alpha_+ < 1/2$ the series $\sum_m m/a_+(m)$ converges, the part is at most $CT_-(j)T_+(N)$, and the ratio is at most $CjT_-(j)^2$, which tends to 0 when $\alpha_- > 1/2$. When $\alpha_- < 1/2$ we use (W$_+$) instead: the part is at most $CT_-(j)\, o(1/(D_N T_+(N)))$, and the ratio is at most $o(1)\, jT_-(j)^2/(D_N T_-(D_N)^2) \le o(1)$, because $xT_-(x)^2$ has the positive index $1 - 2\alpha_-$ and $j \le D_N$ [7, Theorem 1.5.3]. At $\alpha_- = 1/2$ both routes leave a ratio of slowly varying factors that need not be bounded, which is why we exclude the line, except that under (SRT$_+$) with $\alpha_+ < 1/2$ the ratio is at most $C\ell_-(j)^2/(NT_+(N)^2)$ with $T_-(x) = x^{-1/2}\ell_-(x)$, and $NT_+(N)^2$ grows polynomially. For the second term with $\mu_- < \infty$, the part $m > \delta j$ is at most $C_\delta T_+(N) j/a_+(j) = o(T_+(N))$, since $u^+_m(n) \le C(m/a_+(m))T_+(n)$ and $m/a_+(m)$ has a negative index, and the part $m \le \delta j$ is $o(T_+(N))$ as $j \to \infty$ by the count in the proof of (c), which did not use $j \asymp D_N$.

The matching. On $1 \ll j \ll D_N$ we have $\Psi(j/D_N) \to 1$ and $\kappa_+(j) = u_-(j)$ for $j \ge 1$, so the two previous paragraphs give the statement there. On $j \asymp D_N$ it is (c), since $T_+(N) u_-(j) \Psi(j/D_N) = \rho(j/D_N)(1+o(1))/D_N$ in both cases, by (SRT$_-$) and $jT_-(j) \sim y^{1-\alpha_-} D_N T_+(N)$ when $\alpha_- < 1$, and by $u_-(j) \to 1/\mu_-$ when $\mu_- < \infty$, while $T_+(N)\omega(j) \asymp 1/D_N$. For bounded j it is the fixed-j statement with $\Psi(j/D_N) \to 1$. A subsequence argument joins the three ranges. □

Remark 8.2 (initial runs)**.** The first-run decomposition applies unchanged. A minus initial run of length d shifts j by d and adds the defect $T_+(N) \sum_i \nu_-(j-i) u_-(i)$ to the main term. When $\mu_- < \infty$ the kernel u_- is bounded, so the defect is $O(T_+(N)\mathbb{P}(D \in [j/2, j])) = o(1/D_N)$ for every initial run that is finite almost surely, and the bulk of the mixed region needs no condition on a minus initial run, as in the finite-mean case [23]. When $\alpha_- < 1$ the defect is that of Theorem C with $D_N T_+(N)$ in place of $NT_+(N)$, and the conditions of that theorem on ν_- carry over. A plus initial run changes the horizon, and we have not written out the uniform bound for plus initial runs of length close to N.

9 The shape of the law

This section has two parts. The first classifies the critical points of the Lamperti density $L_{\alpha,b}$ for every index and every tail ratio b. The second counts the local extrema of the discrete law. The discrete law has more structure than the density. At the ends it is the tail of a run times a renewal mass function, and these profiles add local extrema of their own. We prove a lower bound on the number of sign changes of $k \mapsto P_N(k+1) - P_N(k)$ in general, and we show what the exact count needs.

9.1 The critical points of the Lamperti density

Let $\alpha_c \in (0,1)$ be the root of $\cos(\pi\alpha/2) = \alpha$ and $\alpha_1 \in (1/2, 1)$ the root of $\sin(\pi\alpha) = \alpha$. The functions $\cos(\pi\alpha/2) - \alpha$ on $[0,1]$ and $\sin(\pi\alpha) - \alpha$ on $[1/2, 1]$ are strictly decreasing from a positive to a negative value, so both roots exist and are unique, with $\alpha_c = 0.594611644\ldots$ and $\alpha_1 = 0.736484448\ldots$, and $\alpha_c < \alpha_1$. Put $c = \cos \pi\alpha$, and for $\alpha_c < \alpha < \alpha_1$ let

$$w_+ = -c + \frac{\alpha \sin \pi\alpha}{\sqrt{1-\alpha^2}}, \qquad s_0 = \operatorname{arcosh} w_+, \qquad \beta_c(\alpha) = 2\alpha \operatorname{artanh}\Big(\frac{\alpha \sinh s_0}{w_+ + c}\Big) - s_0, \quad (9.1)$$

and put $\beta_c(\alpha) = 0$ for $\alpha \le \alpha_c$ and $\beta_c(\alpha) = +\infty$ for $\alpha \ge \alpha_1$. The proof shows that $w_+ > 1$ exactly when $\alpha > \alpha_c$, so s_0 is defined, and that β_c is finite and positive on (α_c, α_1).

Theorem G (the critical points of $L_{\alpha,b}$)**.** *Let $0 < \alpha < 1$, $b > 0$ and $\beta = \log b$.*

(a) If $|\beta| < \beta_c(\alpha)$, then $L_{\alpha,b}$ has exactly 3 critical points, a strict minimum, a strict maximum and a strict minimum, in this order.

(b) If $|\beta| > \beta_c(\alpha)$, then $L_{\alpha,b}$ has exactly one critical point, a strict minimum. For $\alpha \le \alpha_c$ this holds for every b, and for $\alpha \ge \alpha_1$ case (a) holds for every b.

(c) Every critical point is nondegenerate, except in two cases. If $\alpha = \alpha_c$ and $b = 1$, the single critical point $x = 1/2$ has $L''_{\alpha,b}(1/2) = 0$. If $\alpha_c < \alpha < \alpha_1$ and $|\beta| = \beta_c(\alpha)$, there is a strict minimum and one double critical point at which the monotonicity does not change.

For $b = 1$ the theorem says that $L_{\alpha,1}$ is U-shaped for $\alpha \le \alpha_c$ and W-shaped for $\alpha > \alpha_c$, with the maximum at $1/2$. This threshold is known. It appears in Baldassarri, Bouchaud, Dornic and Godrèche [3] and in Godrèche and Luck [16, Section 7], and Margolin and Barkai [21, Section III] quote the value $0.5946\ldots$. What is new is the count for $b \ne 1$, the window $|\log b| < \beta_c(\alpha)$, the constant α_1 beyond which the W shape survives every asymmetry, and the list of degenerate cases. Some values: $\beta_c(0.6) = 0.00589$, $\beta_c(0.65) = 0.2353$, $\beta_c(0.7) = 0.8648$, $\beta_c(0.72) = 1.4765$ and $\beta_c(0.735) = 3.2915$, so at $\alpha = 0.7$ the density is W-shaped for $b < 2.3745$ and U-shaped for $b > 2.3745$. Figure 1(a) shows the density on both sides of α_c for $b = 1$, and Figure 1(c) the curve β_c.

Proof. By (2.4), $L_{\alpha,b}(x) = \frac{\sin \pi\alpha}{\pi} \phi(t)$ with $t = \log((1-x)/x)$ and

$$\phi(t) = \frac{1 + \cosh t}{\cosh(\alpha t - \beta) + c}.$$

The map $x \mapsto t$ is a decreasing diffeomorphism of $(0,1)$ onto $\mathbb{R}$, so the critical points of $L_{\alpha,b}$ and of ϕ correspond, with the same type and the same order of degeneracy. Put $s = \alpha t - \beta$ and $R(s) = \sinh s/(\cosh s + c)$, which is defined because $\cosh s + c > 0$. Then

$$\frac{\phi'(t)}{\phi(t)} = \tanh \frac{t}{2} - \alpha R(s) =: k(t).$$

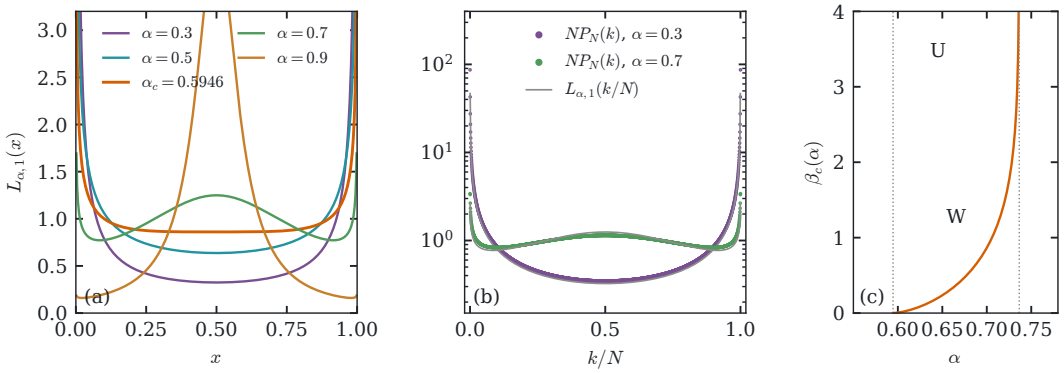


Figure 1: (a) The density $L_{\alpha,1}$ for $\alpha = 0.3$, 0.5, α_c, 0.7 and 0.9. At α_c the minimum at $1/2$ is flat to second order. (b) The exact law $NP_N(k)$ for $N = 2000$, the zeta law $f(n) \propto n^{-\alpha-1}$ for both signs and the fair start, with $\alpha = 0.3$ and $\alpha = 0.7$, on a logarithmic scale, together with $L_{\alpha,1}(k/N)$. The end values are of order $NT(N)$ and lie far above the density for $\alpha = 0.3$. (c) The threshold $\beta_c(\alpha)$ of Theorem G on (α_c, α_1). Below the curve $L_{\alpha,b}$ is W-shaped. Section 11 describes the computations.

Since $R(s) \to \pm 1$ as $s \to \pm\infty$, $k(t) \to \pm(1-\alpha)$ as $t \to \pm\infty$, so ϕ decreases near $-\infty$ and increases near $+\infty$. Hence the number of zeros of k is odd when they are simple, and if there are 3 they are a minimum, a maximum and a minimum.

The function H. The map $t \mapsto s$ is an increasing bijection. Let $D = \{s : \alpha|R(s)| < 1\}$. Off D we have $|\alpha R(s)| \geq 1 > |\tanh(t/2)|$, so k has no zero there. On D write $k(t) = \tanh A - \tanh B$ with $A = (s+\beta)/(2\alpha)$ and $B = \operatorname{artanh}(\alpha R(s))$. Since $\tanh$ is increasing, k has the sign of $A - B$, that is of $\beta - H(s)$, where

$$H(s) = 2\alpha \operatorname{artanh}(\alpha R(s)) - s.$$

So the critical points of $L_{\alpha,b}$ correspond bijectively to the solutions of $H(s) = \beta$ in D. At a solution, $k' = \operatorname{sech}^2(A)(A' - B') = -\operatorname{sech}^2(A)H'(s)/(2\alpha)$ in the variable s, and $\phi'' = \phi k'$ there, so the critical point is nondegenerate exactly when $H'(s) \neq 0$.

The sign of H'. We have $R'(s) = (1 + c\cosh s)/(\cosh s + c)^2$, and with $w = \cosh s$ a direct computation gives

$$2\alpha^2 R'(s) - \big(1 - \alpha^2 R(s)^2\big) = -\frac{Q(w)}{(w+c)^2}, \qquad Q(w) = (1-\alpha^2)(w - w_-)(w - w_+),$$

where $w_\pm = -c \pm \alpha \sin(\pi\alpha)/\sqrt{1-\alpha^2}$ are the roots of $(1-\alpha^2)w^2 + 2c(1-\alpha^2)w + c^2 - \alpha^2$, whose discriminant is $4\alpha^2(1-\alpha^2)(1-c^2)$. Since $H'(s) = [2\alpha^2 R' - (1-\alpha^2 R^2)]/(1-\alpha^2 R^2)$ on D, and $w_- < 1 \leq \cosh s$, we get $H'(s) > 0$ exactly when $\cosh s < w_+$ and $H'(s) < 0$ exactly when $\cosh s > w_+$. With $\sin \pi\alpha = 2\sin(\pi\alpha/2)\cos(\pi\alpha/2)$ and $1 + c = 2\cos^2(\pi\alpha/2)$, the inequality $w_+ > 1$ reads $\alpha \sin(\pi\alpha/2) > \cos(\pi\alpha/2)\sqrt{1-\alpha^2}$, and both sides are nonnegative, so it is $\alpha^2 > \cos^2(\pi\alpha/2)$, that is $\alpha > \alpha_c$.

The domain D. The function R is odd. If $c \geq 0$, that is $\alpha \leq 1/2$, R increases on $[0, \infty)$ to the limit 1, so $D = \mathbb{R}$. If $c < 0$, R increases on $[0, s_m]$ with $\cosh s_m = -1/c$ to the value $R(s_m) = 1/\sin \pi\alpha$, and then decreases to 1. So $D = \mathbb{R}$ exactly when $\alpha < \sin \pi\alpha$, that is $\alpha < \alpha_1$. For $\alpha \geq \alpha_1$ the set D is $\mathbb{R}$ minus two closed intervals $\pm[s_1, s_2]$ around $\pm s_m$, which are single points when $\alpha = \alpha_1$, and $H \to +\infty$ at both edges of $[s_1, s_2]$ and $H \to -\infty$ at both edges of $[-s_2, -s_1]$, because $\alpha R \to 1$ there from below. In every case H is odd with $H(0) = 0$, and $H(s) \to \mp\infty$ as $s \to \pm\infty$, since $\alpha R \to \pm\alpha$ and artanh stays bounded.

Counting. Let $\alpha \le \alpha_c$. Then $D = \mathbb{R}$ and $H' < 0$ except at $s = 0$ when $\alpha = \alpha_c$, so H is a decreasing bijection of $\mathbb{R}$, and $H(s) = \beta$ has exactly one solution. It is simple unless $s = 0$ and $H'(0) = 0$, that is $\alpha = \alpha_c$ and $\beta = 0$. Let $\alpha_c < \alpha < \alpha_1$. Then $D = \mathbb{R}$, $H' > 0$ on $(-s_0, s_0)$ and $H' < 0$ outside, so H decreases from $+\infty$ to a local minimum $-H(s_0)$ at $-s_0$, increases to the local maximum $H(s_0) > H(0) = 0$ at s_0, and decreases to $-\infty$. Hence $H(s) = \beta$ has 3, 2 or 1 solutions according as $|\beta| <$, $=$ or $> H(s_0)$, all simple except the double one at $\pm s_0$ in the case of equality. By the definition of R, $H(s_0) = \beta_c(\alpha)$ of (9.1), which is therefore finite and positive. Finally let $\alpha \ge \alpha_1$. On $(0, s_1)$ the function H rises from 0 to $+\infty$, so H' cannot be negative just below s_1, which forces $s_0 \ge s_1$ and $H' > 0$ on $(-s_1, s_1)$. On (s_2, ∞) it falls from $+\infty$, which forces $s_0 \le s_2$ and $H' < 0$ there, and likewise on $(-\infty, -s_2)$. So each of the three components of D carries exactly one solution of $H = \beta$, all simple, for every β. This proves (a) to (c). □

The monotonicity of H is what closes the count. Descartes' rule for the exponential sum that has the sign of ϕ' gives at most 3 zeros only for $\alpha \le 1/2$. As α increases to α_1 the window β_c opens to ∞, and at $b = 0$ the dominated regime of Section 8 takes over. There the density (8.1) of the ratio Z_-/Z_+ has an interior maximum exactly when $\alpha > \sin \pi\alpha$, that is $\alpha > \alpha_1$, since the numerator of its logarithmic derivative in $r = y^\alpha$ is $-(1+\alpha)r^2 - 2cr - (1-\alpha)$, which has a positive root exactly when $c < 0$ and $c^2 > 1 - \alpha^2$. So the two shape thresholds meet at α_1. Near α_c with $b = 1$ the two minima split off from $1/2$ like a pitchfork, at distance $\frac{1}{4}(3(2\alpha^2 - 1 - \cos\pi\alpha)/(\alpha^2(1-\alpha^2)))^{1/2}(1 + o(1))$ from $1/2$, which follows from the expansion $\chi(t) = \chi(0+) + \frac{1}{3}\alpha^2(1-\alpha^2)t^2(1 + O(t^2))$ of the function $\chi(t) = \cosh \alpha t + c - \alpha \sinh(\alpha t) \coth(t/2)$ that has the sign of k for $b = 1$. At $\alpha = 1/2$ and $b = 1$ the theorem gives the arcsine density, with its minimum at $1/2$.

9.2 Difference forms of the renewal theorem

The end profiles are renewal mass functions. To compare neighbouring values of P_N near an end we need the differences of u_σ, and (DSRT$_\sigma$) asks that they have the size of the derivative of $\frac{\sin \pi\alpha}{\pi}/(nT_\sigma(n))$. It holds for log-convex laws.

Lemma 9.1 (the difference renewal theorem for log-convex laws)**.** *Let f be a log-convex mass function on $\{1, 2, \dots\}$ with $f(1) > 0$ and a tail T that is regularly varying with index $-\alpha \in (-1, 0)$. Then f satisfies (LB) and (SRT), its renewal mass function u is strictly decreasing, and $u(j) - u(j+1) \sim (1-\alpha)u(j)/j$.*

Proof. A log-convex f that tends to 0 is nonincreasing, so $f(n) \le 2T(n/2)/n \le CT(n)/n$, which is (LB), and Doney's theorem (Lemma 3.1(iii)) gives (SRT), so u is regularly varying with index $\alpha - 1$. By de Bruijn and Erdős [11, Theorem 1], u is log-convex, so $r_j = u(j+1)/u(j)$ is nondecreasing, and $r_j < 1$ because $u \to 0$. For $\lambda > 1$ and i large put $j = \lfloor i/\lambda \rfloor$. Then $\log u(i) - \log u(j) = \sum_{l=j}^{i-1} \log r_l$ lies between $(i-j)\log r_j$ and $(i-j)\log r_i$, and tends to $(\alpha-1)\log\lambda$. Since $i - j \sim (\lambda-1)i/\lambda$, this gives $\limsup_i i \log r_i \le (\alpha-1)\lambda\log\lambda/(\lambda-1)$, and applying the other inequality with i in place of j gives $\liminf_j j \log r_j \ge (\alpha-1)\log\lambda/(\lambda-1)$. Let $\lambda \downarrow 1$. So $j \log r_j \to \alpha - 1$, which is the claim. □

For Sibuya laws, $T(n) = \prod_{i<n}(1 - \alpha/i)$, one has $r_j = (j+\alpha)/(j+1)$ exactly. The zeta law $f(n) = n^{-\alpha-1}/\zeta(\alpha+1)$ is strictly log-convex. We do not know whether (LR) and the first-difference condition $|f(n) - f(n+1)| = O(f(n)/n)$ imply (DSRT) when $\alpha \le 1/2$.

We also need the end profile of a prepared start. For an initial run (ν_+, ν_-) with $n\nu_\pm(n) \to 0$, Theorem C(f) gives $P_N(N-j)/T_+(N) \to \kappa_+(j)$ for $j \ge 1$, with κ_+ as in (6.1), and we write $\widetilde{\kappa}_+ = (\widetilde{\kappa}_+(0), \kappa_+(1), \kappa_+(2), \dots)$ for the profile with the end value $\widetilde{\kappa}_+(0) = \lim_N Q_+(N)/T_+(N)$, when this limit exists in $[0, \infty]$. We say that the initial run

satisfies (S) if

$$|\nu_\sigma(n+1)-\nu_\sigma(n)| = o(n^{-2}), \qquad |\nu_\sigma(n+1)-2\nu_\sigma(n)+\nu_\sigma(n-1)| = o(n^{-3}) \tag{S}$$

for both signs, and the limits $\widetilde{\kappa}_\pm(0)$ exist. Fresh and fair starts satisfy (S) under (LR) and (LD2), and so does every smooth regularly varying ν, such as $\nu(n) \propto n^{-1-\gamma}$ for any $\gamma > 0$.

Lemma 9.2 (the prepared profile)**.** *Assume (BAL), (SPAN), (DSRT$_-$) and the first half of (S) for ν_-. Then $\kappa_+(j+1)-\kappa_+(j) = (1+o(1))(u_-(j+1)-u_-(j))$. In particular $\widetilde{\kappa}_+$ is eventually strictly decreasing.*

Proof. Write $u = u_-$, $\Delta u(i) = u(i+1)-u(i)$ and $\Delta u(0) = u(1)-1$. From (6.1), $\Delta\kappa_+(j) = \pi_+\Delta u(j) + \sum_{d\le j}\nu_-(d)\Delta u(j-d) + \nu_-(j+1)$. The part $d \le j/2$ is $(\pi_- + o(1))\Delta u(j)$ by dominated convergence, since Δu is regularly varying by (DSRT$_-$). With $\sum_{i<j/2}\Delta u(i) = u(\lceil j/2\rceil)-1$, the rest is $\sum_{0\le i<j/2}(\nu_-(j-i)-\nu_-(j+1))\Delta u(i) + \nu_-(j+1)u(\lceil j/2\rceil)$. With $\delta_j = \sup_{n\in[j/2,j+1]}|\Delta\nu_-(n)| = o(j^{-2})$ and $\sum_{i\le j}(i+1)|\Delta u(i)| \le Cju(j)$, this is at most $C\delta_j ju(j) + o(1/j)u(j/2) = o(u(j)/j)$, while $|\Delta u(j)| \sim (1-\alpha)u(j)/j$. □

9.3 Sign changes of the discrete law

For a finite or infinite real sequence (s_i) let $V(s)$ be the number of sign changes of the sequence of its nonzero terms. The sign changes of $k \mapsto P_N(k+1)-P_N(k)$ are the strict local extrema of P_N when no two neighbouring values are equal.

Theorem 9.3 (a lower bound on the sign changes)**.** *Assume (BAL), (SPAN) and (SRT$_\pm$), and let the initial run be such that $NP_N(k) = L_{\alpha,b}(k/N) + o(1)$ uniformly on compact subsets of $(0,1)$ (for instance a fresh start, or under (LB$_\pm$) any initial run with $n\nu_\pm(n)\to 0$) and such that $W_N^\pm = o(T_\pm(N))$. Assume that the limits $\widetilde{\kappa}_\pm(0)$ exist and are finite, that $\widetilde{\kappa}_\pm(j+1) \ne \widetilde{\kappa}_\pm(j)$ for all $j \ge 0$, and that $\widetilde{\kappa}_\pm$ is strictly decreasing from some point on. Let $\sigma_\pm = V((\widetilde{\kappa}_\pm(j+1)-\widetilde{\kappa}_\pm(j))_{j\ge0})$, and let σ_L be the number of critical points of $L_{\alpha,b}$, which is 1 or 3 by Theorem G, each of them a strict local extremum. Then for all large N,*

$$V\big((P_N(k+1)-P_N(k))_{0\le k<N}\big) \ge \sigma_L + \sigma_+ + \sigma_-.$$

Proof. If (s_{i_l}) is a subsequence of (s_i), the differences of (s_i) have at least as many sign changes as the differences of (s_{i_l}), since each nonzero difference $s_{i_{l+1}} - s_{i_l}$ is a sum of consecutive differences of (s_i), one of which has its sign. Let $c_1 < \cdots < c_s$ be the critical points of $L_{\alpha,b}$, $s = \sigma_L$. Since $L_{\alpha,b}\to\infty$ at both ends, $L_{\alpha,b}$ decreases strictly on $(0,c_1]$ and increases strictly on $[c_s,1)$, and the critical points alternate between strict minima and strict maxima, starting and ending with a minimum. Choose $z_0 \in (0,c_1)$ and $z_{s+1}\in(c_s,1)$. The differences of $L_{\alpha,b}(z_0), L_{\alpha,b}(c_1), \dots, L_{\alpha,b}(c_s), L_{\alpha,b}(z_{s+1})$ have the signs $-,+,-,\dots,+$, with s sign changes. Choose J such that $\widetilde{\kappa}_+$ and $\widetilde{\kappa}_-$ are strictly decreasing on $[J-1,\infty)$, so $\widetilde{\kappa}_\pm(J) > 0$. Consider the subsequence of $(P_N(k))_{0\le k\le N}$ at the points

$$0,1,\dots,J,\ \lfloor z_0N\rfloor, \lfloor c_1N\rfloor,\dots,\lfloor c_sN\rfloor, \lfloor z_{s+1}N\rfloor,\ N-J,\dots,N.$$

By Theorem C(f), its mirror, and the definition of $\widetilde{\kappa}_\pm(0)$, $P_N(N-j)/T_+(N)\to\widetilde{\kappa}_+(j)$ and $P_N(j)/T_-(N)\to\widetilde{\kappa}_-(j)$ for $0\le j\le J$. Since consecutive values of $\widetilde{\kappa}_\pm$ differ, for large N the differences at the two ends have the signs of the differences of $\widetilde{\kappa}_\pm$, with σ_- and σ_+ sign changes, and the differences next to $j=J$ have the sign of $\widetilde{\kappa}_\pm(J)-\widetilde{\kappa}_\pm(J-1) < 0$. In the bulk, $NP_N(\lfloor zN\rfloor)\to L_{\alpha,b}(z)$ at each chosen point, so for large N the bulk differences have the signs $-,+,\dots,+$ with s sign changes. At the junctions, $P_N(J)\sim T_-(N)\widetilde{\kappa}_-(J)$ and $P_N(N-J)\sim T_+(N)\widetilde{\kappa}_+(J)$, while the bulk values are $O(1/N)$ and $NT_\pm(N)\to\infty$. So the difference from $P_N(J)$ to $P_N(\lfloor z_0N\rfloor)$ is negative and the difference from $P_N(\lfloor z_{s+1}N\rfloor)$ to $P_N(N-J)$ is positive. Reading from left to right, the last difference at the minus end,

the junction and the first bulk difference are all negative, and the last bulk difference, the junction and the first difference at the plus end, which is $\sim T_+(N)(\widetilde{\kappa}_+(J-1)-\widetilde{\kappa}_+(J))$, are all positive. So the junctions add no sign change, and the subsequence has $\sigma_- + s + \sigma_+$ sign changes. □

Example 9.4 (the fair start and the fresh start)**.** Let both run laws be the zeta law, so $b = 1$, and let the start be fair. Then $Q_\pm(N) = \frac{1}{2}T(N)$, so $\widetilde{\kappa}_\pm(0) = \frac{1}{2}$, and (6.1) gives $\kappa_\pm(j) = \frac{1}{2}u(j) + \frac{1}{2}\sum_{d\le j} f(d)u(j-d) = u(j)$ for $j \ge 1$. Since u is strictly decreasing by Lemma 9.1, $\sigma_\pm = 1$ if $u(1) = f(1) > \frac{1}{2}$ and $\sigma_\pm = 0$ if $f(1) < \frac{1}{2}$. Now $f(1) = 1/\zeta(1+\alpha)$, and $\zeta(1.7) = 2.054\ldots$ and $\zeta(1.9) = 1.750\ldots$. So Theorems 9.3 and G give at least 1 sign change for $\alpha = 0.3$ and $\alpha = 0.5$, at least 3 for $\alpha = 0.7$, and at least $3+1+1 = 5$ for $\alpha = 0.9$. For a fresh plus start with different laws, $\widetilde{\kappa}_+ = (1, u_-(1), \dots)$ is strictly decreasing and $\widetilde{\kappa}_- = (0, u_+(1), u_+(2), \dots)$ has one sign change, so the bound is $\sigma_L + 1$.

9.4 The layer at the level of differences

Near the plus end the first differences of P_N follow those of the renewal mass function u_-. The tool is a difference form of Doney's local bound, Lemma 3.1(v), uniform in the number of summands. It is the case $\alpha < 1$ of the difference form of the local big-jump theorem of [23, Section 6], and it needs no centering.

Lemma 9.5 (the difference form of Doney's bound)**.** *Let $\alpha_\sigma < 1$, assume (SPAN), (LR) and (LD) for the sign σ, drop the sign and let $V_m = f^{*m}$. There is C such that*

$$|V_m(n-1) - V_m(n)| \le C\, m\, \frac{f(n)}{n} \qquad \text{for all } m \ge 1 \text{ and } n \ge a_m.$$

Proof. Write $\nabla V_m(n) = V_m(n-1) - V_m(n)$. For $n \ge a_m$ we have $mT(n) \le mT(a_m) \le C$, and (LR) gives $f(n) \asymp T(n)/n$. Fix $\Lambda \ge 1$. If $n \le \Lambda a_m$, then $mT(n) \ge c_\Lambda$ by Potter's bounds, so $mf(n)/n \ge c_\Lambda a_m^{-2}$, and Lemma 3.7(ii) gives the claim. Let $n > \Lambda a_m$ with Λ large, fix an integer $K \ge 2$, put $y = \lceil n/K \rceil$, and let I be the first index i with $X_i \ge y$, with $I = \infty$ if there is none. Then $V_m(n) = \sum_{i=1}^m \mathbb{P}(S_m = n, I = i) + V_m^{(y)}(n)$ in the notation of Lemma 3.7.

A summand of size at least y. Let $h_i(z) = \mathbb{P}(S_m - X_i = z,\ X_1, \dots, X_{i-1} < y)$, a measure of mass at most 1 that does not depend on n. Since X_i is independent of the rest, $\mathbb{P}(S_m = n, I = i) = \sum_{z \le n-y} h_i(z) f(n-z)$, and

$$\nabla\mathbb{P}(S_m = \cdot, I = i)(n) = \sum_{z \le n-1-y} h_i(z)\big[f(n-1-z) - f(n-z)\big] - h_i(n-y)f(y).$$

In the sum $n - z \ge y + 1 \ge n/K$, so (LD) and Potter's bounds for the regularly varying f give $|f(n-1-z) - f(n-z)| \le Cf(n-z)/(n-z) \le C_K f(n)/n$, and the sum is at most $C_K f(n)/n$. In the boundary term $h_i(n-y) \le V_{m-1}(n-y)$ with $n - y \ge n/2 \ge a_m/2$, so Lemma 3.1(v) gives $h_i(n-y) \le CmT(n)/n$, and $f(y) \le C_K f(n)$. Summing over $i \le m$, this part of $\nabla V_m(n)$ is at most $C_K mf(n)/n + C_K m^2 T(n) f(n)/n \le C_K mf(n)/n$.

No summand of size y. Let $m_1 = \lfloor m/2 \rfloor$, $m_2 = m - m_1$ and $g_l = V_{m_l}^{(y)}$, so that $V_m^{(y)} = g_1 * g_2$ and, with n even for notational ease,

$$\begin{aligned}\nabla V_m^{(y)}(n) &= \sum_{z \le n/2} g_1(z)\nabla g_2(n-z) + \sum_{z > n/2} g_1(z)\nabla g_2(n-z) \\ &= \sum_{w \ge n/2} \nabla g_1(n-w) g_2(w) + g_1(n/2)g_2(n/2-1) + \sum_{z > n/2} g_1(z)\nabla g_2(n-z),\end{aligned}$$

where the first sum was rearranged by $w = n - z$ and summed by parts. By Lemma 3.7(ii) and $m_l T(y) \le C_K$, $\sup|\nabla g_l| \le C_K a_m^{-2}$. The summands of g_l are bounded by y, so the

Chernoff bound in the proof of Lemma 3.3, with $\mathbb{E}[X; X < y] \le CyT(y)$, gives

$$\sum_{w \ge n/2} g_l(w) = \mathbb{P}\Big(\sum_{i \le m_l} X_i \mathbf{1}_{\{X_i < y\}} \ge \frac{n}{2} \Big) \le \big(C_K m T(n) \big)^{K/2},$$

and splitting $g_l(x)$ at $x \ge n/2 - 1$ in the same way, with one of two halves of its summands above $n/4$ and Lemma 3.1(i) for the other half, $g_l(x) \le C a_m^{-1} (C_K m T(n))^{K/8}$ for $x \ge n/2 - 1$. Hence

$$|\nabla V_m^{(y)}(n)| \le C_K a_m^{-2} \big(C_K m T(n) \big)^{K/2} + C a_m^{-2} \big(C_K m T(n) \big)^{K/4}.$$

Put $\lambda = n/a_m > \Lambda$. By Potter's bounds $c\lambda^{-\alpha-\delta} \le mT(n) \le C\lambda^{-\alpha+\delta}$, so for Λ large $C_K m T(n) \le 1$, and

$$\frac{a_m^{-2} (C_K m T(n))^{K/4}}{m f(n)/n} \le C \, \frac{(n/a_m)^2 (C_K m T(n))^{K/4}}{mT(n)} \le C_K \lambda^{2+\alpha+\delta-(\alpha-\delta)K/4} \le C_K$$

once $K \ge 4(2+\alpha+\delta)/(\alpha-\delta)$. Fix such a K and the matching Λ. This proves the claim for $n > \Lambda a_m$, and together with the first case for all $n \ge a_m$. □

Lemma 9.6 (consequences)**.** *Assume (BAL) and (SPAN), and (LR) and (LD) for both signs. Write $V_m = u_m^+$, $q_m = q_m^+$, $\delta V_m = V_{m+1} - V_m$, and f, T, a_m for the plus law.*

(i) $|\delta V_m(n)| \le C f(n)(1 + mT(n))$ *for* $n \ge 2a_m$.

(ii) $|q_{m+1}(n) - T(n)| \le C m T(n)^2$ *for* $n \ge 2a_m$.

(iii) For $j \ge 1$, *with* $\Delta u_m^-(j) = u_m^-(j+1) - u_m^-(j)$,

$$\sum_{m \ge 1} m \, |\Delta u_m^-(j)| \le \frac{C}{j^2 T_-(j)^2}, \qquad |u_-(j+1) - u_-(j)| \le \frac{C}{j^2 T_-(j)},$$
$$\sum_{m \ge 1} m \, u_m^-(j) \le \frac{C}{j T_-(j)^2}, \qquad \sum_{m \ge 1} m \, q_m^-(j) \le \frac{C}{T_-(j)}.$$

Proof. (i) Since $\delta V_m = f * V_m - V_m$, $\delta V_m(n) = \sum_x f(x)[V_m(n-x) - V_m(n)]$. For $x \le n/2$, Lemma 9.5 at the levels $n - i \ge n/2 \ge a_m$ gives $|V_m(n-x) - V_m(n)| \le \sum_{i<x} |\nabla V_m(n-i)| \le Cxmf(n)/n$, and $\sum_{x \le n/2} x f(x) \le CnT(n)$ by Karamata's theorem, so this part is at most $CmT(n)f(n)$. The rest is $\sum_{w < n/2} f(n-w) V_m(w) - V_m(n) T(n/2)$, at most $f(n/2) + C(mT(n)/n) T(n/2) \le Cf(n)(1 + mT(n))$ by Lemma 3.1(v) and (LR).

(ii) By (2.6), $q_{m+1}(n) - T(n) = \sum_{x<n} V_m(x)[T(n-x) - T(n)] - T(n)\mathbb{P}(S_m \ge n)$. For $x \le n/2$, $T(n-x) - T(n) = \sum_{i=n-x}^{n-1} f(i) \le CxT(n)/n$ by (LB), so this part is at most $C(T(n)/n)\mathbb{E}[S_m; S_m \le n/2] \le C(T(n)/n)\, m\, \mathbb{E}[X; X \le n] \le CmT(n)^2$. For $x > n/2$, Lemma 3.1(v) gives $V_m(x) \le CmT(n)/n$ and $\sum_{x > n/2} T(n-x) \le CnT(n)$. The last term is at most $CmT(n)^2$ by Lemma 3.2(i).

(iii) Split the first sum at $m_j = c/T_-(j)$ with c small, so that $j \ge a_-(m)$ for $m \le m_j$. There Lemma 9.5 for the minus law gives $|\Delta u_m^-(j)| \le C m f_-(j)/j$, and $\sum_{m \le m_j} m^2 \le C m_j^3$, so this part is at most $C(f_-(j)/j) T_-(j)^{-3} \le C/(j^2 T_-(j)^2)$ by (LR). For $m > m_j$, Lemma 3.7(ii) gives $|\Delta u_m^-(j)| \le C a_-(m)^{-2} e^{-cmT_-(j)}$ with $a_-(m) \ge cj$, and $\sum_m m e^{-cmT_-(j)} \le C/T_-(j)^2$. The second bound is the same computation without the factor m. For the third, $\sum_m m u_m^-(j) = (u_- * u_-)(j) - u_-(j)$, since $\sum_{m,m'} u_{m+m'}^-(j) = \sum_s (s+1) u_s^-(j)$, and $(u_- * u_-)(j) \le 2 \sup_{l \ge j/2} u_-(l) \sum_{l \le j/2} u_-(l) \le C\omega_-(j)/T_-(j)$ by (SRT$_-$), which (LR) gives through (LB), and by $\sum_{l \le x} u_-(l) \le C/T_-(x)$. For the last, $\sum_m m q_m^-(j)$ is the expected index of the minus run that covers level j, which is $1 + \sum_{i \ge 1} \mathbb{P}(S_i^- < j) \le 1 + \sum_{l<j} u_-(l) \le C/T_-(j)$. □

Theorem 9.7 (the layer at the level of differences)**.** *Assume (BAL), (SPAN), and (LR) and (LD) for both signs. Let the start be fresh with either sign or fair, and put $D_N(j) = P_N(N-j-1) - P_N(N-j)$. For every $\eta > 0$ there are $\varepsilon > 0$ and J such that for all large N and all $J \le j \le \varepsilon N$,*

$$\big|D_N(j) - T_+(N)\big(u_-(j+1) - u_-(j)\big)\big| \le \eta\, T_+(N)\,\frac{u_-(j)}{j}.$$

The same holds at the minus end with the signs exchanged. Under (DSRT$_-$), $D_N(j) < 0$ on $J \le j \le \varepsilon N$, so P_N increases strictly toward the plus end through the layer.

Proof. Let the start be fresh plus, write $a_m = u_m^-$, $b_m = q_m^-$, $Q_m = q_{m+1}^+$, $V_m = u_m^+$ and $n = N - j$, and abbreviate $T = T_+$, $f = f_+$ and $u = u_-$. By Proposition 2.1, $P_N(N-j) = \sum_m [a_m(j)Q_m(n) + V_m(n)b_m(j)]$, and

$$D_N(j) = \mathrm{I} + \mathrm{II} + \mathrm{III} + \mathrm{IV}, \qquad \mathrm{I} = \sum_m \Delta a_m(j)\, Q_m(n-1), \quad \mathrm{II} = \sum_m a_m(j)\, \delta V_m(n-1),$$

$$\mathrm{III} = \sum_m V_m(n-1)\, \Delta b_m(j), \qquad \mathrm{IV} = \sum_m \nabla V_m(n)\, b_m(j),$$

where Δ acts on j, $\nabla V_m(n) = V_m(n-1) - V_m(n)$, and we used $Q_m(n-1) - Q_m(n) = \mathbb{P}(S_{m+1}^+ \ge n-1) - \mathbb{P}(S_m^+ \ge n-1) - \mathbb{P}(S_{m+1}^+ \ge n) + \mathbb{P}(S_m^+ \ge n) = \delta V_m(n-1)$. With $B_i(j) = \sum_{m>i} b_m(j) = \mathbb{P}(S_i^- < j)$ and $\Delta B_i(j) = a_i(j)$, summation by parts gives $\mathrm{III} = \sum_{i \ge 1} a_i(j)\delta V_i(n-1) = \mathrm{II}$. Put $\tau_N(j) = T(N)u(j)/j$. By (SRT$_-$), $u(j) \asymp 1/(jT_-(j))$ for $j \ge J$, so $\tau_N(j) \asymp T(N)/(j^2 T_-(j))$ there, and by (R1), $T(n-1) = T(N)(1 + O(\varepsilon) + o(1))$ uniformly in $j \le \varepsilon N$. Fix c_0 small and put $m_0 = c_0/T(N)$, so that $a_+(m) \le N/4 \le (n-1)/2$ for $m \le m_0$ and large N.

The range $m \le m_0$. Write $\mathrm{I} = T(n-1)\sum_{m \le m_0} \Delta a_m(j) + \sum_{m \le m_0} \Delta a_m(j)[Q_m(n-1) - T(n-1)] + \mathrm{I}'$, with I' the part $m > m_0$. By Lemma 9.6(ii) and (iii) the middle sum is at most $CT(N)^2/(j^2 T_-(j)^2) \le C(T(N)/T_-(j))\tau_N(j) \le C\varepsilon^\alpha \tau_N(j)$. By Lemma 9.6(i) and (iii), the part $m \le m_0$ of II and of III is at most $Cf(N)[u(j) + T(N)\sum_m m a_m(j)] \le Cf(N)u(j)(1 + T(N)/T_-(j)) \le C(j/N)\tau_N(j) \le C\varepsilon\tau_N(j)$. By Lemma 9.5 and Lemma 9.6(iii), the part $m \le m_0$ of IV is at most $C(f(N)/N)\sum_m m b_m(j) \le Cf(N)/(NT_-(j)) \le C(j/N)^2\tau_N(j) \le C\varepsilon^2\tau_N(j)$.

The range $m > m_0$. Here $mT_-(j) \ge c_0 T_-(j)/T(N) \ge c\,\varepsilon^{-\alpha}$ for $j \le \varepsilon N$ and large N. The minus factors carry this decay: by Lemma 3.2(iii), Lemma 3.6 and Lemma 3.7(ii), $a_m(j) \le Ca_-(m)^{-1}e^{-cmT_-(j)}$, $b_m(j) \le CT_-(j)e^{-cmT_-(j)}$ and $|\Delta a_m(j)| \le Ca_-(m)^{-2}e^{-cmT_-(j)}$. The plus factors are bounded by Lemma 3.5 and Lemma 3.7: $Q_m \le CT(N)$, $|\delta V_m| \le C/(ma_+(m))$ and $|\nabla V_m| \le Ca_+(m)^{-2}$. Since $a_\pm(m_0) \ge cN$ and the sequences $a_-(m)^{-2}$, $1/(ma_+(m)a_-(m))$ and $a_+(m)^{-2}$ are regularly varying with indices below -1, the sums over $m > m_0$ are at most $Cm_0 a_-(m_0)^{-2} \le C/(T(N)N^2)$, $C/(a_+(m_0)a_-(m_0)) \le C/N^2$ and $Cm_0 a_+(m_0)^{-2} \le C/(T(N)N^2)$. Hence

$$|\mathrm{I}'| \le Ce^{-c\varepsilon^{-\alpha}}N^{-2}, \qquad |\mathrm{II}'| + |\mathrm{III}'| \le Ce^{-c\varepsilon^{-\alpha}}N^{-2}, \qquad |\mathrm{IV}'| \le Ce^{-c\varepsilon^{-\alpha}}\,\frac{T_-(j)}{T(N)N^2},$$

and also $T(n-1)|\sum_{m>m_0}\Delta a_m(j)| \le Ce^{-c\varepsilon^{-\alpha}}N^{-2}$. Since $x^2T_-(x)$ and $x^2T_-(x)^2$ have positive indices, $j^2T_-(j) \le C\varepsilon^{2-\alpha}N^2T(N)$ and $j^2T_-(j)^2 \le C\varepsilon^{2-2\alpha}N^2T(N)^2$ for $j \le \varepsilon N$, so $\tau_N(j) \ge c\,\varepsilon^{\alpha-2}N^{-2}$ and all four bounds are at most $Ce^{-c\varepsilon^{-\alpha}}\varepsilon^{2-2\alpha}\tau_N(j)$.

Conclusion. Collecting the terms, with $\sum_{m \ge 1}\Delta a_m(j) = u(j+1) - u(j)$ and $|u(j+1) - u(j)| \le C/(j^2T_-(j)) \le C\tau_N(j)/T(N)$ by Lemma 9.6(iii),

$$|D_N(j) - T(N)(u(j+1) - u(j))| \le C\big(\varepsilon^\alpha + \varepsilon + \varepsilon^2 + e^{-c\varepsilon^{-\alpha}}\varepsilon^{2-2\alpha} + o(1)\big)\tau_N(j)$$

uniformly in $J \le j \le \varepsilon N$, where the $o(1)$ is that of (R1). Choose ε small, then N large. For a fresh minus start the identity has $Q_m = q_m^+$ and $b_m = q_{m+1}^-$. The same computation applies, with the one extra boundary term $V_1(n-1)f_-(j) = f(n-1)f_-(j) \le C\varepsilon\tau_N(j)$ in III, and gives the same limit, since $\sum_{m\ge1} \Delta a_m(j) = u(j+1) - u(j)$ for $j \ge 1$. The fair start is the average of the two. Under (DSRT$_-$), $u(j+1) - u(j) \le -(1-\alpha-o(1))u(j)/j$, so $D_N(j) < 0$ for $\eta < 1-\alpha$. □

9.5 The exact count

In every case of Section 11 with a fresh, a fair or a smooth start, the computed number of sign changes equals the lower bound of Theorem 9.3 for all large N. Theorem 9.7 settles the sign of the differences in the layers. One statement about the bulk turns the bound into an equality, and we state it as a hypothesis.

(B) Uniformly for k/N in compact subsets of $(0,1)$, $N^2(P_N(k+1) - P_N(k)) \to L'_{\alpha,b}(k/N)$ and $N^3(P_N(k+1) - 2P_N(k) + P_N(k-1)) \to L''_{\alpha,b}(k/N)$.

Proposition 9.8 (the exact count)**.** *Assume (BAL), (SPAN), (LR), (LD2) and (DSRT$_\pm$), let the start be fresh or fair, assume that (α, b) is not one of the degenerate pairs of Theorem G(c), and assume (B). Let $J_\pm$ be such that $\widetilde\kappa_\pm$ is strictly decreasing on $[J_\pm, \infty)$, assume $\widetilde\kappa_\pm(j+1) \ne \widetilde\kappa_\pm(j)$ for $j < J_\pm$, and let $M_\pm$ be the number of local maxima of $\widetilde\kappa_\pm$. Let $m_L \in \{0,1\}$ be the number of interior maxima of $L_{\alpha,b}$. Then for all large N, P_N has exactly $M_+ + M_- + m_L$ modes, M_+ of them within J_+ of N, M_- within J_- of 0, and the interior one, when $m_L = 1$, at $(x_{\max} + o(1))N$ with $x_{\max}$ the maximum of $L_{\alpha,b}$. Between the end blocks lie exactly $1 + m_L$ further local minima, at $(c + o(1))N$ for the minima c of $L_{\alpha,b}$, and no antimode in the layers. For strictly log-convex run laws $M_+ = M_- = 1$, so the count is $2 + m_L$.*

Proof. We check the sign of $\Delta P_N(k) = P_N(k+1) - P_N(k)$ on five ranges. For $0 \le j < J_+$, Theorem C(f) gives $D_N(j)$ the sign of $\widetilde\kappa_+(j+1) - \widetilde\kappa_+(j)$, since $n\nu_\pm(n) \to 0$ for fresh and fair starts. For $J_+ \le j \le \varepsilon N$, Theorem 9.7 with (DSRT$_-$) gives $D_N(j) < 0$, so P_N increases strictly toward the plus end through the layer, and (LD) holds by (LR) and (LD2). By (B) and Theorem G, on $[\varepsilon N, (1-\varepsilon)N]$ the sign of $\Delta P_N(k)$ is that of $L'_{\alpha,b}(k/N)$ away from the critical points, and near each critical point c the second difference has the sign of $L''_{\alpha,b}(c) \ne 0$, so ΔP_N is strictly monotone on a window of width δN around cN and changes sign exactly once there. The minus end is the mirror. Taking ε below the first critical point and above 1 minus the last one makes the ranges overlap, and the signs agree on the overlaps, since $L'_{\alpha,b} > 0$ near 1 and $D_N < 0$ in the plus layer. Reading the signs of ΔP_N from $k = 0$ to $k = N$, the pattern is that of the minus profile, then a block of negative signs, then the bulk block, which is empty when $m_L = 0$ and $(+)^a(-)^b$ when $m_L = 1$, then a block of positive signs, then the mirror of the plus profile. The modes are the changes from $+$ to $-$, a leading $-$ and a trailing $+$, which gives $M_- + m_L + M_+$, and the minima are the changes from $-$ to $+$, which are the minima of $L_{\alpha,b}$. For log-convex laws $u_\mp$ is strictly decreasing with limit 0, and the profiles $(1, u(1), \dots)$, $(0, u(1), \dots)$ and $(\frac12, u(1), \dots)$ have one local maximum each, at $j = 0$ or $j = 1$ according as $u(1)$ is below or above the end value, with no tie when $f(1) \ne \frac12$ for the fair start. □

What is proved toward (B). The first-difference limit follows by taking a difference inside the proof of Theorem A, with Lemma 3.7 for the differences of the convolution powers in space and in the index. The second-difference limit is the part that needs work: the range of few runs contributes at the order N^{-3} through the second differences of the convolution powers, and a second-difference form of Lemma 9.5 under (LD2) is needed there. For starts with (S) other than fresh and fair, Theorem 9.7 needs in addition a

bound on the differences of the remainder walk over initial runs of length above $N/2$. The computations of Section 11 agree with (B) and the count in every case.

10 Open problems

We list what the paper leaves open, in the order of the sections.

(1) (The lower half of the matching.) Theorem B(b) matches the end layer to the bulk uniformly under (LB$_+$). Lemma 5.1 gives its upper half under (SRT$_\pm$) for $\alpha \neq 1/2$. Whether the lower half holds under (SRT$_\pm$) alone is open (Remark 5.3).

(2) (Necessity in the aged law.) Theorem A needs (SRT$_\pm$) for a fresh start. In the aged decomposition the plus part sees $F^-_{N-d}(k-d)$ with the same number $N-k$ of minus steps for every d, so we expect a failure of (SRT$_-$) to survive the average over d and break Theorem E(a). This is not proved. Whether (SRT$_\pm$) alone implies (U) when $\alpha \leq 1/2$, the uniform matching of the aged layer to L_θ, and the slowly varying factor at $\alpha = 1/2$ are open as well.

(3) (The line $\alpha_- = 1/2$.) The uniform matching of Theorem F(d) is stated for $\alpha_- \neq 1/2$. On the line, both routes of the proof lose a ratio of slowly varying factors of T_-, which need not be bounded.

(4) (The big-jump zone of the dominated regime.) Between the bulk at the scale D_N and the lighter end we expect the two-term kernel of [23] with the signs exchanged, and a window of width $(\log N)^{1-\alpha_+} D_N$ when the lighter mean is finite. Nothing is proved there, and the initial-run theorem of the dominated regime is written only for minus initial runs (Remark 8.2).

(5) (The exact count.) Proposition 9.8 rests on the hypothesis (B), a first- and second-difference local limit theorem in the bulk. The first-difference half follows from the proof of Theorem A. The second-difference half needs a second-difference form of Lemma 9.5 on the range of few runs. The count in the dominated regime is open as well. We expect $M_+ + M_- + m$ modes there, with $m = 1$ exactly when the density ρ has an interior maximum, which for $\mu_- < \infty$ is $\alpha_+ > 1/2$, and the threshold curve in the (α_+, α_-) square runs from (α_1, α_1) to $(1/2, 1)$. Every exact law we computed agrees with this.

(6) (Every N.) All counts here are for large N. For completely monotone run laws with a fresh start, every exact law we computed has at most 3 local maxima in $\{1, \dots, N-1\}$ for every N. For the fair start no such bound holds: there are completely monotone laws, the same for both signs, whose fair law has 5 modes at $N = 39$ while both fresh laws are monotone. We know no proof of a bound for every N.

(7) (The line $\alpha = 1$.) When one or both indices equal 1 the periods may or may not have a finite mean, and the scales carry slowly varying corrections. The local tools of Berger [5] for the Cauchy domain should give the local law at the scale $N\ell(N)/m(N)$, with m the truncated mean, and the dominated regime with a lighter law of index 1 converges at the rate $1/\log N$. We have not worked out either line.

11 Computations

No proof uses a computed number. The computations check the exact identities and test the theorems at moderate N. Except where we say otherwise they use the zeta law $f(n) = n^{-\alpha-1}/\zeta(\alpha+1)$ for both signs, so $b = 1$, and the fair start. The exact law comes

α	N	ratio at $k = N/2$	ratio at $k = N/20$	sign changes
0.3	1000, 2000, 4000	1.0852, 1.0687, 1.0555	1.1023, 1.0845, 1.0694	1, 1, 1
0.5	1000, 2000, 4000	1.0149, 1.0105, 1.0075	1.0045, 1.0036, 1.0028	1, 1, 1
0.58	1000, 2000, 4000	0.9846, 0.9863, 0.9883	1.0142, 1.0114, 1.0090	1, 1, 1
0.61	1000, 2000, 4000	0.9696, 0.9739, 0.9781	1.0282, 1.0230, 1.0185	3, 3, 3
0.7	1000, 2000, 4000	0.9023, 0.9159, 0.9285	1.1202, 1.1004, 1.0833	3, 3, 3
0.9	1000, 2000, 4000	0.4903, 0.5168, 0.5433	2.2892, 2.1277, 1.9905	5, 5, 5

Table 1: The ratio $NP_N(k)/L_{\alpha,1}(k/N)$ and the number of sign changes of the first differences of P_N, for the zeta law and the fair start.

from (2.10), or from the first-run decomposition with the convolution powers computed by FFT. Every entry is a sum of nonnegative terms, so the relative rounding error stays small, and the total mass is 1 to 12 digits in every case. The identities of Section 2.5 and (2.10) agree with a sum over all sign strings for $N \le 12$, for two different laws of the two signs and a general initial run, to $2 \cdot 10^{-16}$, and the aged law of Section 7.2 agrees with a sum over all run sequences for $t_a + N \le 12$ to $6 \cdot 10^{-16}$.

The local law and the ends. Table 1 shows the ratio $NP_N(k)/L_{\alpha,1}(k/N)$ at $k = N/2$ and $k = N/20$, and the number of sign changes of $k \mapsto P_N(k+1) - P_N(k)$. At $\alpha = 1/2$ the central ratio is 1.0149, 1.0105 and 1.0075 for $N = 1000$, 2000 and 4000, and the distance to 1 shrinks by a factor close to $\sqrt{2}$ when N doubles. The convergence is slower away from $\alpha = 1/2$ and much slower as $\alpha \to 1$. The sign changes sit at $k = N/2$ for $\alpha \le 0.58$. At $\alpha = 0.7$ and $N = 1000$ they sit at $k = 104$, 500 and 896, near the minima of $L_{0.7,1}$. At $\alpha = 0.9$ and $N = 4000$ they sit at $k = 1$, 142, 2000, 3858 and 3999, and the two outer ones come from the ends, as in Example 9.4. On both sides of α_c the count is 1 at $\alpha = 0.58$ and 3 at $\alpha = 0.61$. Every count in the table equals the lower bound of Theorem 9.3. At the plus end, for $\alpha = 1/2$, the ratio $P_N(N-j)/(T(N)u(j))$ lies in $[1.0009, 1.0040]$ for $1 \le j \le 5$ at $N = 1000$ and in $[1.0004, 1.0020]$ at $N = 2000$, and the ratio of $P_N(N-j)$ to $T(N)u(j)\Phi_1(j/N)$ over $1 \le j \le 0.8N$ lies in $[1.0004, 1.0148]$ at $N = 1000$ and in $[1.0002, 1.0105]$ at $N = 2000$, as Theorem B(b) predicts. For unequal laws, the zeta law for the plus runs and $f_-(n) \propto (n+3)^{-\alpha-1}$ for the minus runs with a fresh plus start, the ratios to $L_{\alpha,b}$ at $x = 0.1, 0.3, \dots, 0.9$ lie in $[1.0121, 1.0397]$ at $N = 1000$ and in $[1.0069, 1.0199]$ at $N = 4000$ for $\alpha = 1/2$, where $b = 2.4498$.

The initial run. For $\alpha = 1/2$, a fair sign and $\nu_\pm(d) = d^{-1.2}/(2\zeta(1.2))$, the central ratio is 0.815, 0.836 and 0.855 for $N = 500$, 1000 and 2000. The gap to 1 decays like $N^{-0.2}$, the order of $\mathbb{P}(D \ge N)$, which enters through the first run with $\varepsilon N \le D < N$ in the proof of Theorem C(a). For D uniform on $\{1, \dots, N/2\}$ with a fair sign, $NP_N(k)$ at $k/N = 0.3$ and 0.5 is 0.898 and 1.325 at $N = 500$ and 0.884 and 1.299 at $N = 2000$, against the values 0.8694 and 1.2732 of the mixture (7.1) with $\rho_\pm = 1$ on $(0, \frac{1}{2})$ and $\pi^0 = 0$, and against the values 0.6946 and 0.6366 of $L_{1/2,1}$ for a fresh start.

The aged law. For Theorem E with $\theta \in \{\frac{1}{2}, 1, 2\}$ and $N = 500$, 1000, 2000, the aged law is computed from the first-run decomposition with the initial run (7.2). At $\alpha = 1/2$ and $\theta = 1$ the largest relative error of $NP_N(\lfloor xN \rfloor)$ against $L_\theta(x)$ on $x \in [0.1, 0.9]$ is below 5% at $N = 2000$ and decreases with N, and at $\alpha = 0.7$ it is below 15%. The atom $P_N(N)$ is 1.0282, 1.0198 and 1.0139 times $Q_+^\infty(1)$ at the three values of N for $\alpha = 1/2$, and 1.0418, 1.0293 and 1.0206 times $Q_+^\infty(\frac{1}{2})$. The ratio $P_N(N-j)/(T_+(N)c_+(\theta)u_-(j))$ for $j = 1, \dots, 5$ lies in $[1.050, 1.089]$ at $N = 2000$ for $\theta = 1$, and after removal of the term $T_+(N)\sum_{d \le j} \nu_-(d)u_-(j-d)$, which is $o(T_+(N))$ but only by a factor of order $1/(NT(N))$, it lies in $[1.0161, 1.0167]$. Each deviation shrinks by about $1/\sqrt{2}$ per doubling of N. The mass identity $\int_0^1 L_\theta + Q_+^\infty + Q_-^\infty = 1$ and the closed forms of Q_σ^∞ and c_σ agree with

quadrature to 10 digits.

The dominated regime. For Theorem F we take zeta laws with a fresh plus start, $N = 1000, 2000, 4000$, the mixed pair $(\alpha_+, \alpha_-) = (0.6, 1.5)$ and the infinite pair $(0.5, 0.8)$. The largest relative error of $D_N P_N(N-j)$ against $\rho(j/D_N)$ on $y \in [0.05, 1]$ is 0.304, 0.247 and 0.201 in the mixed case, decreasing like $N^{-0.3}$, and on $y \in [0.1, 3]$ it is 0.588, 0.533 and 0.478 in the infinite case, where the lighter sums carry a drift correction of relative size m^{1-1/α_-} with only about 83 cycles at $N = 4000$. The finite-N main term of the proof, $T_+(N-j)\sum_m u_m^-(j)\phi(mT_+(N-j))$, is closer, within 0.098 and 0.078 at $N = 4000$. At the plus end the ratio $P_N(N-j)/(T_+(N)u_-(j))$ for $j \le 5$ lies in $[1.0000, 1.0172]$ and $[1.0000, 1.0003]$ at $N = 4000$, and at the minus end $P_N(i)/(T_-(N)u_+(i))$ for $1 \le i \le 5$ is within 0.9% of 1 in both cases. In the mixed case the law has exactly 3 modes at every N, at $k = 1$, at $k = N$ and at an interior point near $N - y^* D_N$ with y^* the maximum of ϕ, since $\alpha_+ > 1/2$; for $(\alpha_+, \alpha_-) = (0.4, 1.5)$ it has exactly 2, at $k = 1$ and $k = N$. At $\alpha = 1/2$ the density ϕ agrees with $e^{-\pi x^2/4}$ to 8 digits, and the ratio density at $\alpha = 0.7$ agrees with (8.1) to 10 digits.

The shape. The roots are $\alpha_c = 0.594611644056836$ and $\alpha_1 = 0.736484448241517$. On a grid of α from 0.02 to 0.98 and $|\beta| \le 6$ in steps of 0.05, the number of zeros of the exponential sum that has the sign of $\frac{d}{dt}\log L_{\alpha,b}$ agrees with Theorem G, and it flips between $\beta_c(0.7) - 10^{-3}$ and $\beta_c(0.7) + 10^{-3}$. The transform (4.1) agrees with direct quadrature in 30-digit arithmetic, and at $\alpha = 1/2$ the double integral (2.3) agrees with $L_{1/2,1}$ to a relative error below 10^{-13} at $x = 0.2$, 0.5 and 0.7. For Proposition 9.8 we use Sibuya laws, $T(n) = \prod_{i<n}(1 - \alpha/i)$, which are log-convex, and $N = 250, 500, \dots, 3000$. With $b = 1$ and a fresh plus or a fair start the law has exactly 2 modes at every N for $\alpha = 0.3$ and 0.5, and exactly 3 for $\alpha = 0.65$, 0.7 and 0.9, with the bulk minima at $0.136N$, $0.085N$ and $0.021N$ against the minima 0.1388, 0.085 and 0.0216 of $L_{\alpha,b}$. For the law $f(n) \propto n^{-1.7}$ on $n \ge 2$ the profiles have maxima at $j = 0, 2, 4$ and the counts are 6 for the fresh plus start and 7 for the fair start at every N. For unequal tails of the same index, the minus law a mixture of a Sibuya law and a geometric law so that $b = 0.6$ at $\alpha = 0.7$, inside the window, the count is 3 at every N; for $b = 0.5$ at $\alpha = 0.65$, outside the window, it is 3 up to $N = 1500$ and 2 from $N = 2000$ on; and for $b = 0.25$ at $\alpha = 0.7$ it is still 3 at $N = 3000$, where the local law itself is far from its limit because the light component adds $O(N^\alpha)$ steps of finite mean. For Theorem 9.7, the ratio $D_N(j)/(T_+(N)(u_-(j+1) - u_-(j)))$ at $j = 5$, 20, 50 and $N = 3000$ is 0.952, 0.897, 0.847 for $\alpha = 0.3$ and 0.969, 0.917, 0.819 for $\alpha = 0.7$, rising with N at fixed j. For (B), $N^2(P_N(k+1) - P_N(k))$ at $x = 0.2$, 0.3, 0.4, 0.45 and $N = 3000$ is -1.688, -0.681, -0.266, -0.125 against $L'_{\alpha,b} = -1.571$, -0.635, -0.248, -0.117 for $\alpha = 0.3$, and 1.435, 1.729, 1.261, 0.697 against 1.413, 1.681, 1.208, 0.666 for $\alpha = 0.7$. The scans decide signs from the floating-point differences, whose smallest relative gap at a counted extremum was $7 \cdot 10^{-10}$ against a rounding error of order 10^{-13}.

The scripts and their outputs, each headed by the hash of its script, are in the folder kagey-problems/problem131/paperG of https://github.com/apemm/Kagey-Problems. The command `python -I verify_all.py` runs the main checks in a few minutes, and README.md lists what each script tests and how to run the longer scans.

A Routine facts

This appendix proves the routine facts used in Sections 2 to 4.

A.1 Regular variation

Lemma A.1. *Let T be regularly varying with index $-\alpha \in (-1, 0)$, nonincreasing and positive.*

(i) (R4) of Section 3.1 holds.

(ii) If a is increasing with $tT(a(t)) \to 1$, then $\sup_{m\ge 1} mT(a_m) < \infty$.

(iii) With $m_N(t) = \lfloor t/T(N) \rfloor$, $a_{m_N(t)}/N \to t^{1/\alpha}$ uniformly for t in compact subsets of $(0,\infty)$.

Proof. (i) The function $\phi(w) = wT(w)$ is regularly varying with index $1-\alpha > 0$, and it is positive, so $\inf_{w\ge 1}\phi(w) > 0$ because $\phi(w) \to \infty$. By Potter's theorem [7, Theorem 1.5.6] there is x_0 with $\phi(y)/\phi(x) \le 2(y/x)^{(1-\alpha)/2}$ for $x_0 \le y \le x$. By (R2), $2^l T(2^{l-1}) \le C\phi(2^l)$. The terms with $2^l \ge x_0$ add up to at most $C\phi(x)\sum_{l\ge 0} 2^{-l(1-\alpha)/2}$, and the finitely many terms with $2^l < x_0$ add up to a constant, which is at most $C\phi(x)$. For the second sum let $x \ge x_0$. The first term with $2^l > x$ has $2^{l-1} \in (x/2, x]$, so it is at most $T(x/2) \le CT(x)$ by (R2). The later terms have $2^{l-1} \ge x$, and Potter's bounds give $T(y)/T(x) \le 2(y/x)^{-\alpha/2}$ for $y \ge x \ge x_0$. So the sum is at most $CT(x)\sum_{l\ge 0} 2^{-l\alpha/2}$. For $1 \le x < x_0$ the sum is at most $\sum_{l\ge 1} T(2^{l-1}) < \infty$, which is at most $CT(x_0) \le CT(x)$.

(ii) The sequence $mT(a_m)$ converges to 1 and each term is finite.

(iii) Fix $t > 0$ and write $m = m_N(t)$. Then $mT(N) \to t$ and $mT(a_m) \to 1$, so $T(a_m)/T(N) \to 1/t$. Take a subsequence along which $a_m/N \to \lambda \in [0,\infty]$. If $\lambda \in (0,\infty)$, (R1) gives $T(a_m)/T(N) \to \lambda^{-\alpha}$, so $\lambda = t^{1/\alpha}$. If $\lambda = 0$ or $\lambda = \infty$, Potter's bounds give $T(a_m)/T(N) \to \infty$ or $\to 0$, which is impossible. So $a_m/N \to t^{1/\alpha}$. The functions $t \mapsto a_{m_N(t)}/N$ are nondecreasing and the limit is continuous and increasing, so the convergence is uniform on compact sets. □

A.2 The Poisson integral

Lemma A.2. *Let $\Lambda \ge 0$ be integrable on $(0,1)$ and continuous at $x \in (0,1)$, and let $G(z) = \int_0^1 \Lambda(v)\,dv/(z-v)$ for $z \notin [0,1]$. Then $-\frac{1}{\pi}\operatorname{Im} G(x+i\varepsilon) \to \Lambda(x)$ as $\varepsilon \downarrow 0$.*

Proof. Since $\operatorname{Im}(x+i\varepsilon-v)^{-1} = -\varepsilon/((x-v)^2+\varepsilon^2)$,

$$-\frac{1}{\pi}\operatorname{Im} G(x+i\varepsilon) = \int_0^1 \Lambda(v)\,P_\varepsilon(x-v)\,dv, \qquad P_\varepsilon(u) = \frac{\varepsilon}{\pi(u^2+\varepsilon^2)}.$$

The kernel P_ε is positive with total mass 1 on $\mathbb{R}$, and its mass outside $[-\eta,\eta]$ and its supremum there tend to 0 for each $\eta > 0$. Split the integral at $|x-v| \le \eta$. The near part differs from $\Lambda(x)\int_{|u|\le\eta} P_\varepsilon$ by at most $\sup_{|v-x|\le\eta}|\Lambda(v)-\Lambda(x)|$, and the far part is at most $\sup_{|u|>\eta} P_\varepsilon(u)\int_0^1 \Lambda$. Let $\varepsilon \to 0$ and then $\eta \to 0$. □

A.3 The normalization of the minus sums

Under (RV$_\alpha$), $mT_-(a_m z) \to bz^{-\alpha}$ for every $z > 0$, by (R1) and $mT_+(a_m) \to 1$. So the sum of m minus runs, divided by a_m, converges in law to the positive stable variable whose Lévy measure has tail $bz^{-\alpha}$, which is $b^{1/\alpha}Z$. Gnedenko's local limit theorem [17, Theorem 4.2.1] holds for every norming sequence under which the sums converge, so it applies with the norming a_m and the density g_b. This is the form used in Lemma 3.1(i).

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